# MCP2518FD

## External CAN FD Controller with SPI Interface

### Features

**General**

- External CAN FD Controller with Serial Peripheral Interface (SPI)
- Arbitration Bit Rate up to 1 Mbps
- Data Bit Rate up to 8 Mbps
- CAN FD Controller modes
  - Mixed CAN 2.0B and CAN FD Mode
  - CAN 2.0B Mode
- Conforms to ISO 11898-1:2015

**Message FIFOs**

- 31 FIFOs, configurable as Transmit or Receive FIFOs
- One Transmit Queue (TXQ)
- Transmit Event FIFO (TEF) with 32 bit time stamp

**Message Transmission**

- Message transmission prioritization:
  - Based on priority bit field
  - Message with lowest ID gets transmitted first using the Transmit Queue (TXQ)
- Programmable automatic retransmission attempts: unlimited, 3 attempts or disabled

**Message Reception**

- 32 Flexible Filter and Mask Objects
- Each object can be configured to filter either:
  - Standard ID + first 18 data bits, or
  - Extended ID
- 32-bit Time Stamp

**Special Features**

- $V_{DD}$: 2.7 to 5.5V
- Active Current: maximum 20 mA at 5.5 V, 40 MHz CAN clock
- Sleep Current: 15 μA, typical
- Low Power Mode current: maximum 10 μA from -40°C to +150°C
- Message Objects are located in RAM: 2 KB
- Up to 3 Configurable Interrupt Pins
- Bus Health Diagnostics and Error Counters
- Transceiver Standby Control
- Start of frame pin for indicating the beginning of messages on the bus
- AEC-Q100 Qualified
- Temperature Ranges:
  - Extended (E): -40°C to +125°C
  - High (H): -40°C to +150°C

**Oscillator Options**

- 40, 20 or 4 MHz Crystal or Ceramic Resonator; External Clock Input
- Clock Output with Prescaler

**SPI Interface**

- Up to 20 MHz SPI clock speed
- Supports SPI Modes `0,0` and `1,1`
- Registers and bit fields are arranged in a way to enable efficient access through SPI

**Safety Critical Systems**

- SPI commands with CRC to detect noise on SPI interface
- Error Correction Code (ECC) protected RAM

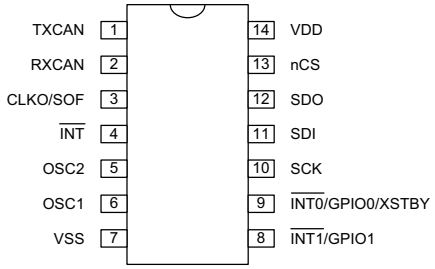
**Additional Features**

- GPIO pins: $\overline{INT0}$ and $\overline{INT1}$ can be configured as general purpose I/O
- Open drain outputs: TXCAN, $\overline{INT}$, $\overline{INT0}$, and $\overline{INT1}$ pins can be configured as push/pull or open drain outputs
- ISO 26262 Functional Safety ready

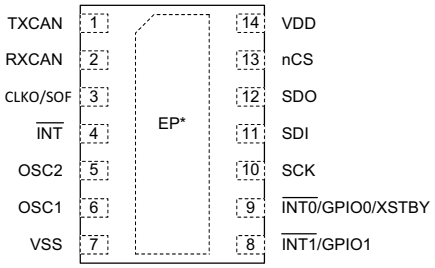
© 2019-2020 Microchip Technology Inc.

## Package Types

**MCP2518FD**
SOIC14




**MCP2518FD**
VDFN14 with wettable flanks*




VDFN14 includes an Exposed Thermal Pad (EP); see Table 1-1


© 2019-2020 Microchip Technology Inc.

## 1.0 DEVICE OVERVIEW

The MCP2518FD device is a cost-effective and small-footprint CAN FD controller that can be easily added to a microcontroller with an available SPI interface. A CAN FD channel can be easily added to a microcontroller that is either lacking a CAN FD peripheral or does not have enough CAN FD channels.

MCP2518FD supports both CAN frames in the Classical format (CAN2.0B) and CAN Flexible Data Rate (CAN FD) format, as specified in ISO 11898-1:2015.

The MCP2518FD device was improved as follows:

- Added Low Power Mode (LPM), in order to reduce leakage current to 10 μA over the full temperature range.
- Extended SEQ field in Transmit Message Object and Transmit Event FIFO Object from 7 to 23 bits.
- Added DEVID register to distinguish between future members of the device family.
- Switched to saw cut DFN package with wettable flanks.

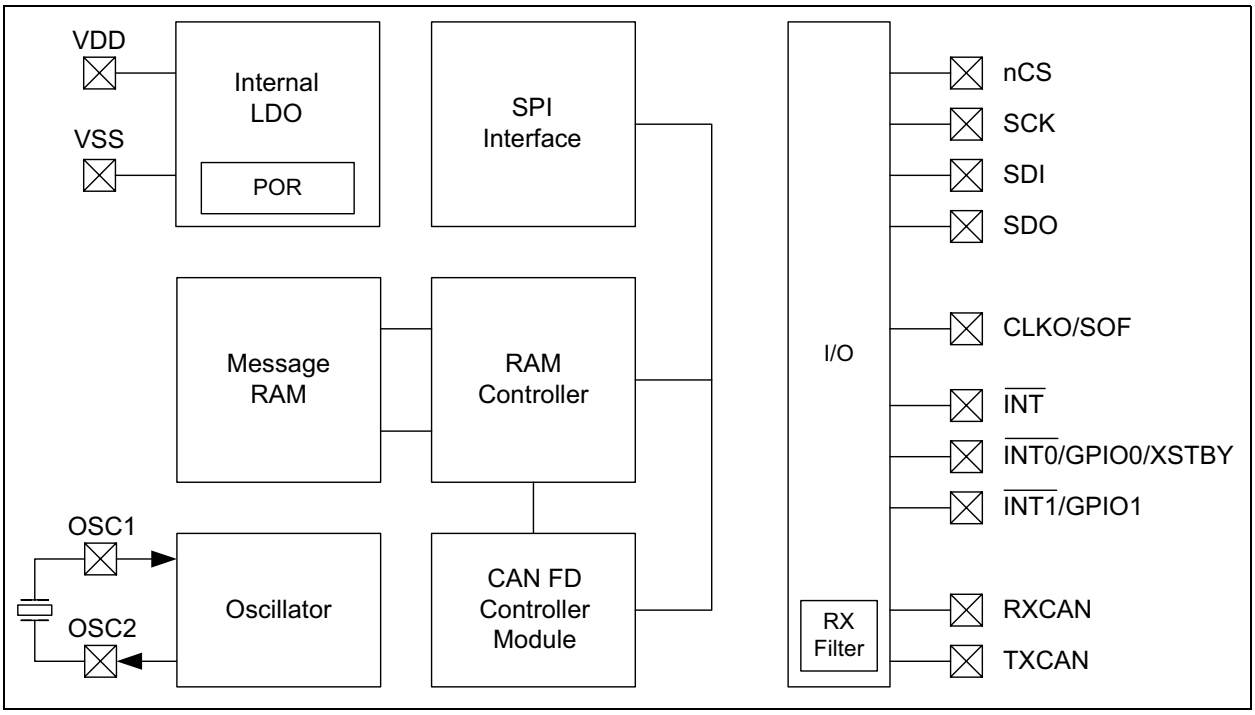
## 1.1 Block Diagram

Figure 1.1 shows the block diagram of the MCP2518FD device. MCP2518FD contains the following main blocks:

- The CAN FD Controller module implements the CAN FD protocol, and contains the FIFOs and Filters.
- The SPI interface is used to control the device by accessing Special Function Registers (SFR) and RAM.
- The RAM controller arbitrates the RAM accesses between the SPI and CAN FD Controller module.
- The Message RAM is used to store the data of the Message Objects.
- The oscillator generates the CAN clock.
- The Internal LDO and POR circuit.
- The I/O control.

**Note 1:** This data sheet summarizes the features of the MCP2518FD device. It is not intended to be a comprehensive reference source. To complement the information in this data sheet, refer to the related section of the “MCP25xxFD Family Reference Manual”.

**FIGURE 1-1: MCP2518FD BLOCK DIAGRAM**



## 1.2 Pinout Description

Table 1-1 describes the functions of the pins.

**TABLE 1-1: MCP2518FD STANDARD PINOUT VERSION**

| Pin Name | SOIC | VDFN | Pin Type | Description |
|---|---|---|---|---|
| TXCAN | 1 | 1 | O | Transmit output to CAN FD transceiver |
| RXCAN | 2 | 2 | I | Receive input from CAN FD transceiver |
| CLKO/SOF | 3 | 3 | O | Clock output/Start of Frame output |
| $\overline{\text{INT}}$ | 4 | 4 | O | Interrupt output (active low) |
| OSC2 | 5 | 5 | O | External oscillator output |
| OSC1 | 6 | 6 | I | External oscillator input |
| VSS | 7 | 7 | P | Ground |
| $\overline{\text{INT1}}$/GPIO1 | 8 | 8 | I/O | RX Interrupt output (active low)/GPIO |
| $\overline{\text{INT0}}$/GPIO0/ XSTBY | 9 | 9 | I/O | TX Interrupt output (active low)/GPIO/ Transceiver Standby output |
| SCK | 10 | 10 | I | SPI clock input |
| SDI | 11 | 11 | I | SPI data input |
| SDO | 12 | 12 | O | SPI data output |
| nCS | 13 | 13 | I | SPI chip select input |
| VDD | 14 | 14 | P | Positive Supply |
| EP | - | 15 | P | Exposed Pad; connect to VSS |

**Legend:** P = Power, I = Input, O = Output


© 2019-2020 Microchip Technology Inc.

## 1.3 Typical Application

Figure 1-2 shows an example of a typical application of the MCP2518FD device. In this example, the microcontroller operates at 3.3V.

The MCP2518FD device interfaces directly with microcontrollers operating at 2.7V to 5.5V. In addition, the MCP2518FD device connects directly to high-speed CAN FD transceivers. There are no external level shifters required when connecting VDD of the MCP2518FD and the microcontroller to VIO of the transceiver.

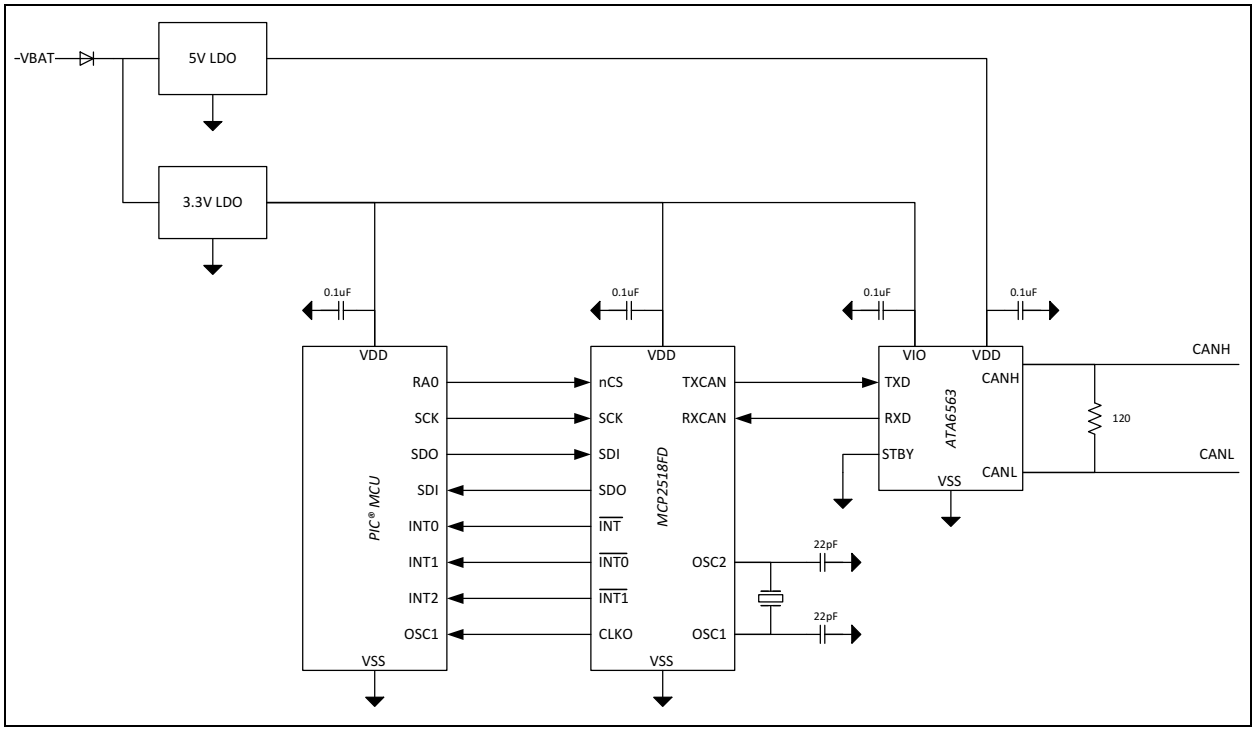
The VDD of the CAN FD transceiver is connected to 5V.

The SPI interface is used to configure and control the CAN FD controller.

The MCP2518FD device signals interrupts to the microcontroller by using $\overline{\text{INT}}$, $\overline{\text{INT0}}$ and $\overline{\text{INT1}}$. Interrupts need to be cleared by the microcontroller through SPI.

The CLKO pin provides the clock to the microcontroller.

**FIGURE 1-2: MCP2518FD INTERFACING WITH A 3.3V MICROCONTROLLER**



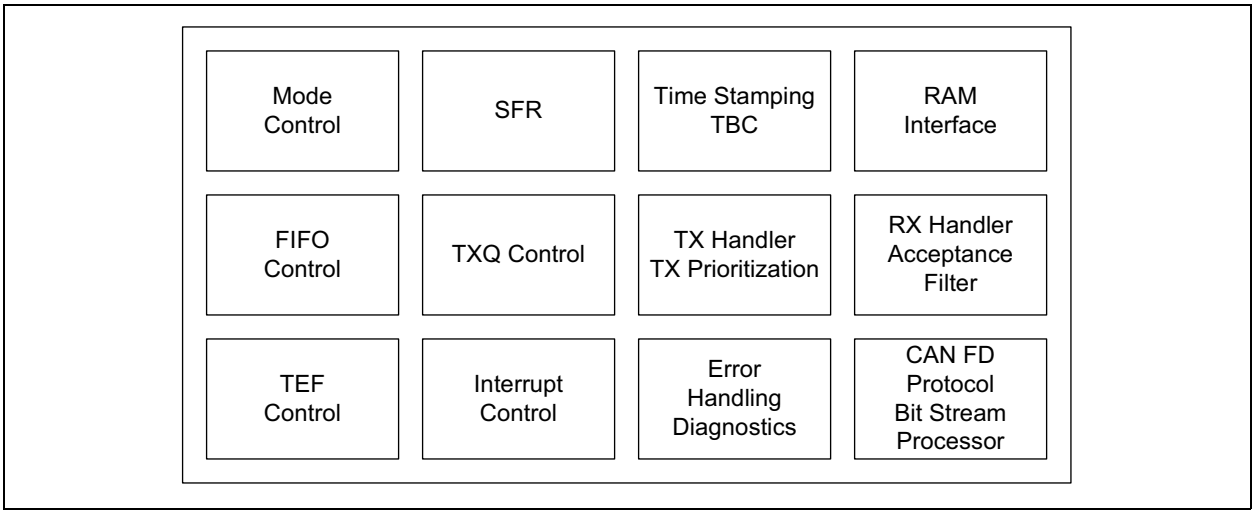
## 2.0 CAN FD CONTROLLER MODULE

Figure 2-1 shows the main blocks of the CAN FD Controller module:

- The CAN FD Controller module has multiple modes:
  - Configuration
  - Normal CAN FD
  - Normal CAN 2.0
  - Sleep (normal Sleep mode and Low Power Mode)
  - Listen Only
  - Restricted Operation
  - Internal and External Loop back modes
- The CAN FD Bit Stream Processor (BSP) implements the Medium Access Control of the CAN FD protocol described in ISO 11898-1:2015. It serializes and de-serializes the bit stream, encodes and decodes the CAN FD frames, manages the medium access, acknowledges frames and detects and signals errors.
- The TX Handler prioritizes the messages that are requested for transmission by the Transmit FIFOs. It uses the RAM Interface to fetch the transmit data from RAM and provides it to the BSP for transmission.
- The BSP provides received messages to the RX Handler. The RX Handler uses the Acceptance Filter to filter out messages that shall be stored into Receive FIFOs. It uses the RAM Interface to store received data into RAM.
- Each FIFO can be configured either as a Transmit or Receive FIFO. The FIFO Control keeps track of the FIFO Head and Tail, and calculates the User Address. For a TX FIFO, the User Address points to the address in RAM where the data for the next transmit message shall be stored. For a RX FIFO, the User Address points to the address in RAM where the data of the next receive message shall be read. The User notifies the FIFO that a message was written to or read from RAM by incrementing the Head/Tail of the FIFO.
- The Transmit Queue (TXQ) is a special transmit FIFO that transmits the messages based on the ID of the messages stored in the queue.
- The Transmit Event FIFO (TEF) stores the message IDs of the transmitted messages.
- A free-running Time Base Counter is used to time stamp received messages. Messages in the TEF can also be time stamped.
- The CAN FD Controller module generates interrupts when new messages are received or when messages were transmitted successfully.
- The SFR are used to control and to read the status of the CAN FD Controller module.

**Note 1:** This data sheet summarizes the features of the CAN FD Controller module. It is not intended to be a comprehensive reference source. To complement the information in this data sheet, refer to the related section of the “MCP25xxFD Family Reference Manual”.

**FIGURE 2-1: CAN FD CONTROLLER MODULE BLOCK DIAGRAM**





© 2019-2020 Microchip Technology Inc.

## 3.0 MEMORY ORGANIZATION

Figure 3-1 illustrates the main sections of the memory and its address ranges:

- MCP2518FD Special Function Registers
- CAN FD Controller module SFR
- Message Memory (RAM)

The SFR are 32-bit wide. The LSB is located at the lower address, for example, the LSB of C1CON is located at address 0x000, while its MSB is located at address 0x003.

Table 3-1 lists the MCP2518FD specific registers. The first column contains the address of the SFR.

Table 3-2 lists the registers of the CAN FD Controller module. The first column contains the address of the SFR.

**FIGURE 3-1: MEMORY MAP**

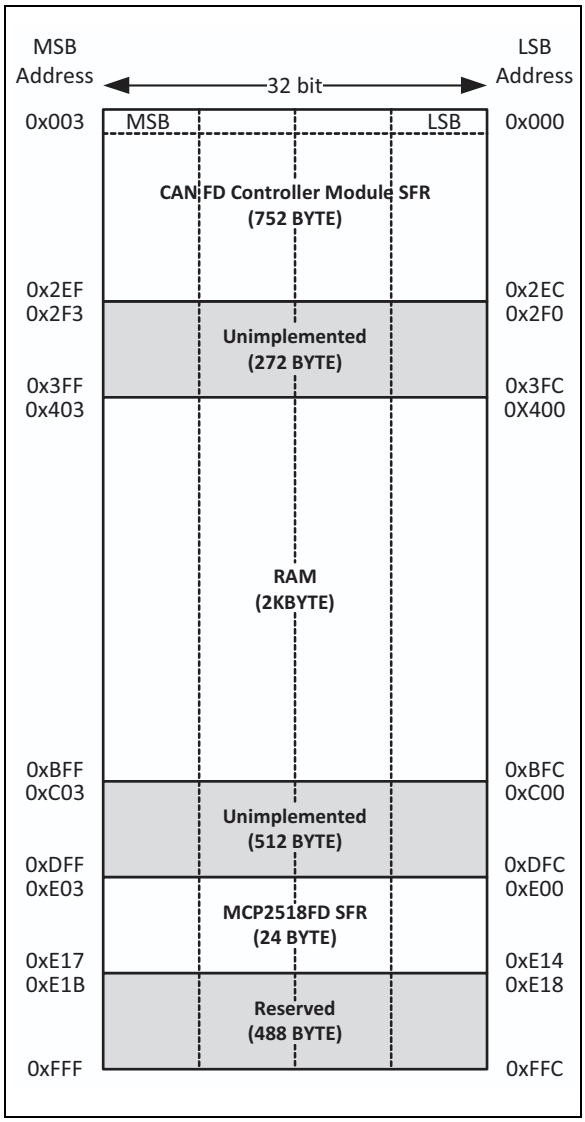


**TABLE 3-1: MCP2518FD REGISTER SUMMARY**

| Address | Name | | Bit 31/23/15/7 | Bit 30/22/14/6 | Bit 29/21/13/5 | Bit 28/20/12/4 | Bit 27/19/11/3 | Bit 26/18/10/2 | Bit 25/17/9/1 | Bit 24/16/8/0 |
|---|---|---|---|---|---|---|---|---|---|---|
| E03 | OSC | 31:24 | — | — | — | — | — | — | — | — |
| E02 | | 23:16 | — | — | — | — | — | — | — | — |
| E01 | | 15:8 | — | — | — | SCLKRDY | — | OSCRDY | — | PLLRDY |
| E00(1) | | 7:0 | — | CLKODIV[1:0] | | SCLKDIV | LPMEN | OSCDIS | — | PLLEN |
| | IOCON | 31:24 | — | INTOD | SOF | TXCANOD | — | — | PM1 | PM0 |
| | | 23:16 | — | — | — | — | — | — | GPIO1 | GPIO0 |
| | | 15:8 | — | — | — | — | — | — | LAT1 | LAT0 |
| E04 | | 7:0 | — | XSTBYEN | — | — | — | — | TRIS1 | TRIS0 |
| | CRC | 31:24 | — | — | — | — | — | — | FERRIE | CRCERRIE |
| | | 23:16 | — | — | — | — | — | — | FERRIF | CRCERRIF |
| | | 15:8 | CRC[15:8] | | | | | | | |
| E08 | | 7:0 | CRC[7:0] | | | | | | | |
| | ECCCON | 31:24 | — | — | — | — | — | — | — | — |
| | | 23:16 | — | — | — | — | — | — | — | — |
| | | 15:8 | — | PARITY[6:0] | | | | | | |
| E0C | | 7:0 | — | — | — | — | — | DEDIE | SECIE | ECCEN |
| | ECCSTAT | 31:24 | — | — | — | — | ERRADDR[11:8] | | | |
| | | 23:16 | ERRADDR[7:0] | | | | | | | |
| | | 15:8 | — | — | — | — | — | — | — | — |
| E10 | | 7:0 | — | — | — | — | — | DEDIF | SECIF | — |
| | DEVID | 31:24 | — | — | — | — | — | — | — | — |
| | | 23:16 | — | — | — | — | — | — | — | — |
| | | 15:8 | — | — | — | — | — | — | — | — |
| E14 | | 7:0 | ID[3:0] | | | | REV[3:0] | | | |

**Note 1:** The lower order byte of the 32-bit register resides at the low-order address.

**2:** The bit fields in the IOCON register must be written using single data byte SFR WRITE instructions.


© 2019-2020 Microchip Technology Inc.

**TABLE 3-2: CAN FD CONTROLLER MODULE REGISTER SUMMARY**

| Addr. | Name | | Bit 31/23/15/7 | Bit 30/22/14/6 | Bit 29/21/13/5 | Bit 28/20/12/4 | Bit 27/19/11/3 | Bit 26/18/10/2 | Bit 25/17/9/1 | Bit 24/16/8/0 |
|---|---|---|---|---|---|---|---|---|---|---|
| 03 | C1CON | 31:24 | TXBWS[3:0] | | | | ABAT | REQOP[2:0] | | |
| 02 | | 23:16 | OPMOD[2:0] | | | TXQEN | STEF | SERR2LOM | ESIGM | RTXAT |
| 01 | | 15:8 | — | — | — | BRSDIS | BUSY | WFT[1:0] | | WAKFIL |
| 00[1] | | 7:0 | — | PXEDIS | ISOCRCEN | DNCNT[4:0] | | | | |
| | C1NBTCFG | 31:24 | BRP[7:0] | | | | | | | |
| | | 23:16 | TSEG1[7:0] | | | | | | | |
| | | 15:8 | — | TSEG2[6:0] | | | | | | |
| 04 | | 7:0 | — | SJW[6:0] | | | | | | |
| | C1DBTCFG | 31:24 | BRP[7:0] | | | | | | | |
| | | 23:16 | — | — | — | TSEG1[4:0] | | | | |
| | | 15:8 | — | — | — | — | TSEG2[3:0] | | | |
| 08 | | 7:0 | — | — | — | — | SJW[3:0] | | | |
| | C1TDC | 31:24 | — | — | — | — | — | — | EDGFLTEN | SID11EN |
| | | 23:16 | — | — | — | — | — | — | TDCMOD[1:0] | |
| | | 15:8 | — | TDCO[6:0] | | | | | | |
| 0C | | 7:0 | — | — | TDCV[5:0] | | | | | |
| | C1TBC | 31:24 | TBC[31:24] | | | | | | | |
| | | 23:16 | TBC[23:16] | | | | | | | |
| | | 15:8 | TBC[15:8] | | | | | | | |
| 10 | | 7:0 | TBC[7:0] | | | | | | | |
| | C1TSCON | 31:24 | — | — | — | — | — | — | — | — |
| | | 23:16 | — | — | — | — | — | TSRES | TSEOF | TBCEN |
| | | 15:8 | — | — | — | — | — | — | TBCPRE[9:8] | |
| 14 | | 7:0 | TBCPRE[7:0] | | | | | | | |
| | C1VEC | 31:24 | — | RXCODE[6:0] | | | | | | |
| | | 23:16 | — | TXCODE[6:0] | | | | | | |
| | | 15:8 | — | — | — | FILHIT[4:0] | | | | |
| 18 | | 7:0 | — | ICODE[6:0] | | | | | | |
| | C1INT | 31:24 | IVMIE | WAKIE | CERRIE | SERRIE | RXOVIE | TXATIE | SPICRCIE | ECCIE |
| | | 23:16 | — | — | — | TEFIE | MODIE | TBCIE | RXIE | TXIE |
| | | 15:8 | IVMIF | WAKIF | CERRIF | SERRIF | RXOVIF | TXATIF | SPICRCIF | ECCIF |
| 1C | | 7:0 | — | — | — | TEFIF | MODIF | TBCIF | RXIF | TXIF |
| | C1RXIF | 31:24 | RFIF[31:24] | | | | | | | |
| | | 23:16 | RFIF[23:16] | | | | | | | |
| | | 15:8 | RFIF[15:8] | | | | | | | |
| 20 | | 7:0 | RFIF[7:1] | | | | | | | — |
| | C1TXIF | 31:24 | TFIF[31:24] | | | | | | | |
| | | 23:16 | TFIF[23:16] | | | | | | | |
| | | 15:8 | TFIF[15:8] | | | | | | | |
| 24 | | 7:0 | TFIF[7:0] | | | | | | | |
| | C1RXOVIF | 31:24 | RFOVIF[31:24] | | | | | | | |
| | | 23:16 | RFOVIF[23:16] | | | | | | | |
| | | 15:8 | RFOVIF[15:8] | | | | | | | |
| 28 | | 7:0 | RFOVIF[7:1] | | | | | | | — |
| | C1TXATIF | 31:24 | TFATIF[31:24] | | | | | | | |
| | | 23:16 | TFATIF[23:16] | | | | | | | |
| | | 15:8 | TFATIF[15:8] | | | | | | | |
| 2C | | 7:0 | TFATIF[7:0] | | | | | | | |

**Note 1:** The lower order byte of the 32-bit register resides at the low-order address.

**2:** Reserved register reads 0.

**TABLE 3-2: CAN FD CONTROLLER MODULE REGISTER SUMMARY (CONTINUED)**

| Addr. | Name | | Bit 31/23/15/7 | Bit 30/22/14/6 | Bit 29/21/13/5 | Bit 28/20/12/4 | Bit 27/19/11/3 | Bit 26/18/10/2 | Bit 25/17/9/1 | Bit 24/16/8/0 |
|---|---|---|---|---|---|---|---|---|---|---|
| | C1TXREQ | 31:24 | TXREQ[31:24] | | | | | | | |
| | | 23:16 | TXREQ[23:16] | | | | | | | |
| | | 15:8 | TXREQ[15:8] | | | | | | | |
| 30 | | 7:0 | TXREQ[7:0] | | | | | | | |
| | C1TREC | 31:24 | — | — | — | — | — | — | — | — |
| | | 23:16 | — | — | TXBO | TXBP | RXBP | TXWARN | RXWARN | EWARN |
| | | 15:8 | TEC[7:0] | | | | | | | |
| 34 | | 7:0 | REC[7:0] | | | | | | | |
| | C1BDIAG0 | 31:24 | DTERRCNT[7:0] | | | | | | | |
| | | 23:16 | DRERRCNT[7:0] | | | | | | | |
| | | 15:8 | NTERRCNT[7:0] | | | | | | | |
| 38 | | 7:0 | NRERRCNT[7:0] | | | | | | | |
| | C1BDIAG1 | 31:24 | DLCMM | ESI | DCRCERR | DSTUFERR | DFORMERR | — | DBIT1ERR | DBIT0ERR |
| | | 23:16 | TXBOERR | — | NCRCERR | NSTUFERR | NFORMERR | NACKERR | NBIT1ERR | NBIT0ERR |
| | | 15:8 | EFMSGCNT[15:8] | | | | | | | |
| 3C | | 7:0 | EFMSGCNT[7:0] | | | | | | | |
| | C1TEFCON | 31:24 | — | — | — | FSIZE[4:0] | | | | |
| | | 23:16 | — | — | — | — | — | — | — | — |
| | | 15:8 | — | — | — | — | — | FRESET | — | UINC |
| 40 | | 7:0 | — | — | TEFTSEN | — | TEFOVIE | TEFFIE | TEFHIE | TEFNEIE |
| | C1TEFSTA | 31:24 | — | — | — | — | — | — | — | — |
| | | 23:16 | — | — | — | — | — | — | — | — |
| | | 15:8 | — | — | — | — | — | — | — | — |
| 44 | | 7:0 | — | — | — | — | TEFOVIF | TEFFIF | TEFHIF | TEFNEIF |
| | C1TEFUA | 31:24 | TEFUA[31:24] | | | | | | | |
| | | 23:16 | TEFUA[23:16] | | | | | | | |
| | | 15:8 | TEFUA[15:8] | | | | | | | |
| 48 | | 7:0 | TEFUA[7:0] | | | | | | | |
| | Reserved(2) | 31:24 | Reserved[31:24] | | | | | | | |
| | | 23:16 | Reserved[23:16] | | | | | | | |
| | | 15:8 | Reserved[15:8] | | | | | | | |
| 4C | | 7:0 | Reserved[7:0] | | | | | | | |
| | C1TXQCON | 31:24 | PLSIZE[2:0] | | | FSIZE[4:0] | | | | |
| | | 23:16 | — | TXAT[1:0] | | TXPRI[4:0] | | | | |
| | | 15:8 | — | — | — | — | — | FRESET | TXREQ | UINC |
| 50 | | 7:0 | TXEN | — | — | TXATIE | — | TXQEIE | — | TXQNIE |
| | C1TXQSTA | 31:24 | — | — | — | — | — | — | — | — |
| | | 23:16 | — | — | — | — | — | — | — | — |
| | | 15:8 | — | — | — | TXQCI[4:0] | | | | |
| 54 | | 7:0 | TXABT | TXLARB | TXERR | TXATIF | — | TXQEIF | — | TXQNIF |
| | C1TXQUA | 31:24 | TXQUA[31:24] | | | | | | | |
| | | 23:16 | TXQUA[23:16] | | | | | | | |
| | | 15:8 | TXQUA[15:8] | | | | | | | |
| 58 | | 7:0 | TXQUA[7:0] | | | | | | | |

**Note 1:** The lower order byte of the 32-bit register resides at the low-order address.

**2:** Reserved register reads 0.


© 2019-2020 Microchip Technology Inc.

**TABLE 3-2: CAN FD CONTROLLER MODULE REGISTER SUMMARY (CONTINUED)**

| Addr. | Name | | Bit 31/23/15/7 | Bit 30/22/14/6 | Bit 29/21/13/5 | Bit 28/20/12/4 | Bit 27/19/11/3 | Bit 26/18/10/2 | Bit 25/17/9/1 | Bit 24/16/8/0 |
|---|---|---|---|---|---|---|---|---|---|---|
| | C1FIFOCON1 | 31:24 | PLSIZE[2:0] | | | FSIZE[4:0] | | | | |
| | | 23:16 | — | TXAT[1:0] | | TXPRI[4:0] | | | | |
| | | 15:8 | — | — | — | — | — | FRESET | TXREQ | UINC |
| 5C | | 7:0 | TXEN | RTREN | RXTSEN | TXATIE | RXOVIE | TFERFFIE | TFHRFHIE | TFNRFNIE |
| | C1FIFOSTA1 | 31:24 | — | — | — | — | — | — | — | — |
| | | 23:16 | — | — | — | — | — | — | — | — |
| | | 15:8 | — | — | — | FIFOCI[4:0] | | | | |
| 60 | | 7:0 | TXABT | TXLARB | TXERR | TXATIF | RXOVIF | TFERFFIF | TFHRFHIF | TFNRFNIF |
| | C1FIFOUA1 | 31:24 | FIFOUA[31:24] | | | | | | | |
| | | 23:16 | FIFOUA[23:16] | | | | | | | |
| | | 15:8 | FIFOUA[15:8] | | | | | | | |
| 64 | | 7:0 | FIFOUA[7:0] | | | | | | | |
| 68 | C1FIFOCON2 | 31:0 | same as C1FIFOCON1 | | | | | | | |
| 6C | C1FIFOSTA2 | 31:0 | same as C1FIFOSTA1 | | | | | | | |
| 70 | C1FIFOUA2 | 31:0 | same as C1FIFOUA1 | | | | | | | |
| 74 | C1FIFOCON3 | 31:0 | same as C1FIFOCON1 | | | | | | | |
| 78 | C1FIFOSTA3 | 31:0 | same as C1FIFOSTA1 | | | | | | | |
| 7C | C1FIFOUA3 | 31:0 | same as C1FIFOUA1 | | | | | | | |
| 80 | C1FIFOCON4 | 31:0 | same as C1FIFOCON1 | | | | | | | |
| 84 | C1FIFOSTA4 | 31:0 | same as C1FIFOSTA1 | | | | | | | |
| 88 | C1FIFOUA4 | 31:0 | same as C1FIFOUA1 | | | | | | | |
| 8C | C1FIFOCON5 | 31:0 | same as C1FIFOCON1 | | | | | | | |
| 90 | C1FIFOSTA5 | 31:0 | same as C1FIFOSTA1 | | | | | | | |
| 94 | C1FIFOUA5 | 31:0 | same as C1FIFOUA1 | | | | | | | |
| 98 | C1FIFOCON6 | 31:0 | same as C1FIFOCON1 | | | | | | | |
| 9C | C1FIFOSTA6 | 31:0 | same as C1FIFOSTA1 | | | | | | | |
| A0 | C1FIFOUA6 | 31:0 | same as C1FIFOUA1 | | | | | | | |
| A4 | C1FIFOCON7 | 31:0 | same as C1FIFOCON1 | | | | | | | |
| A8 | C1FIFOSTA7 | 31:0 | same as C1FIFOSTA1 | | | | | | | |
| AC | C1FIFOUA7 | 31:0 | same as C1FIFOUA1 | | | | | | | |
| B0 | C1FIFOCON8 | 31:0 | same as C1FIFOCON1 | | | | | | | |
| B4 | C1FIFOSTA8 | 31:0 | same as C1FIFOSTA1 | | | | | | | |
| B8 | C1FIFOUA8 | 31:0 | same as C1FIFOUA1 | | | | | | | |
| BC | C1FIFOCON9 | 31:0 | same as C1FIFOCON1 | | | | | | | |
| C0 | C1FIFOSTA9 | 31:0 | same as C1FIFOSTA1 | | | | | | | |
| C4 | C1FIFOUA9 | 31:0 | same as C1FIFOUA1 | | | | | | | |
| C8 | C1FIFOCON10 | 31:0 | same as C1FIFOCON1 | | | | | | | |
| CC | C1FIFOSTA10 | 31:0 | same as C1FIFOSTA1 | | | | | | | |
| D0 | C1FIFOUA10 | 31:0 | same as C1FIFOUA1 | | | | | | | |
| D4 | C1FIFOCON11 | 31:0 | same as C1FIFOCON1 | | | | | | | |
| D8 | C1FIFOSTA11 | 31:0 | same as C1FIFOSTA1 | | | | | | | |
| DC | C1FIFOUA11 | 31:0 | same as C1FIFOUA1 | | | | | | | |
| E0 | C1FIFOCON12 | 31:0 | same as C1FIFOCON1 | | | | | | | |
| E4 | C1FIFOSTA12 | 31:0 | same as C1FIFOSTA1 | | | | | | | |
| E8 | C1FIFOUA12 | 31:0 | same as C1FIFOUA1 | | | | | | | |
| EC | C1FIFOCON13 | 31:0 | same as C1FIFOCON1 | | | | | | | |
| F0 | C1FIFOSTA13 | 31:0 | same as C1FIFOSTA1 | | | | | | | |
| F4 | C1FIFOUA13 | 31:0 | same as C1FIFOUA1 | | | | | | | |
| F8 | C1FIFOCON14 | 31:0 | same as C1FIFOCON1 | | | | | | | |
| FC | C1FIFOSTA14 | 31:0 | same as C1FIFOSTA1 | | | | | | | |
| 100 | C1FIFOUA14 | 31:0 | same as C1FIFOUA1 | | | | | | | |

**Note 1:** The lower order byte of the 32-bit register resides at the low-order address.

**2:** Reserved register reads 0.

**TABLE 3-2: CAN FD CONTROLLER MODULE REGISTER SUMMARY (CONTINUED)**

| Addr. | Name | | Bit 31/23/15/7 | Bit 30/22/14/6 | Bit 29/21/13/5 | Bit 28/20/12/4 | Bit 27/19/11/3 | Bit 26/18/10/2 | Bit 25/17/9/1 | Bit 24/16/8/0 |
|---|---|---|---|---|---|---|---|---|---|---|
| 104 | C1FIFOCON15 | 31:0 | same as C1FIFOCON1 | | | | | | | |
| 108 | C1FIFOSTA15 | 31:0 | same as C1FIFOSTA1 | | | | | | | |
| 10C | C1FIFOUA15 | 31:0 | same as C1FIFOUA1 | | | | | | | |
| 110 | C1FIFOCON16 | 31:0 | same as C1FIFOCON1 | | | | | | | |
| 114 | C1FIFOSTA16 | 31:0 | same as C1FIFOSTA1 | | | | | | | |
| 118 | C1FIFOUA16 | 31:0 | same as C1FIFOUA1 | | | | | | | |
| 11C | C1FIFOCON17 | 31:0 | same as C1FIFOCON1 | | | | | | | |
| 120 | C1FIFOSTA17 | 31:0 | same as C1FIFOSTA1 | | | | | | | |
| 124 | C1FIFOUA17 | 31:0 | same as C1FIFOUA1 | | | | | | | |
| 128 | C1FIFOCON18 | 31:0 | same as C1FIFOCON1 | | | | | | | |
| 12C | C1FIFOSTA18 | 31:0 | same as C1FIFOSTA1 | | | | | | | |
| 130 | C1FIFOUA18 | 31:0 | same as C1FIFOUA1 | | | | | | | |
| 134 | C1FIFOCON19 | 31:0 | same as C1FIFOCON1 | | | | | | | |
| 138 | C1FIFOSTA19 | 31:0 | same as C1FIFOSTA1 | | | | | | | |
| 13C | C1FIFOUA19 | 31:0 | same as C1FIFOUA1 | | | | | | | |
| 140 | C1FIFOCON20 | 31:0 | same as C1FIFOCON1 | | | | | | | |
| 144 | C1FIFOSTA20 | 31:0 | same as C1FIFOSTA1 | | | | | | | |
| 148 | C1FIFOUA20 | 31:0 | same as C1FIFOUA1 | | | | | | | |
| 14C | C1FIFOCON21 | 31:0 | same as C1FIFOCON1 | | | | | | | |
| 150 | C1FIFOSTA21 | 31:0 | same as C1FIFOSTA1 | | | | | | | |
| 154 | C1FIFOUA21 | 31:0 | same as C1FIFOUA1 | | | | | | | |
| 158 | C1FIFOCON22 | 31:0 | same as C1FIFOCON1 | | | | | | | |
| 15C | C1FIFOSTA22 | 31:0 | same as C1FIFOSTA1 | | | | | | | |
| 160 | C1FIFOUA22 | 31:0 | same as C1FIFOUA1 | | | | | | | |
| 164 | C1FIFOCON23 | 31:0 | same as C1FIFOCON1 | | | | | | | |
| 168 | C1FIFOSTA23 | 31:0 | same as C1FIFOSTA1 | | | | | | | |
| 16C | C1FIFOUA23 | 31:0 | same as C1FIFOUA1 | | | | | | | |
| 170 | C1FIFOCON24 | 31:0 | same as C1FIFOCON1 | | | | | | | |
| 174 | C1FIFOSTA24 | 31:0 | same as C1FIFOSTA1 | | | | | | | |
| 178 | C1FIFOUA24 | 31:0 | same as C1FIFOUA1 | | | | | | | |
| 17C | C1FIFOCON25 | 31:0 | same as C1FIFOCON1 | | | | | | | |
| 180 | C1FIFOSTA25 | 31:0 | same as C1FIFOSTA1 | | | | | | | |
| 184 | C1FIFOUA25 | 31:0 | same as C1FIFOUA1 | | | | | | | |
| 188 | C1FIFOCON26 | 31:0 | same as C1FIFOCON1 | | | | | | | |
| 18C | C1FIFOSTA26 | 31:0 | same as C1FIFOSTA1 | | | | | | | |
| 190 | C1FIFOUA26 | 31:0 | same as C1FIFOUA1 | | | | | | | |
| 194 | C1FIFOCON27 | 31:0 | same as C1FIFOCON1 | | | | | | | |
| 198 | C1FIFOSTA27 | 31:0 | same as C1FIFOSTA1 | | | | | | | |
| 19C | C1FIFOUA27 | 31:0 | same as C1FIFOUA1 | | | | | | | |
| 1A0 | C1FIFOCON28 | 31:0 | same as C1FIFOCON1 | | | | | | | |
| 1A4 | C1FIFOSTA28 | 31:0 | same as C1FIFOSTA1 | | | | | | | |
| 1A8 | C1FIFOUA28 | 31:0 | same as C1FIFOUA1 | | | | | | | |
| 1AC | C1FIFOCON29 | 31:0 | same as C1FIFOCON1 | | | | | | | |
| 1B0 | C1FIFOSTA29 | 31:0 | same as C1FIFOSTA1 | | | | | | | |
| 1B4 | C1FIFOUA29 | 31:0 | same as C1FIFOUA1 | | | | | | | |
| 1B8 | C1FIFOCON30 | 31:0 | same as C1FIFOCON1 | | | | | | | |
| 1BC | C1FIFOSTA30 | 31:0 | same as C1FIFOSTA1 | | | | | | | |
| 1C0 | C1FIFOUA30 | 31:0 | same as C1FIFOUA1 | | | | | | | |
| 1C4 | C1FIFOCON31 | 31:0 | same as C1FIFOCON1 | | | | | | | |
| 1C8 | C1FIFOSTA31 | 31:0 | same as C1FIFOSTA1 | | | | | | | |
| 1CC | C1FIFOUA31 | 31:0 | same as C1FIFOUA1 | | | | | | | |

**Note 1:** The lower order byte of the 32-bit register resides at the low-order address.

**2:** Reserved register reads 0.


© 2019-2020 Microchip Technology Inc.

**TABLE 3-2: CAN FD CONTROLLER MODULE REGISTER SUMMARY (CONTINUED)**

| Addr. | Name | | Bit 31/23/15/7 | Bit 30/22/14/6 | Bit 29/21/13/5 | Bit 28/20/12/4 | Bit 27/19/11/3 | Bit 26/18/10/2 | Bit 25/17/9/1 | Bit 24/16/8/0 |
|---|---|---|---|---|---|---|---|---|---|---|
| | C1FLTCON0 | 31:24 | FLTEN3 | — | — | F3BP[4:0] | | | | |
| | | 23:16 | FLTEN2 | — | — | F2BP[4:0] | | | | |
| | | 15:8 | FLTEN1 | — | — | F1BP[4:0] | | | | |
| 1D0 | | 7:0 | FLTEN0 | — | — | F0BP[4:0] | | | | |
| | C1FLTCON1 | 31:24 | FLTEN7 | — | — | F7BP[4:0] | | | | |
| | | 23:16 | FLTEN6 | — | — | F6BP[4:0] | | | | |
| | | 15:8 | FLTEN5 | — | — | F5BP[4:0] | | | | |
| 1D4 | | 7:0 | FLTEN4 | — | — | F4BP[4:0] | | | | |
| | C1FLTCON2 | 31:24 | FLTEN11 | — | — | F11BP[4:0] | | | | |
| | | 23:16 | FLTEN10 | — | — | F10BP[4:0] | | | | |
| | | 15:8 | FLTEN9 | — | — | F9BP[4:0] | | | | |
| 1D8 | | 7:0 | FLTEN8 | — | — | F8BP[4:0] | | | | |
| | C1FLTCON3 | 31:24 | FLTEN15 | — | — | F15BP[4:0] | | | | |
| | | 23:16 | FLTEN14 | — | — | F14BP[4:0] | | | | |
| | | 15:8 | FLTEN13 | — | — | F13BP[4:0] | | | | |
| 1DC | | 7:0 | FLTEN12 | — | — | F12BP[4:0] | | | | |
| | C1FLTCON4 | 31:24 | FLTEN19 | — | — | F19BP[4:0] | | | | |
| | | 23:16 | FLTEN18 | — | — | F18BP[4:0] | | | | |
| | | 15:8 | FLTEN17 | — | — | F17BP[4:0] | | | | |
| 1E0 | | 7:0 | FLTEN16 | — | — | F16BP[4:0] | | | | |
| | C1FLTCON5 | 31:24 | FLTEN23 | — | — | F23BP[4:0] | | | | |
| | | 23:16 | FLTEN22 | — | — | F22BP[4:0] | | | | |
| | | 15:8 | FLTEN21 | — | — | F21BP[4:0] | | | | |
| 1E4 | | 7:0 | FLTEN20 | — | — | F20BP[4:0] | | | | |
| | C1FLTCON6 | 31:24 | FLTEN27 | — | — | F27BP[4:0] | | | | |
| | | 23:16 | FLTEN26 | — | — | F26BP[4:0] | | | | |
| | | 15:8 | FLTEN25 | — | — | F25BP[4:0] | | | | |
| 1E8 | | 7:0 | FLTEN24 | — | — | F24BP[4:0] | | | | |
| | C1FLTCON7 | 31:24 | FLTEN31 | — | — | F31BP[4:0] | | | | |
| | | 23:16 | FLTEN30 | — | — | F30BP[4:0] | | | | |
| | | 15:8 | FLTEN29 | — | — | F29BP[4:0] | | | | |
| 1EC | | 7:0 | FLTEN28 | — | — | F28BP[4:0] | | | | |
| | C1FLTOBJ0 | 31:24 | — | EXIDE | SID11 | EID[17:6] | | | | |
| | | 23:16 | EID[12:5] | | | | | | | |
| | | 15:8 | EID[4:0] | | | | | SID[10:8] | | |
| 1F0 | | 7:0 | SID[7:0] | | | | | | | |
| | C1MASK0 | 31:24 | — | MIDE | MSID11 | MEID[17:6] | | | | |
| | | 23:16 | MEID[12:5] | | | | | | | |
| | | 15:8 | MEID[4:0] | | | | | MSID[10:8] | | |
| 1F4 | | 7:0 | MSID[7:0] | | | | | | | |
| 1F8 | C1FLTOBJ1 | 31:0 | same as C1FLTOBJ0 | | | | | | | |
| 1FC | C1MASK1 | 31:0 | same as C1MASK0 | | | | | | | |
| 200 | C1FLTOBJ2 | 31:0 | same as C1FLTOBJ0 | | | | | | | |
| 204 | C1MASK2 | 31:0 | same as C1MASK0 | | | | | | | |
| 208 | C1FLTOBJ3 | 31:0 | same as C1FLTOBJ0 | | | | | | | |
| 20C | C1MASK3 | 31:0 | same as C1MASK0 | | | | | | | |
| 210 | C1FLTOBJ4 | 31:0 | same as C1FLTOBJ0 | | | | | | | |
| 214 | C1MASK4 | 31:0 | same as C1MASK0 | | | | | | | |
| 218 | C1FLTOBJ5 | 31:0 | same as C1FLTOBJ0 | | | | | | | |
| 21C | C1MASK5 | 31:0 | same as C1MASK0 | | | | | | | |

**Note 1:** The lower order byte of the 32-bit register resides at the low-order address.

**2:** Reserved register reads 0.

**TABLE 3-2: CAN FD CONTROLLER MODULE REGISTER SUMMARY (CONTINUED)**

| Addr. | Name | | Bit 31/23/15/7 | Bit 30/22/14/6 | Bit 29/21/13/5 | Bit 28/20/12/4 | Bit 27/19/11/3 | Bit 26/18/10/2 | Bit 25/17/9/1 | Bit 24/16/8/0 |
|---|---|---|---|---|---|---|---|---|---|---|
| 220 | C1FLTOBJ6 | 31:0 | same as C1FLTOBJ0 | | | | | | | |
| 224 | C1MASK6 | 31:0 | same as C1MASK0 | | | | | | | |
| 228 | C1FLTOBJ7 | 31:0 | same as C1FLTOBJ0 | | | | | | | |
| 22C | C1MASK7 | 31:0 | same as C1MASK0 | | | | | | | |
| 230 | C1FLTOBJ8 | 31:0 | same as C1FLTOBJ0 | | | | | | | |
| 234 | C1MASK8 | 31:0 | same as C1MASK0 | | | | | | | |
| 238 | C1FLTOBJ9 | 31:0 | same as C1FLTOBJ0 | | | | | | | |
| 23C | C1MASK9 | 31:0 | same as C1MASK0 | | | | | | | |
| 240 | C1FLTOBJ10 | 31:0 | same as C1FLTOBJ0 | | | | | | | |
| 244 | C1MASK10 | 31:0 | same as C1MASK0 | | | | | | | |
| 248 | C1FLTOBJ11 | 31:0 | same as C1FLTOBJ0 | | | | | | | |
| 24C | C1MASK11 | 31:0 | same as C1MASK0 | | | | | | | |
| 250 | C1FLTOBJ12 | 31:0 | same as C1FLTOBJ0 | | | | | | | |
| 254 | C1MASK12 | 31:0 | same as C1MASK0 | | | | | | | |
| 258 | C1FLTOBJ13 | 31:0 | same as C1FLTOBJ0 | | | | | | | |
| 25C | C1MASK13 | 31:0 | same as C1MASK0 | | | | | | | |
| 260 | C1FLTOBJ14 | 31:0 | same as C1FLTOBJ0 | | | | | | | |
| 264 | C1MASK14 | 31:0 | same as C1MASK0 | | | | | | | |
| 268 | C1FLTOBJ15 | 31:0 | same as C1FLTOBJ0 | | | | | | | |
| 26C | C1MASK15 | 31:0 | same as C1MASK0 | | | | | | | |
| 270 | C1FLTOBJ16 | 31:0 | same as C1FLTOBJ0 | | | | | | | |
| 274 | C1MASK16 | 31:0 | same as C1MASK0 | | | | | | | |
| 278 | C1FLTOBJ17 | 31:0 | same as C1FLTOBJ0 | | | | | | | |
| 27C | C1MASK17 | 31:0 | same as C1MASK0 | | | | | | | |
| 280 | C1FLTOBJ18 | 31:0 | same as C1FLTOBJ0 | | | | | | | |
| 284 | C1MASK18 | 31:0 | same as C1MASK0 | | | | | | | |
| 288 | C1FLTOBJ19 | 31:0 | same as C1FLTOBJ0 | | | | | | | |
| 28C | C1MASK19 | 31:0 | same as C1MASK0 | | | | | | | |
| 290 | C1FLTOBJ20 | 31:0 | same as C1FLTOBJ0 | | | | | | | |
| 294 | C1MASK20 | 31:0 | same as C1MASK0 | | | | | | | |
| 298 | C1FLTOBJ21 | 31:0 | same as C1FLTOBJ0 | | | | | | | |
| 29C | C1MASK21 | 31:0 | same as C1MASK0 | | | | | | | |
| 2A0 | C1FLTOBJ22 | 31:0 | same as C1FLTOBJ0 | | | | | | | |
| 2A4 | C1MASK22 | 31:0 | same as C1MASK0 | | | | | | | |
| 2A8 | C1FLTOBJ23 | 31:0 | same as C1FLTOBJ0 | | | | | | | |
| 2AC | C1MASK23 | 31:0 | same as C1MASK0 | | | | | | | |
| 2B0 | C1FLTOBJ24 | 31:0 | same as C1FLTOBJ0 | | | | | | | |
| 2B4 | C1MASK24 | 31:0 | same as C1MASK0 | | | | | | | |
| 2B8 | C1FLTOBJ25 | 31:0 | same as C1FLTOBJ0 | | | | | | | |
| 2BC | C1MASK25 | 31:0 | same as C1MASK0 | | | | | | | |
| 2C0 | C1FLTOBJ26 | 31:0 | same as C1FLTOBJ0 | | | | | | | |
| 2C4 | C1MASK26 | 31:0 | same as C1MASK0 | | | | | | | |
| 2C8 | C1FLTOBJ27 | 31:0 | same as C1FLTOBJ0 | | | | | | | |
| 2CC | C1MASK27 | 31:0 | same as C1MASK0 | | | | | | | |
| 2D0 | C1FLTOBJ28 | 31:0 | same as C1FLTOBJ0 | | | | | | | |
| 2D4 | C1MASK28 | 31:0 | same as C1MASK0 | | | | | | | |
| 2D8 | C1FLTOBJ29 | 31:0 | same as C1FLTOBJ0 | | | | | | | |
| 2DC | C1MASK29 | 31:0 | same as C1MASK0 | | | | | | | |
| 2E0 | C1FLTOBJ30 | 31:0 | same as C1FLTOBJ0 | | | | | | | |
| 2E4 | C1MASK30 | 31:0 | same as C1MASK0 | | | | | | | |
| 2E8 | C1FLTOBJ31 | 31:0 | same as C1FLTOBJ0 | | | | | | | |
| 2EC | C1MASK31 | 31:0 | same as C1MASK0 | | | | | | | |

**Note 1:** The lower order byte of the 32-bit register resides at the low-order address.

**2:** Reserved register reads 0.


© 2019-2020 Microchip Technology Inc.

## 3.1 MCP2518FD Specific Registers

- Register 3-1: OSC
- Register 3-2: IOCON
- Register 3-3: CRC
- Register 3-4: ECCCON
- Register 3-5: ECCSTAT
- Register 3-6: DEVID

**TABLE 3-3: REGISTER LEGEND**

| Symbol | Description | Symbol | Description |
|---|---|---|---|
| R | Readable bit | HC | Cleared by Hardware only |
| W | Writable bit | HS | Set by Hardware only |
| U | Unimplemented bit, read as '0' | 1 | Bit is set at Reset |
| S | Settable bit | 0 | Bit is cleared at Reset |
| C | Clearable bit | x | Bit is unknown at Reset |

**EXAMPLE 3-1:**

| R/W - 0 indicates the bit is both readable and writable, and reads '0' after a Reset. |
|---|

**REGISTER 3-1: OSC – MCP2518FD OSCILLATOR CONTROL REGISTER**

| U-0 | U-0 | U-0 | U-0 | U-0 | U-0 | U-0 | U-0 |
|---|---|---|---|---|---|---|---|
| — | — | — | — | — | — | — | — |
| bit 31 | | | | | | | bit 24 |

| U-0 | U-0 | U-0 | U-0 | U-0 | U-0 | U-0 | U-0 |
|---|---|---|---|---|---|---|---|
| — | — | — | — | — | — | — | — |
| bit 23 | | | | | | | bit 16 |

| U-0 | U-0 | U-0 | R-0 | U-0 | R-0 | U-0 | R-0 |
|---|---|---|---|---|---|---|---|
| — | — | — | SCLKRDY | — | OSCRDY | — | PLLRDY |
| bit 15 | | | | | | | bit 8 |

| U-0 | R/W-1 | R/W-1 | R/W-0 | R/W-0 | HS/C-0 | U-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| — | CLKODIV[1:0] | | SCLKDIV(1) | LPMEN(3) | OSCDIS(2) | — | PLLEN(1) |
| bit 7 | | | | | | | bit 0 |

| **Legend:** | | | |
|---|---|---|---|
| R = Readable bit | W = Writable bit | U = Unimplemented bit, read as '0' | |
| -n = Value at POR | '1' = Bit is set | '0' = Bit is cleared | x = Bit is unknown |

bit 31-13 **Unimplemented:** Read as '0'

bit 12 **SCLKRDY:** Synchronized SCLKDIV bit
1 = SCLKDIV 1
0 = SCLKDIV 0

bit 11 **Unimplemented:** Read as '0'

bit 10 **OSCRDY:** Clock Ready
1 = Clock is running and stable
0 = Clock not ready or off

bit 9 **Unimplemented:** Read as '0'

bit 8 **PLLRDY:** PLL Ready
1 = PLL Locked
0 = PLL not ready

bit 7 **Unimplemented:** Read as '0'

bit 6-5 **CLKODIV[1:0]:** Clock Output Divisor
11 =CLKO is divided by 10
10 =CLKO is divided by 4
01 =CLKO is divided by 2
00 =CLKO is divided by 1

bit 4 **SCLKDIV:** System Clock Divisor(1)
1 = SCLK is divided by 2
0 = SCLK is divided by 1

**Note 1:** This bit can only be modified in Configuration mode.

**2:** Clearing OSCDIS while in Sleep mode will wake-up the device and put it back in Configuration mode.

**3:** Setting LPMEN does not actually put the device in LPM. It selects which Sleep mode will be entered after requesting Sleep mode using CiCON.REQOP. In order to wake up on RXCAN activity, CiINT.WAKIE must be set.


© 2019-2020 Microchip Technology Inc.

**REGISTER 3-1: OSC – MCP2518FD OSCILLATOR CONTROL REGISTER (CONTINUED)**

bit 3 **LPMEN:** Low Power Mode (LPM) Enable(3)

1 = When in LPM, the device will stop the clock and power down the majority of the chip. Register and RAM values will be lost. The device will wake-up due to asserting nCS, or due to RXCAN activity.
0 = When in Sleep mode, the device will stop the clock, and retain it's register and RAM values. It will wake-up due to clearing the OSCDIS bit, or due to RXCAN activity.

bit 2 **OSCDIS:** Clock (Oscillator) Disable(2)

1 = Clock disabled, the device is in Sleep mode.
0 = Enable Clock

bit 1 **Unimplemented:** Read as '0'

bit 0 **PLLEN:** PLL Enable(1)

1 = System Clock from 10x PLL
0 = System Clock comes directly from XTAL oscillator

**Note 1:** This bit can only be modified in Configuration mode.

**2:** Clearing OSCDIS while in Sleep mode will wake-up the device and put it back in Configuration mode.

**3:** Setting LPMEN does not actually put the device in LPM. It selects which Sleep mode will be entered after requesting Sleep mode using CiCON.REQOP. In order to wake up on RXCAN activity, CiINT.WAKIE must be set.

**REGISTER 3-2: IOCON – INPUT/OUTPUT CONTROL REGISTER**

| U-0 | R/W-0 | R/W-0 | R/W-0 | U-0 | U-0 | R/W-1 | R/W-1 |
|---|---|---|---|---|---|---|---|
| — | INTOD | SOF | TXCANOD | — | — | PM1 | PM0 |
| bit 31 | | | | | | | bit 24 |

| U-0 | U-0 | U-0 | U-0 | U-0 | U-0 | R/W-x | R/W-x |
|---|---|---|---|---|---|---|---|
| — | — | — | — | — | — | GPIO1 | GPIO0 |
| bit 23 | | | | | | | bit 16 |

| U-0 | U-0 | U-0 | U-0 | U-0 | U-0 | R/W-x | R/W-x |
|---|---|---|---|---|---|---|---|
| — | — | — | — | — | — | LAT1 | LAT0 |
| bit 15 | | | | | | | bit 8 |

| U-0 | R/W-0 | U-0 | U-0 | U-0 | U-0 | R/W-1 | R/W-1 |
|---|---|---|---|---|---|---|---|
| — | XSTBYEN | — | — | — | — | TRIS1(1) | TRIS0(1) |
| bit 7 | | | | | | | bit 0 |

| **Legend:** | | | |
|---|---|---|---|
| R = Readable bit | W = Writable bit | U = Unimplemented bit, read as '0' | |
| -n = Value at POR | '1' = Bit is set | '0' = Bit is cleared | x = Bit is unknown |

bit 31 **Unimplemented:** Read as '0'

bit 30 **INTOD:** Interrupt pins Open Drain Mode
- 1 = Open Drain Output
- 0 = Push/Pull Output

bit 29 **SOF:** Start-Of-Frame signal
- 1 = SOF signal on CLKO pin
- 0 = Clock on CLKO pin

bit 28 **TXCANOD:** TXCAN Open Drain Mode
- 1 = Open Drain Output
- 0 = Push/Pull Output

bit 27-26 **Unimplemented:** Read as '0'

bit 25 **PM1:** GPIO Pin Mode
- 1 = Pin is used as GPIO1
- 0 = Interrupt Pin $\overline{INT1}$, asserted when CiINT.RXIF and RXIE are set

bit 24 **PM0:** GPIO Pin Mode
- 1 = Pin is used as GPIO0
- 0 = Interrupt Pin $\overline{INT0}$, asserted when CiINT.TXIF and TXIE are set

bit 23-18 **Unimplemented:** Read as '0'

bit 17 **GPIO1:** GPIO1 Status
- 1 = $V_{GPIO1} > V_{IH}$
- 0 = $V_{GPIO1} < V_{IL}$

bit 16 **GPIO0:** GPIO0 Status
- 1 = $V_{GPIO0} > V_{IH}$
- 0 = $V_{GPIO0} < V_{IL}$

bit 15-10 **Unimplemented:** Read as '0'

**Note 1:** If PMx = 0, TRISx will be ignored and the pin will be an output.

**2:** The bit fields in the IOCON register must be written using single data byte SFR WRITE instructions.


© 2019-2020 Microchip Technology Inc.

**REGISTER 3-2: IOCON – INPUT/OUTPUT CONTROL REGISTER (CONTINUED)**

| | |
|---|---|
| bit 9 | **LAT1:** GPIO1 Latch<br>1 = Drive Pin High<br>0 = Drive Pin Low |
| bit 8 | **LAT0:** GPIO0 Latch<br>1 = Drive Pin High<br>0 = Drive Pin Low |
| bit 7 | **Unimplemented:** Read as '0' |
| bit 6 | **XSTBYEN:** Enable Transceiver Standby Pin Control<br>1 = XSTBY control enabled<br>0 = XSTBY control disabled |
| bit 5-2 | **Unimplemented:** Read as '0' |
| bit 1 | **TRIS1:** GPIO1 Data Direction(1)<br>1 = Input Pin<br>0 = Output Pin |
| bit 0 | **TRIS0:** GPIO0 Data Direction(1)<br>1 = Input Pin<br>0 = Output Pin |

**Note 1:** If PMx = 0, TRISx will be ignored and the pin will be an output.

**2:** The bit fields in the IOCON register must be written using single data byte SFR WRITE instructions.

**REGISTER 3-3: CRC – CRC REGISTER**

| U-0 | U-0 | U-0 | U-0 | U-0 | U-0 | R/W-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| — | — | — | — | — | — | FERRIE | CRCERRIE |
| bit 31 | | | | | | | bit 24 |

| U-0 | U-0 | U-0 | U-0 | U-0 | U-0 | HS/C-0 | HS/C-0 |
|---|---|---|---|---|---|---|---|
| — | — | — | — | — | — | FERRIF | CRCERRIF |
| bit 23 | | | | | | | bit 16 |

| R-0 | R-0 | R-0 | R-0 | R-0 | R-0 | R-0 | R-0 |
|---|---|---|---|---|---|---|---|
| CRC[15:8] | | | | | | | |
| bit 15 | | | | | | | bit 8 |

| R-0 | R-0 | R-0 | R-0 | R-0 | R-0 | R-0 | R-0 |
|---|---|---|---|---|---|---|---|
| CRC[7:0] | | | | | | | |
| bit 7 | | | | | | | bit 0 |

| **Legend:** | | | |
|---|---|---|---|
| R = Readable bit | W = Writable bit | U = Unimplemented bit, read as ‘0’ | |
| -n = Value at POR | ‘1’ = Bit is set | ‘0’ = Bit is cleared | x = Bit is unknown |

bit 31-26 **Unimplemented:** Read as ‘0’

bit 25 **FERRIE:** CRC Command Format Error Interrupt Enable

bit 24 **CRCERRIE:** CRC Error Interrupt Enable

bit 23-18 **Unimplemented:** Read as ‘0’

bit 17 **FERRIF:** CRC Command Format Error Interrupt Flag

1 = Number of Bytes mismatch during “SPI with CRC” command occurred
0 = No SPI CRC command format error occurred

bit 16 **CRCERRIF:** CRC Error Interrupt Flag

1 = CRC mismatch occurred
0 = No CRC error has occurred

bit 15-0 **CRC[15:0]:** Cycle Redundancy Check from last CRC mismatch


© 2019-2020 Microchip Technology Inc.

**REGISTER 3-4: ECCCON – ECC CONTROL REGISTER**

| U-0 | U-0 | U-0 | U-0 | U-0 | U-0 | U-0 | U-0 |
|---|---|---|---|---|---|---|---|
| — | — | — | — | — | — | — | — |
| bit 31 | | | | | | | bit 24 |

| U-0 | U-0 | U-0 | U-0 | U-0 | U-0 | U-0 | U-0 |
|---|---|---|---|---|---|---|---|
| — | — | — | — | — | — | — | — |
| bit 23 | | | | | | | bit 16 |

| U-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| — | PARITY[6:0] | | | | | | |
| bit 15 | | | | | | | bit 8 |

| U-0 | U-0 | U-0 | U-0 | U-0 | R/W-0 | R/W-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| — | — | — | — | — | DEDIE | SECIE | ECCEN |
| bit 7 | | | | | | | bit 0 |

| **Legend:** | | | |
|---|---|---|---|
| R = Readable bit | W = Writable bit | U = Unimplemented bit, read as ‘0’ | |
| -n = Value at POR | ‘1’ = Bit is set | ‘0’ = Bit is cleared | x = Bit is unknown |

bit 31-15 **Unimplemented:** Read as ‘0’

bit 14-8 **PARITY[6:0]:** Parity bits used during write to RAM when ECC is disabled

bit 7-3 **Unimplemented:** Read as ‘0’

bit 2 **DEDIE:** Double Error Detection Interrupt Enable Flag

bit 1 **SECIE:** Single Error Detection Interrupt Enable Flag

bit 0 **ECCEN:** ECC Enable

1 = ECC enabled
0 = ECC disabled

**REGISTER 3-5: ECCSTAT – ECC STATUS REGISTER**

| U-0 | U-0 | U-0 | U-0 | R-0 | R-0 | R-0 | R-0 |
|---|---|---|---|---|---|---|---|
| — | — | — | — | ERRADDR[11:8] | | | |
| bit 31 | | | | | | | bit 24 |

| R-0 | R-0 | R-0 | R-0 | R-0 | R-0 | R-0 | R-0 |
|---|---|---|---|---|---|---|---|
| ERRADDR[7:0] | | | | | | | |
| bit 23 | | | | | | | bit 16 |

| U-0 | U-0 | U-0 | U-0 | U-0 | U-0 | U-0 | U-0 |
|---|---|---|---|---|---|---|---|
| — | — | — | — | — | — | — | — |
| bit 15 | | | | | | | bit 8 |

| U-0 | U-0 | U-0 | U-0 | U-0 | HS/C-0 | HS/C-0 | U-0 |
|---|---|---|---|---|---|---|---|
| — | — | — | — | — | DEDIF | SECIF | — |
| bit 7 | | | | | | | bit 0 |

| **Legend:** | | | |
|---|---|---|---|
| R = Readable bit | W = Writable bit | U = Unimplemented bit, read as ‘0’ | |
| -n = Value at POR | ‘1’ = Bit is set | ‘0’ = Bit is cleared | x = Bit is unknown |

bit 31-28 **Unimplemented:** Read as ‘0’

bit 27-16 **ERRADDR[11:0]:** Address where last ECC error occurred

bit 15-3 **Unimplemented:** Read as ‘0’

bit 2 **DEDIF:** Double Error Detection Interrupt Flag
- 1 = Double Error was detected
- 0 = No Double Error Detection occurred

bit 1 **SECIF:** Single Error Detection Interrupt Flag
- 1 = Single Error was detected
- 0 = No Single Error occurred

bit 0 **Unimplemented:** Read as ‘0’


© 2019-2020 Microchip Technology Inc.

**REGISTER 3-6: DEVID – DEVICE ID REGISTER**

| U-0 | U-0 | U-0 | U-0 | U-0 | U-0 | U-0 | U-0 |
|---|---|---|---|---|---|---|---|
| — | — | — | — | — | — | — | — |
| bit 31 | | | | | | | bit 24 |

| U-0 | U-0 | U-0 | U-0 | U-0 | U-0 | U-0 | U-0 |
|---|---|---|---|---|---|---|---|
| — | — | — | — | — | — | — | — |
| bit 23 | | | | | | | bit 16 |

| U-0 | U-0 | U-0 | U-0 | U-0 | U-0 | U-0 | U-0 |
|---|---|---|---|---|---|---|---|
| — | — | — | — | — | — | — | — |
| bit 15 | | | | | | | bit 8 |

| R-0 | R-0 | R-0 | R-0 | R-0 | R-0 | R-0 | R-0 |
|---|---|---|---|---|---|---|---|
| ID[3:0] | | | | REV[3:0] | | | |
| bit 7 | | | | | | | bit 0 |

| **Legend:** | | | |
|---|---|---|---|
| R = Readable bit | W = Writable bit | U = Unimplemented bit, read as ‘0’ | |
| -n = Value at POR | ‘1’ = Bit is set | ‘0’ = Bit is cleared | x = Bit is unknown |

bit 31-8 **Unimplemented:** Read as ‘0’

bit 7-4 **ID[3:0]:** Device ID

bit 3-0 **REV[3:0]:** Silicon Revision

**NOTES:**


© 2019-2020 Microchip Technology Inc.

## 3.2 CAN FD Controller Module Registers

Configuration Registers

- Register 3-7: CiCON
- Register 3-8: CiNBTCFG
- Register 3-9: CiDBTCFG
- Register 3-10: CiTDC
- Register 3-11: CiTBC
- Register 3-12: CiTSCON

Interrupt and Status Registers

- Register 3-13: CiVEC
- Register 3-14: CiINT
- Register 3-15: CiRXIF
- Register 3-16: CiRXOVIF
- Register 3-17: CiTXIF
- Register 3-18: CiTXATIF
- Register 3-19: CiTXREQ

Error and Diagnostic Registers

- Register 3-20: CiTREC
- Register 3-21: CiBDIAG0
- Register 3-22: CiBDIAG1

Fifo Control and Status Registers

- Register 3-23: CiTEFCON
- Register 3-24: CiTEFSTA
- Register 3-25: CiTEFUA
- Register 3-26: CiTXQCON
- Register 3-27: CiTXQSTA
- Register 3-28: CiTXQUA
- Register 3-29: CiFIFOCONm – m = 1 to 31
- Register 3-30: CiFIFOSTAm – m = 1 to 31
- Register 3-31: CiFIFOUAm – m = 1 to 31

Filter Configuration and Control Registers

- Register 3-32: CiFLTCONm – m = 0 to 7
- Register 3-33: CiFLTOBJm – m = 0 to 31
- Register 3-34: CiMASKm – m = 0 to 31

**Note:** The ‘i’ shown in the register identifier denotes CANi, for example, C1CON. The MCP2518FD device contains one CAN FD Controller Module.

**TABLE 3-4: REGISTER LEGEND**

| Sym | Description | Sym | Description |
|---|---|---|---|
| R | Readable bit | HC | Cleared by Hardware only |
| W | Writable bit | HS | Set by Hardware only |
| U | Unimplemented bit, read as ‘0’ | 1 | Bit is set at Reset |
| S | Settable bit | 0 | Bit is cleared at Reset |
| C | Clearable bit | x | Bit is unknown at Reset |

**EXAMPLE 3-2:**

R/W - 0 indicates the bit is both readable and writable, and reads ‘0’ after a Reset.

**REGISTER 3-7: CiCON – CAN CONTROL REGISTER**

| R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-1 | R/W-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| TXBWS[3:0] | | | | ABAT | REQOP[2:0] | | |
| bit 31 | | | | | | | bit 24 |

| R-1 | R-0 | R-0 | R/W-1 | R/W-1 | R/W-0 | R/W-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| OPMOD[2:0] | | | TXQEN(1) | STEF(1) | SERR2LOM(1) | ESIGM(1) | RTXAT(1) |
| bit 23 | | | | | | | bit 16 |

| U-0 | U-0 | U-0 | R/W-0 | R-0 | R/W-1 | R/W-1 | R/W-1 |
|---|---|---|---|---|---|---|---|
| — | — | — | BRSDIS | BUSY | WFT[1:0] | | WAKFIL(1) |
| bit 15 | | | | | | | bit 8 |

| U-0 | R/W-1 | R/W-1 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| — | PXEDIS(1) | ISOCRCEN(1) | DNCNT[4:0] | | | | |
| bit 7 | | | | | | | bit 0 |

| **Legend:** | | | |
|---|---|---|---|
| R = Readable bit | W = Writable bit | U = Unimplemented bit, read as '0' | |
| -n = Value at POR | '1' = Bit is set | '0' = Bit is cleared | x = Bit is unknown |

bit 31-28 **TXBWS[3:0]**: Transmit Bandwidth Sharing bits
Delay between two consecutive transmissions (in arbitration bit times)
0000 = No delay
0001 = 2
0010 = 4
0011 = 8
0100 = 16
0101 = 32
0110 = 64
0111 = 128
1000 = 256
1001 = 512
1010 = 1024
1011 = 2048
1111-1100 = 4096

bit 27 **ABAT**: Abort All Pending Transmissions bit
1 = Signal all transmit FIFOs to abort transmission
0 = Module will clear this bit when all transmissions were aborted

**Note 1:** These bits can only be modified in Configuration mode.

**2:** In Sleep mode, the OPMOD bits indicate Configuration mode (OPMOD = 100) and OSC. OSCDIS will read as '1'. The application software should use these bit fields as a handshake indication for the Sleep mode request.


© 2019-2020 Microchip Technology Inc.

**REGISTER 3-7: CiCON – CAN CONTROL REGISTER (CONTINUED)**

bit 26-24 **REQOP[2:0]**: Request Operation Mode bits
000 = Set Normal CAN FD mode; supports mixing of CAN FD and Classic CAN 2.0 frames
001 = Set Sleep mode
010 = Set Internal Loopback mode
011 = Set Listen Only mode
100 = Set Configuration mode
101 = Set External Loopback mode
110 = Set Normal CAN 2.0 mode; possible error frames on CAN FD frames
111 = Set Restricted Operation mode

bit 23-21 **OPMOD[2:0]**: Operation Mode Status bits(2)
000 = Module is in Normal CAN FD mode; supports mixing of CAN FD and Classic CAN 2.0 frames
001 = Module is in Sleep mode
010 = Module is in Internal Loopback mode
011 = Module is in Listen Only mode
100 = Module is in Configuration mode
101 = Module is in External Loopback mode
110 = Module is Normal CAN 2.0 mode; possible error frames on CAN FD frames
111 = Module is Restricted Operation mode

bit 20 **TXQEN**: Enable Transmit Queue bit(1)
1 = Enables TXQ and reserves space in RAM
0 = Do not reserve space in RAM for TXQ

bit 19 **STEF**: Store in Transmit Event FIFO bit(1)
1 = Saves transmitted messages in TEF and reserves space in RAM
0 = Do not save transmitted messages in TEF

bit 18 **SERR2LOM**: Transition to Listen Only Mode on System Error bit(1)
1 = Transition to Listen Only Mode
0 = Transition to Restricted Operation Mode

bit 17 **ESIGM**: Transmit ESI in Gateway Mode bit(1)
1 = ESI is transmitted recessive when ESI of message is high or CAN controller error passive
0 = ESI reflects error status of CAN controller

bit 16 **RTXAT**: Restrict Retransmission Attempts bit(1)
1 = Restricted retransmission attempts, CiFIFOCONm.TXAT is used
0 = Unlimited number of retransmission attempts, CiFIFOCONm.TXAT will be ignored

bit 15-13 **Unimplemented**: Read as '0'

bit 12 **BRSDIS**: Bit Rate Switching Disable bit
1 = Bit Rate Switching is Disabled, regardless of BRS in the Transmit Message Object
0 = Bit Rate Switching depends on BRS in the Transmit Message Object

bit 11 **BUSY**: CAN Module is Busy bit
1 = The CAN module is transmitting or receiving a message
0 = The CAN module is inactive

bit 10-9 **WFT[1:0]**: Selectable Wake-up Filter Time bits
00 = T00FILTER
01 = T01FILTER
10 = T10FILTER
11 = T11FILTER

**Note:** Please refer to Table 7-5.

bit 8 **WAKFIL**: Enable CAN Bus Line Wake-up Filter bit(1)
1 = Use CAN bus line filter for wake-up
0 = CAN bus line filter is not used for wake-up

**Note 1:** These bits can only be modified in Configuration mode.

**2:** In Sleep mode, the OPMOD bits indicate Configuration mode (OPMOD = 100) and OSC. OSCDIS will read as '1'. The application software should use these bit fields as a handshake indication for the Sleep mode request.

**REGISTER 3-7: CiCON – CAN CONTROL REGISTER (CONTINUED)**

bit 7 **Unimplemented**: Read as '0'

bit 6 **PXEDIS**: Protocol Exception Event Detection Disabled bit(1)
A recessive "res bit" following a recessive FDF bit is called a Protocol Exception.
1 = Protocol Exception is treated as a Form Error.
0 = If a Protocol Exception is detected, the CAN FD Controller Module will enter Bus Integrating state.

bit 5 **ISOCRCEN**: Enable ISO CRC in CAN FD Frames bit(1)
1 = Include Stuff Bit Count in CRC Field and use Non-Zero CRC Initialization Vector according to ISO 11898-1:2015
0 = Do NOT include Stuff Bit Count in CRC Field and use CRC Initialization Vector with all zeros

bit 4-0 **DNCNT[4:0]**: Device Net Filter Bit Number bits
10011-11111 = Invalid Selection (compare up to 18-bits of data with EID)
10010 = Compare up to data byte 2 bit 6 with EID17
...
00001 = Compare up to data byte 0 bit 7 with EID0
00000 = Do not compare data bytes

**Note 1:** These bits can only be modified in Configuration mode.
**2:** In Sleep mode, the OPMOD bits indicate Configuration mode (OPMOD = 100) and OSC. OSCDIS will read as '1'. The application software should use these bit fields as a handshake indication for the Sleep mode request.

**REGISTER 3-8: CiNBTCFG – NOMINAL BIT TIME CONFIGURATION REGISTER**

| R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| BRP[7:0] | | | | | | | |
| bit 31 | | | | | | | bit 24 |

| R/W-0 | R/W-0 | R/W-1 | R/W-1 | R/W-1 | R/W-1 | R/W-1 | R/W-0 |
|---|---|---|---|---|---|---|---|
| TSEG1[7:0] | | | | | | | |
| bit 23 | | | | | | | bit 16 |

| U-0 | R/W-0 | R/W-0 | R/W-0 | R/W-1 | R/W-1 | R/W-1 | R/W-1 |
|---|---|---|---|---|---|---|---|
| — | TSEG2[6:0] | | | | | | |
| bit 15 | | | | | | | bit 8 |

| U-0 | R/W-0 | R/W-0 | R/W-0 | R/W-1 | R/W-1 | R/W-1 | R/W-1 |
|---|---|---|---|---|---|---|---|
| — | SJW[6:0] | | | | | | |
| bit 7 | | | | | | | bit 0 |

| **Legend:** | | | |
|---|---|---|---|
| R = Readable bit | W = Writable bit | U = Unimplemented bit, read as '0' | |
| -n = Value at POR | '1' = Bit is set | '0' = Bit is cleared | x = Bit is unknown |

bit 31-24 **BRP[7:0]**: Baud Rate Prescaler bits
1111 1111 = TQ = 256/Fsys
...
0000 0000 = TQ = 1/Fsys

**Note 1:** This register can only be modified in Configuration mode.


© 2019-2020 Microchip Technology Inc.

**REGISTER 3-8: CiNBTCFG – NOMINAL BIT TIME CONFIGURATION REGISTER (CONTINUED)**

| | |
|---|---|
| bit 23-16 | **TSEG1[7:0]**: Time Segment 1 bits (Propagation Segment + Phase Segment 1)<br>1111 1111 = Length is 256 x TQ<br>...<br>0000 0000 = Length is 1 x TQ |
| bit 15 | **Unimplemented**: Read as '0' |
| bit 14-8 | **TSEG2[6:0]**: Time Segment 2 bits (Phase Segment 2)<br>111 1111 = Length is 128 x TQ<br>...<br>000 0000 = Length is 1 x TQ |
| bit 7 | **Unimplemented**: Read as '0' |
| bit 6-0 | **SJW[6:0]**: Synchronization Jump Width bits<br>111 1111 = Length is 128 x TQ<br>...<br>000 0000 = Length is 1 x TQ |

**Note 1:** This register can only be modified in Configuration mode.

**REGISTER 3-9: CiDBTCFG – DATA BIT TIME CONFIGURATION REGISTER**

| R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| BRP[7:0] | | | | | | | |
| bit 31 | | | | | | | bit 24 |

| U-0 | U-0 | U-0 | R/W-0 | R/W-1 | R/W-1 | R/W-1 | R/W-0 |
|---|---|---|---|---|---|---|---|
| — | — | — | TSEG1[4:0] | | | | |
| bit 23 | | | | | | | bit 16 |

| U-0 | U-0 | U-0 | U-0 | R/W-0 | R/W-0 | R/W-1 | R/W-1 |
|---|---|---|---|---|---|---|---|
| — | — | — | — | TSEG2[3:0] | | | |
| bit 15 | | | | | | | bit 8 |

| U-0 | U-0 | U-0 | U-0 | R/W-0 | R/W-0 | R/W-1 | R/W-1 |
|---|---|---|---|---|---|---|---|
| — | — | — | — | SJW[3:0] | | | |
| bit 7 | | | | | | | bit 0 |

| **Legend:** | | | |
|---|---|---|---|
| R = Readable bit | W = Writable bit | U = Unimplemented bit, read as ‘0’ | |
| -n = Value at POR | ‘1’ = Bit is set | ‘0’ = Bit is cleared | x = Bit is unknown |

bit 31-24 **BRP[7:0]**: Baud Rate Prescaler bits
1111 1111 = TQ = 256/Fsys
...
0000 0000 = TQ = 1/Fsys

bit 23-21 **Unimplemented**: Read as ‘0’

bit 20-16 **TSEG1[4:0]**: Time Segment 1 bits (Propagation Segment + Phase Segment 1)
1 1111 = Length is 32 x TQ
...
0 0000 = Length is 1 x TQ

bit 15-12 **Unimplemented**: Read as ‘0’

bit 11-8 **TSEG2[3:0]**: Time Segment 2 bits (Phase Segment 2)
1111 = Length is 16 x TQ
...
0000 = Length is 1 x TQ

bit 7-4 **Unimplemented**: Read as ‘0’

bit 3-0 **SJW[3:0]**: Synchronization Jump Width bits
1111 = Length is 16 x TQ
...
0000 = Length is 1 x TQ

**Note 1:** This register can only be modified in Configuration mode.


© 2019-2020 Microchip Technology Inc.

**REGISTER 3-10: CiTDC – TRANSMITTER DELAY COMPENSATION REGISTER**

| U-0 | U-0 | U-0 | U-0 | U-0 | U-0 | R/W-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| — | — | — | — | — | — | EDGFLTEN | SID11EN |
| bit 31 | | | | | | | bit 24 |

| U-0 | U-0 | U-0 | U-0 | U-0 | U-0 | R/W-1 | R/W-0 |
|---|---|---|---|---|---|---|---|
| — | — | — | — | — | — | TDCMOD[1:0] | |
| bit 23 | | | | | | | bit 16 |

| U-0 | R/W-0 | R/W-0 | R/W-1 | R/W-0 | R/W-0 | R/W-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| — | TDCO[6:0] | | | | | | |
| bit 15 | | | | | | | bit 8 |

| U-0 | U-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| — | — | TDCV[5:0] | | | | | |
| bit 7 | | | | | | | bit 0 |

| **Legend:** | | | |
|---|---|---|---|
| R = Readable bit | W = Writable bit | U = Unimplemented bit, read as ‘0’ | |
| -n = Value at POR | ‘1’ = Bit is set | ‘0’ = Bit is cleared | x = Bit is unknown |

bit 31-26 **Unimplemented**: Read as ‘0’

bit 25 **EDGFLTEN**: Enable Edge Filtering during Bus Integration state bit
1 = Edge Filtering enabled, according to ISO 11898-1:2015
0 = Edge Filtering disabled

bit 24 **SID11EN**: Enable 12-Bit SID in CAN FD Base Format Messages bit
1 = RRS is used as SID11 in CAN FD base format messages: SID[11:0] = {SID[10:0], SID11}
0 = Do not use RRS; SID[10:0] according to ISO 11898-1:2015

bit 23-18 **Unimplemented**: Read as ‘0’

bit 17-16 **TDCMOD[1:0]**: Transmitter Delay Compensation Mode bits; Secondary Sample Point (SSP)
10-11 = Auto; measure delay and add TDCO.
01 = Manual; Do not measure, use TDCV + TDCO from register
00 = TDC Disabled

bit 15 **Unimplemented**: Read as ‘0’

bit 14-8 **TDCO[6:0]**: Transmitter Delay Compensation Offset bits; Secondary Sample Point (SSP)
Two’s complement; offset can be positive, zero, or negative.
011 1111 = 63 x TSYSCLK
...
000 0000 = 0 x TSYSCLK
...
111 1111 = –64 x TSYSCLK

bit 7-6 **Unimplemented**: Read as ‘0’

bit 5-0 **TDCV[5:0]**: Transmitter Delay Compensation Value bits; Secondary Sample Point (SSP)
11 1111 = 63 x TSYSCLK
...
00 0000 = 0 x TSYSCLK

**Note 1:** This register can only be modified in Configuration mode.

**REGISTER 3-11: CiTBC – TIME BASE COUNTER REGISTER**

| R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| TBC[31:24] | | | | | | | |
| bit 31 | | | | | | | bit 24 |

| R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| TBC[23:16] | | | | | | | |
| bit 23 | | | | | | | bit 16 |

| R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| TBC[15:8] | | | | | | | |
| bit 15 | | | | | | | bit 8 |

| R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| TBC[7:0] | | | | | | | |
| bit 7 | | | | | | | bit 0 |

| **Legend:** | | | |
|---|---|---|---|
| R = Readable bit | W = Writable bit | U = Unimplemented bit, read as '0' | |
| -n = Value at POR | '1' = Bit is set | '0' = Bit is cleared | x = Bit is unknown |

bit 31-0 **TBC[31:0]**: Time Base Counter bits
This is a free running timer that increments every TBCPRE clocks when TBCEN is set

**Note 1:** The TBC will be stopped and reset when TBCEN = 0.

**2:** The TBC prescaler count will be reset on any write to CiTBC (CiTSCON.TBCPRE will be unaffected).


© 2019-2020 Microchip Technology Inc.

**REGISTER 3-12: CiTSCON – TIME STAMP CONTROL REGISTER**

| U-0 | U-0 | U-0 | U-0 | U-0 | U-0 | U-0 | U-0 |
|---|---|---|---|---|---|---|---|
| — | — | — | — | — | — | — | — |
| bit 31 | | | | | | | bit 24 |

| U-0 | U-0 | U-0 | U-0 | U-0 | R/W-0 | R/W-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| — | — | — | — | — | TSRES | TSEOF | TBCEN |
| bit 23 | | | | | | | bit 16 |

| U-0 | U-0 | U-0 | U-0 | U-0 | U-0 | R/W-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| — | — | — | — | — | — | TBCPRE[9:8] | |
| bit 15 | | | | | | | bit 8 |

| R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| TBCPRE[7:0] | | | | | | | |
| bit 7 | | | | | | | bit 0 |

| **Legend:** | | | |
|---|---|---|---|
| R = Readable bit | W = Writable bit | U = Unimplemented bit, read as '0' | |
| -n = Value at POR | '1' = Bit is set | '0' = Bit is cleared | x = Bit is unknown |

bit 31-19 **Unimplemented**: Read as '0'

bit 18 **TSRES**: Time Stamp res bit (FD Frames only)
1 = at sample point of the bit following the FDF bit.
0 = at sample point of SOF

bit 17 **TSEOF**: Time Stamp EOF bit
1 = Time Stamp when frame is taken valid:
- RX no error until last but one bit of EOF
- TX no error until the end of EOF

0 = Time Stamp at "beginning" of Frame:
- Classical Frame: at sample point of SOF
- FD Frame: see TSRES bit.

bit 16 **TBCEN**: Time Base Counter Enable bit
1 = Enable TBC
0 = Stop and reset TBC

bit 15-10 **Unimplemented**: Read as '0'

bit 9-0 **TBCPRE[9:0]**: Time Base Counter Prescaler bits
1023 = TBC increments every 1024 clocks
...
0 = TBC increments every 1 clock

**REGISTER 3-13: CiVEC – INTERRUPT CODE REGISTER**

| U-0 | R-1 | R-0 | R-0 | R-0 | R-0 | R-0 | R-0 |
|---|---|---|---|---|---|---|---|
| — | RXCODE[6:0](1) | | | | | | |
| bit 31 | | | | | | | bit 24 |

| U-0 | R-1 | R-0 | R-0 | R-0 | R-0 | R-0 | R-0 |
|---|---|---|---|---|---|---|---|
| — | TXCODE[6:0](1) | | | | | | |
| bit 23 | | | | | | | bit 16 |

| U-0 | U-0 | U-0 | R-0 | R-0 | R-0 | R-0 | R-0 |
|---|---|---|---|---|---|---|---|
| — | — | — | FILHIT[4:0](1) | | | | |
| bit 15 | | | | | | | bit 8 |

| U-0 | R-1 | R-0 | R-0 | R-0 | R-0 | R-0 | R-0 |
|---|---|---|---|---|---|---|---|
| — | ICODE[6:0](1) | | | | | | |
| bit 7 | | | | | | | bit 0 |

| **Legend:** | | | |
|---|---|---|---|
| R = Readable bit | W = Writable bit | U = Unimplemented bit, read as '0' | |
| -n = Value at POR | '1' = Bit is set | '0' = Bit is cleared | x = Bit is unknown |

bit 31 **Unimplemented**: Read as '0'

bit 30-24 **RXCODE[6:0]**: Receive Interrupt Flag Code bits(1)
1000001-1111111 = Reserved
1000000 = No interrupt
0100000-0111111 = Reserved

0011111 = FIFO 31 Interrupt (RFIF[31] set)
...
0000010 = FIFO 2 Interrupt (RFIF[2] set)
0000001 = FIFO 1 Interrupt (RFIF[1] set)
0000000 = Reserved. FIFO 0 cannot receive.

bit 23 **Unimplemented**: Read as '0'

bit 22-16 **TXCODE[6:0]**: Transmit Interrupt Flag Code bits(1)
1000001-1111111 = Reserved
1000000 = No interrupt
0100000-0111111 = Reserved

0011111 = FIFO 31 Interrupt (TFIF[31] set)
...
0000001 = FIFO 1 Interrupt (TFIF[1] set)
0000000 = TXQ Interrupt (TFIF[0] set)

bit 15-13 **Unimplemented**: Read as '0'

bit 12-8 **FILHIT[4:0]**: Filter Hit Number bits(1)
11111 = Filter 31
11110 = Filter 30
...
00001 = Filter 1
00000 = Filter 0

**Note 1:** If multiple interrupts are pending, the interrupt with the highest number will be indicated.


© 2019-2020 Microchip Technology Inc.

**REGISTER 3-13: CiVEC – INTERRUPT CODE REGISTER (CONTINUED)**

bit 7 **Unimplemented**: Read as '0'

bit 6-0 **ICODE[6:0]**: Interrupt Flag Code bits(1)

1001011-1111111 = Reserved
1001010 = Transmit Attempt Interrupt (any bit in CiTXATIF set)
1001001 = Transmit Event FIFO Interrupt (any bit in CiTEFIF set)
1001000 = Invalid Message Occurred (IVMIF/IE)
1000111 = Operation Mode Change Occurred (MODIF/IE)
1000110 = TBC Overflow (TBCIF/IE)
1000101 = RX/TX MAB Overflow/Underflow (RX: message received before previous message was saved to memory; TX: can't feed TX MAB fast enough to transmit consistent data.) (SERRIF/IE)
1000100 = Address Error Interrupt (illegal FIFO address presented to system) (SERRIF/IE)
1000011 = Receive FIFO Overflow Interrupt (any bit in CiRXOVIF set)
1000010 = Wake-up interrupt (WAKIF/WAKIE)
1000001 = Error Interrupt (CERRIF/IE)
1000000 = No interrupt
0100000-0111111 = Reserved
0011111 = FIFO 31 Interrupt (TFIF[31] or RFIF[31] set)
...
0000001 = FIFO 1 Interrupt (TFIF[1] or RFIF[1] set)
0000000 = TXQ Interrupt (TFIF[0] set)

**Note 1:** If multiple interrupts are pending, the interrupt with the highest number will be indicated.

**REGISTER 3-14: CiINT – INTERRUPT REGISTER**

| R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| IVMIE | WAKIE | CERRIE | SERRIE | RXOVIE | TXATIE | SPICRCIE | ECCIE |
| bit 31 | | | | | | | bit 24 |

| U-0 | U-0 | U-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| — | — | — | TEFIE | MODIE | TBCIE | RXIE | TXIE |
| bit 23 | | | | | | | bit 16 |

| HS/C-0 | HS/C-0 | HS/C-0 | HS/C-0 | R-0 | R-0 | R-0 | R-0 |
|---|---|---|---|---|---|---|---|
| IVMIF(1) | WAKIF(1) | CERRIF(1) | SERRIF(1) | RXOVIF | TXATIF | SPICRCIF | ECCIF |
| bit 15 | | | | | | | bit 8 |

| U-0 | U-0 | U-0 | R-0 | HS/C-0 | HS/C-0 | R-0 | R-0 |
|---|---|---|---|---|---|---|---|
| — | — | — | TEFIF | MODIF(1) | TBCIF(1) | RXIF | TXIF |
| bit 7 | | | | | | | bit 0 |

| **Legend:** | | | |
|---|---|---|---|
| R = Readable bit | W = Writable bit | U = Unimplemented bit, read as '0' | |
| -n = Value at POR | '1' = Bit is set | '0' = Bit is cleared | x = Bit is unknown |

bit 31 **IVMIE**: Invalid Message Interrupt Enable bit
bit 30 **WAKIE**: Bus Wake Up Interrupt Enable bit
bit 29 **CERRIE**: CAN Bus Error Interrupt Enable bit
bit 28 **SERRIE**: System Error Interrupt Enable bit
bit 27 **RXOVIE**: Receive FIFO Overflow Interrupt Enable bit
bit 26 **TXATIE**: Transmit Attempt Interrupt Enable bit
bit 25 **SPICRCIE**: SPI CRC Error Interrupt Enable bit
bit 24 **ECCIE**: ECC Error Interrupt Enable bit
bit 23-21 **Unimplemented**: Read as '0'
bit 20 **TEFIE**: Transmit Event FIFO Interrupt Enable bit
bit 19 **MODIE**: Mode Change Interrupt Enable bit
bit 18 **TBCIE**: Time Base Counter Interrupt Enable bit
bit 17 **RXIE**: Receive FIFO Interrupt Enable bit
bit 16 **TXIE**: Transmit FIFO Interrupt Enable bit
bit 15 **IVMIF**: Invalid Message Interrupt Flag bit(1)
bit 14 **WAKIF**: Bus Wake Up Interrupt Flag bit(1)
bit 13 **CERRIF**: CAN Bus Error Interrupt Flag bit(1)
bit 12 **SERRIF**: System Error Interrupt Flag bit(1)
1 = A system error occurred
0 = No system error occurred
bit 11 **RXOVIF**: Receive Object Overflow Interrupt Flag bit
1 = Receive FIFO overflow occurred
0 = No receive FIFO overflow has occurred
bit 10 **TXATIF**: Transmit Attempt Interrupt Flag bit

**Note 1:** Flags are set by hardware and cleared by application.


© 2019-2020 Microchip Technology Inc.

**REGISTER 3-14: CiINT – INTERRUPT REGISTER (CONTINUED)**

| | |
|---|---|
| bit 9 | **SPICRCIF**: SPI CRC Error Interrupt Flag bit |
| bit 8 | **ECCIF**: ECC Error Interrupt Flag bit |
| bit 7-5 | **Unimplemented**: Read as '0' |
| bit 4 | **TEFIF**: Transmit Event FIFO Interrupt Flag bit<br>1 = TEF interrupt pending<br>0 = No TEF interrupts pending |
| bit 3 | **MODIF**: Operation Mode Change Interrupt Flag bit(1)<br>1 = Operation mode change occurred (OPMOD has changed)<br>0 = No mode change occurred |
| bit 2 | **TBCIF**: Time Base Counter Overflow Interrupt Flag bit(1)<br>1 = TBC has overflowed<br>0 = TBC did not overflow |
| bit 1 | **RXIF**: Receive FIFO Interrupt Flag bit<br>1 = Receive FIFO interrupt pending<br>0 = No receive FIFO interrupts pending |
| bit 0 | **TXIF**: Transmit FIFO Interrupt Flag bit<br>1 = Transmit FIFO interrupt pending<br>0 = No transmit FIFO interrupts pending |

**Note 1:** Flags are set by hardware and cleared by application.

**REGISTER 3-15: CiRXIF – RECEIVE INTERRUPT STATUS REGISTER**

| R-0 | R-0 | R-0 | R-0 | R-0 | R-0 | R-0 | R-0 |
|---|---|---|---|---|---|---|---|
| RFIF[31:24] | | | | | | | |
| bit 31 | | | | | | | bit 24 |

| R-0 | R-0 | R-0 | R-0 | R-0 | R-0 | R-0 | R-0 |
|---|---|---|---|---|---|---|---|
| RFIF[23:16] | | | | | | | |
| bit 23 | | | | | | | bit 16 |

| R-0 | R-0 | R-0 | R-0 | R-0 | R-0 | R-0 | R-0 |
|---|---|---|---|---|---|---|---|
| RFIF[15:8] | | | | | | | |
| bit 15 | | | | | | | bit 8 |

| R-0 | R-0 | R-0 | R-0 | R-0 | R-0 | R-0 | U-0 |
|---|---|---|---|---|---|---|---|
| RFIF[7:1] | | | | | | | — |
| bit 7 | | | | | | | bit 0 |

| **Legend:** | | | |
|---|---|---|---|
| R = Readable bit | W = Writable bit | U = Unimplemented bit, read as '0' | |
| -n = Value at POR | '1' = Bit is set | '0' = Bit is cleared | x = Bit is unknown |

bit 31-1 **RFIF[31:1]**: Receive FIFO Interrupt Pending bits(1)
1 = One or more enabled receive FIFO interrupts are pending
0 = No enabled receive FIFO interrupts are pending

bit 0 **Unimplemented**: Read as '0'

**Note 1:** RFIF = 'or' of enabled RXFIFO flags; flags will be cleared when the condition of the FIFO terminates.


© 2019-2020 Microchip Technology Inc.

**REGISTER 3-16: CiRXOVIF – RECEIVE OVERFLOW INTERRUPT STATUS REGISTER**

| R-0 | R-0 | R-0 | R-0 | R-0 | R-0 | R-0 | R-0 |
|---|---|---|---|---|---|---|---|
| RFOVIF[31:24] | | | | | | | |
| bit 31 | | | | | | | bit 24 |

| R-0 | R-0 | R-0 | R-0 | R-0 | R-0 | R-0 | R-0 |
|---|---|---|---|---|---|---|---|
| RFOVIF[23:16] | | | | | | | |
| bit 23 | | | | | | | bit 16 |

| R-0 | R-0 | R-0 | R-0 | R-0 | R-0 | R-0 | R-0 |
|---|---|---|---|---|---|---|---|
| RFOVIF[15:8] | | | | | | | |
| bit 15 | | | | | | | bit 8 |

| R-0 | R-0 | R-0 | R-0 | R-0 | R-0 | R-0 | U-0 |
|---|---|---|---|---|---|---|---|
| RFOVIF[7:1] | | | | | | | — |
| bit 7 | | | | | | | bit 0 |

| **Legend:** | | | |
|---|---|---|---|
| R = Readable bit | W = Writable bit | U = Unimplemented bit, read as '0' | |
| -n = Value at POR | '1' = Bit is set | '0' = Bit is cleared | x = Bit is unknown |

bit 31-1 **RFOVIF[31:1]**: Receive FIFO Overflow Interrupt Pending bits
1 = Interrupt is pending
0 = Interrupt not pending

bit 0 **Unimplemented**: Read as '0'

**Note 1:** Flags need to be cleared in FIFO register

**REGISTER 3-17: CiTXIF – TRANSMIT INTERRUPT STATUS REGISTER**

| R-0 | R-0 | R-0 | R-0 | R-0 | R-0 | R-0 | R-0 |
|---|---|---|---|---|---|---|---|
| TFIF[31:24] | | | | | | | |
| bit 31 | | | | | | | bit 24 |

| R-0 | R-0 | R-0 | R-0 | R-0 | R-0 | R-0 | R-0 |
|---|---|---|---|---|---|---|---|
| TFIF[23:16](1) | | | | | | | |
| bit 23 | | | | | | | bit 16 |

| R-0 | R-0 | R-0 | R-0 | R-0 | R-0 | R-0 | R-0 |
|---|---|---|---|---|---|---|---|
| TFIF[15:8](1) | | | | | | | |
| bit 15 | | | | | | | bit 8 |

| R-0 | R-0 | R-0 | R-0 | R-0 | R-0 | R-0 | R-0 |
|---|---|---|---|---|---|---|---|
| TFIF[7:0](1) | | | | | | | |
| bit 7 | | | | | | | bit 0 |

| **Legend:** | | | |
|---|---|---|---|
| R = Readable bit | W = Writable bit | U = Unimplemented bit, read as '0' | |
| -n = Value at POR | '1' = Bit is set | '0' = Bit is cleared | x = Bit is unknown |

bit 31-0 **TFIF[31:0]**: Transmit FIFO/TXQ (2) Interrupt Pending bits(1)
1 = One or more enabled transmit FIFO/TXQ interrupts are pending
0 = No enabled transmit FIFO/TXQ interrupt are pending

**Note 1:** TFIF = 'or' of the enabled TXFIFO flags; flags will be cleared when the condition of the FIFO terminates.
**2:** TFIF[0] is for the Transmit Queue.


© 2019-2020 Microchip Technology Inc.

**REGISTER 3-18: CiTXATIF – TRANSMIT ATTEMPT INTERRUPT STATUS REGISTER**

| R-0 | R-0 | R-0 | R-0 | R-0 | R-0 | R-0 | R-0 |
|---|---|---|---|---|---|---|---|
| TFATIF[31:24](1) | | | | | | | |
| bit 31 | | | | | | | bit 24 |

| R-0 | R-0 | R-0 | R-0 | R-0 | R-0 | R-0 | R-0 |
|---|---|---|---|---|---|---|---|
| TFATIF[23:16](1) | | | | | | | |
| bit 23 | | | | | | | bit 16 |

| R-0 | R-0 | R-0 | R-0 | R-0 | R-0 | R-0 | R-0 |
|---|---|---|---|---|---|---|---|
| TFATIF[15:8](1) | | | | | | | |
| bit 15 | | | | | | | bit 8 |

| R-0 | R-0 | R-0 | R-0 | R-0 | R-0 | R-0 | R-0 |
|---|---|---|---|---|---|---|---|
| TFATIF[7:0](1) | | | | | | | |
| bit 7 | | | | | | | bit 0 |

| **Legend:** | | | |
|---|---|---|---|
| R = Readable bit | W = Writable bit | U = Unimplemented bit, read as '0' | |
| -n = Value at POR | '1' = Bit is set | '0' = Bit is cleared | x = Bit is unknown |

bit 31-0 **TFATIF[31:0]**: Transmit FIFO/TXQ (2) Attempt Interrupt Pending bits(1)
1 = Interrupt is pending
0 = Interrupt not pending

**Note 1:** Flags need to be cleared in FIFO register
**2:** TFATIF[0] is for the Transmit Queue.

**REGISTER 3-19: CiTXREQ – TRANSMIT REQUEST REGISTER**

| S/HC-0 | S/HC-0 | S/HC-0 | S/HC-0 | S/HC-0 | S/HC-0 | S/HC-0 | S/HC-0 |
|---|---|---|---|---|---|---|---|
| TXREQ[31:24] | | | | | | | |
| bit 31 | | | | | | | bit 24 |

| S/HC-0 | S/HC-0 | S/HC-0 | S/HC-0 | S/HC-0 | S/HC-0 | S/HC-0 | S/HC-0 |
|---|---|---|---|---|---|---|---|
| TXREQ[23:16] | | | | | | | |
| bit 23 | | | | | | | bit 16 |

| S/HC-0 | S/HC-0 | S/HC-0 | S/HC-0 | S/HC-0 | S/HC-0 | S/HC-0 | S/HC-0 |
|---|---|---|---|---|---|---|---|
| TXREQ[15:8] | | | | | | | |
| bit 15 | | | | | | | bit 8 |

| S/HC-0 | S/HC-0 | S/HC-0 | S/HC-0 | S/HC-0 | S/HC-0 | S/HC-0 | S/HC-0 |
|---|---|---|---|---|---|---|---|
| TXREQ[7:0] | | | | | | | |
| bit 7 | | | | | | | bit 0 |

| **Legend:** | | | |
|---|---|---|---|
| R = Readable bit | W = Writable bit | U = Unimplemented bit, read as '0' | |
| -n = Value at POR | '1' = Bit is set | '0' = Bit is cleared | x = Bit is unknown |

bit 31-1 **TXREQ[31:1]**: Message Send Request bits
TXEN= 1 (Object configured as a Transmit Object)
Setting this bit to '1' requests sending a message.
The bit will automatically clear when the message(s) queued in the object is (are) successfully sent.
**This bit can NOT be used for aborting a transmission.**
TXEN= 0 (Object configured as a Receive Object)
This bit has no effect

bit 0 **TXREQ[0]**: Transmit Queue Message Send Request bit
Setting this bit to '1' requests sending a message.
The bit will automatically clear when the message(s) queued in the object is (are) successfully sent.
**This bit can NOT be used for aborting a transmission.**


© 2019-2020 Microchip Technology Inc.

**REGISTER 3-20: CiTREC – TRANSMIT/RECEIVE ERROR COUNT REGISTER**

| U-0 | U-0 | U-0 | U-0 | U-0 | U-0 | U-0 | U-0 |
|---|---|---|---|---|---|---|---|
| — | — | — | — | — | — | — | — |
| bit 31 | | | | | | | bit 24 |

| U-0 | U-0 | R-1 | R-0 | R-0 | R-0 | R-0 | R-0 |
|---|---|---|---|---|---|---|---|
| — | — | TXBO | TXBP | RXBP | TXWARN | RXWARN | EWARN |
| bit 23 | | | | | | | bit 16 |

| R-0 | R-0 | R-0 | R-0 | R-0 | R-0 | R-0 | R-0 |
|---|---|---|---|---|---|---|---|
| TEC[7:0] | | | | | | | |
| bit 15 | | | | | | | bit 8 |

| R-0 | R-0 | R-0 | R-0 | R-0 | R-0 | R-0 | R-0 |
|---|---|---|---|---|---|---|---|
| REC[7:0] | | | | | | | |
| bit 7 | | | | | | | bit 0 |

| **Legend:** | | | |
|---|---|---|---|
| R = Readable bit | W = Writable bit | U = Unimplemented bit, read as '0' | |
| -n = Value at POR | '1' = Bit is set | '0' = Bit is cleared | x = Bit is unknown |

bit 31-22 **Unimplemented**: Read as '0'

bit 21 **TXBO**: Transmitter in Bus Off State bit (TEC > 255)
In Configuration mode, TXBO is set, since the module is not on the bus.

bit 20 **TXBP**: Transmitter in Error Passive State bit (TEC > 127)

bit 19 **RXBP**: Receiver in Error Passive State bit (REC > 127)

bit 18 **TXWARN**: Transmitter in Error Warning State bit (128 > TEC > 95)

bit 17 **RXWARN**: Receiver in Error Warning State bit (128 > REC > 95)

bit 16 **EWARN**: Transmitter or Receiver is in Error Warning State bit

bit 15-8 **TEC[7:0]**: Transmit Error Counter bits

bit 7-0 **REC[7:0]**: Receive Error Counter bits

**REGISTER 3-21: CiBDIAG0 – BUS DIAGNOSTIC REGISTER 0**

| R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| DTERRCNT[7:0] | | | | | | | |
| bit 31 | | | | | | | bit 24 |

| R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| DRERRCNT[7:0] | | | | | | | |
| bit 23 | | | | | | | bit 16 |

| R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| NTERRCNT[7:0] | | | | | | | |
| bit 15 | | | | | | | bit 8 |

| R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| NRERRCNT[7:0] | | | | | | | |
| bit 7 | | | | | | | bit 0 |

| **Legend:** | | | |
|---|---|---|---|
| R = Readable bit | W = Writable bit | U = Unimplemented bit, read as '0' | |
| -n = Value at POR | '1' = Bit is set | '0' = Bit is cleared | x = Bit is unknown |

bit 31-24 **DTERRCNT[7:0]**: Data Bit Rate Transmit Error Counter bits

bit 23-16 **DRERRCNT[7:0]**: Data Bit Rate Receive Error Counter bits

bit 15-8 **NTERRCNT[7:0]**: Nominal Bit Rate Transmit Error Counter bits

bit 7-0 **NRERRCNT[7:0]**: Nominal Bit Rate Receive Error Counter bits


© 2019-2020 Microchip Technology Inc.

**REGISTER 3-22: CiBDIAG1 – BUS DIAGNOSTICS REGISTER 1**

| R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | U-0 | R/W-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| DLCMM | ESI | DCRCERR | DSTUFERR | DFORMERR | — | DBIT1ERR | DBIT0ERR |
| bit 31 | | | | | | | bit 24 |

| R/W-0 | U-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| TXBOERR | — | NCRCERR | NSTUFERR | NFORMERR | NACKERR | NBIT1ERR | NBIT0ERR |
| bit 23 | | | | | | | bit 16 |

| R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| EFMSGCNT[15:8] | | | | | | | |
| bit 15 | | | | | | | bit 8 |

| R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| EFMSGCNT[7:0] | | | | | | | |
| bit 7 | | | | | | | bit 0 |

| **Legend:** | | | |
|---|---|---|---|
| R = Readable bit | W = Writable bit | U = Unimplemented bit, read as '0' | |
| -n = Value at POR | '1' = Bit is set | '0' = Bit is cleared | x = Bit is unknown |

bit 31 **DLCMM**: DLC Mismatch bit
During a transmission or reception, the specified DLC is larger than the PLSIZE of the FIFO element.

bit 30 **ESI**: ESI flag of a received CAN FD message was set.

bit 29 **DCRCERR**: Same as for nominal bit rate (see below).

bit 28 **DSTUFERR**: Same as for nominal bit rate (see below).

bit 27 **DFORMERR**: Same as for nominal bit rate (see below).

bit 26 **Unimplemented**: Read as '0'

bit 25 **DBIT1ERR**: Same as for nominal bit rate (see below).

bit 24 **DBIT0ERR**: Same as for nominal bit rate (see below).

bit 23 **TXBOERR**: Device went to bus-off (and auto-recovered).

bit 22 **Unimplemented**: Read as '0'

bit 21 **NCRCERR**: The CRC check sum of a received message was incorrect. The CRC of an incoming message does not match with the CRC calculated from the received data.

bit 20 **NSTUFERR**: More than 5 equal bits in a sequence have occurred in a part of a received message where this is not allowed.

bit 19 **NFORMERR**: A fixed format part of a received frame has the wrong format.

bit 18 **NACKERR**: Transmitted message was not acknowledged.

bit 17 **NBIT1ERR**: During the transmission of a message (with the exception of the arbitration field), the device wanted to send a recessive level (bit of logical value '1'), but the monitored bus value was dominant.

bit 16 **NBIT0ERR**: During the transmission of a message (or acknowledge bit, or active error flag, or overload flag), the device wanted to send a dominant level (data or identifier bit logical value '0'), but the monitored bus value was recessive.

bit 15-0 **EFMSGCNT[15:0]**: Error Free Message Counter bits

**REGISTER 3-23: CiTEFCON – TRANSMIT EVENT FIFO CONTROL REGISTER**

| U-0 | U-0 | U-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| — | — | — | FSIZE[4:0](1) | | | | |
| bit 31 | | | | | | | bit 24 |

| U-0 | U-0 | U-0 | U-0 | U-0 | U-0 | U-0 | U-0 |
|---|---|---|---|---|---|---|---|
| — | — | — | — | — | — | — | — |
| bit 23 | | | | | | | bit 16 |

| U-0 | U-0 | U-0 | U-0 | U-0 | S/HC-1 | U-0 | S/HC-0 |
|---|---|---|---|---|---|---|---|
| — | — | — | — | — | FRESET(2) | — | UINC |
| bit 15 | | | | | | | bit 8 |

| U-0 | U-0 | R/W-0 | U-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| — | — | TEFTSEN(1) | — | TEFOVIE | TEFFIE | TEFHIE | TEFNEIE |
| bit 7 | | | | | | | bit 0 |

| **Legend:** | | | |
|---|---|---|---|
| R = Readable bit | W = Writable bit | U = Unimplemented bit, read as '0' | |
| -n = Value at POR | '1' = Bit is set | '0' = Bit is cleared | x = Bit is unknown |

bit 31-29 **Unimplemented:** Read as '0'

bit 28-24 **FSIZE[4:0]:** FIFO Size bits(1)
0_0000 = FIFO is 1 Message deep
0_0001 = FIFO is 2 Messages deep
0_0010 = FIFO is 3 Messages deep
...
1_1111 = FIFO is 32 Messages deep

bit 23-11 **Unimplemented:** Read as '0'

bit 10 **FRESET**: FIFO Reset bit(2)
1 = FIFO will be reset when bit is set, cleared by hardware when FIFO was reset. The user should wait for this bit to clear before taking any action.
0 = No effect

bit 9 **Unimplemented:** Read as '0'

bit 8 **UINC**: Increment Tail bit
When this bit is set, the FIFO tail will increment by a single message.

bit 7-6 **Unimplemented:** Read as '0'

bit 5 **TEFTSEN**: Transmit Event FIFO Time Stamp Enable bit(1)
1 = Time Stamp objects in TEF
0 = Do not Time Stamp objects in TEF

bit 4 **Unimplemented:** Read as '0'

bit 3 **TEFOVIE**: Transmit Event FIFO Overflow Interrupt Enable bit
1 = Interrupt enabled for overflow event
0 = Interrupt disabled for overflow event

bit 2 **TEFFIE**: Transmit Event FIFO Full Interrupt Enable bit
1 = Interrupt enabled for FIFO full
0 = Interrupt disabled for FIFO full

**Note 1:** These bits can only be modified in Configuration mode.
**2:** FRESET is set while in Configuration mode and is automatically cleared in Normal mode.


© 2019-2020 Microchip Technology Inc.

**REGISTER 3-23: CiTEFCON – TRANSMIT EVENT FIFO CONTROL REGISTER (CONTINUED)**

bit 1 **TEFHIE**: Transmit Event FIFO Half Full Interrupt Enable bit
1 = Interrupt enabled for FIFO half full
0 = Interrupt disabled for FIFO half full

bit 0 **TEFNEIE**: Transmit Event FIFO Not Empty Interrupt Enable bit
1 = Interrupt enabled for FIFO not empty
0 = Interrupt disabled for FIFO not empty

**Note 1:** These bits can only be modified in Configuration mode.
**2:** FRESET is set while in Configuration mode and is automatically cleared in Normal mode.

**REGISTER 3-24: CiTEFSTA – TRANSMIT EVENT FIFO STATUS REGISTER**

| U-0 | U-0 | U-0 | U-0 | U-0 | U-0 | U-0 | U-0 |
|---|---|---|---|---|---|---|---|
| — | — | — | — | — | — | — | — |
| bit 31 | | | | | | | bit 24 |

| U-0 | U-0 | U-0 | U-0 | U-0 | U-0 | U-0 | U-0 |
|---|---|---|---|---|---|---|---|
| — | — | — | — | — | — | — | — |
| bit 23 | | | | | | | bit 16 |

| U-0 | U-0 | U-0 | U-0 | U-0 | U-0 | U-0 | U-0 |
|---|---|---|---|---|---|---|---|
| — | — | — | — | — | — | — | — |
| bit 15 | | | | | | | bit 8 |

| U-0 | U-0 | U-0 | U-0 | HS/C-0 | R-0 | R-0 | R-0 |
|---|---|---|---|---|---|---|---|
| — | — | — | — | TEFOVIF | TEFFIF(1) | TEFHIF(1) | TEFNEIF(1) |
| bit 7 | | | | | | | bit 0 |

| **Legend:** | | | |
|---|---|---|---|
| R = Readable bit | W = Writable bit | U = Unimplemented bit, read as '0' | |
| -n = Value at POR | '1' = Bit is set | '0' = Bit is cleared | x = Bit is unknown |

bit 31-4 **Unimplemented:** Read as '0'

bit 3 **TEFOVIF**: Transmit Event FIFO Overflow Interrupt Flag bit
- 1 = Overflow event has occurred
- 0 = No overflow event occurred

bit 2 **TEFFIF**: Transmit Event FIFO Full Interrupt Flag bit(1)
- 1 = FIFO is full
- 0 = FIFO is not full

bit 1 **TEFHIF**: Transmit Event FIFO Half Full Interrupt Flag bit(1)
- 1 = FIFO is ≥ half full
- 0 = FIFO is < half full

bit 0 **TEFNEIF**: Transmit Event FIFO Not Empty Interrupt Flag bit(1)
- 1 = FIFO is not empty, contains at least one message
- 0 = FIFO is empty

**Note 1:** This bit is read only and reflects the status of the FIFO.


© 2019-2020 Microchip Technology Inc.

**REGISTER 3-25: CiTEFUA – TRANSMIT EVENT FIFO USER ADDRESS REGISTER**

| R-x | R-x | R-x | R-x | R-x | R-x | R-x | R-x |
|---|---|---|---|---|---|---|---|
| TEFUA[31:24] | | | | | | | |
| bit 31 | | | | | | | bit 24 |

| R-x | R-x | R-x | R-x | R-x | R-x | R-x | R-x |
|---|---|---|---|---|---|---|---|
| TEFUA[23:16] | | | | | | | |
| bit 23 | | | | | | | bit 16 |

| R-x | R-x | R-x | R-x | R-x | R-x | R-x | R-x |
|---|---|---|---|---|---|---|---|
| TEFUA[15:8] | | | | | | | |
| bit 15 | | | | | | | bit 8 |

| R-x | R-x | R-x | R-x | R-x | R-x | R-x | R-x |
|---|---|---|---|---|---|---|---|
| TEFUA[7:0] | | | | | | | |
| bit 7 | | | | | | | bit 0 |

| **Legend:** | | | |
|---|---|---|---|
| R = Readable bit | W = Writable bit | U = Unimplemented bit, read as ‘0’ | |
| -n = Value at POR | ‘1’ = Bit is set | ‘0’ = Bit is cleared | x = Bit is unknown |

bit 31-0 **TEFUA[31:0]:** Transmit Event FIFO User Address bits
A read of this register will return the address where the next object is to be read (FIFO tail).

**Note 1:** This register is not guaranteed to read correctly in Configuration mode and should only be accessed when the module is not in Configuration mode.

**REGISTER 3-26: CiTXQCON – TRANSMIT QUEUE CONTROL REGISTER**

| R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| PLSIZE[2:0](1) | | | FSIZE[4:0](1) | | | | |
| bit 31 | | | | | | | bit 24 |

| U-0 | R/W-1 | R/W-1 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| — | TXAT[1:0] | | TXPRI[4:0] | | | | |
| bit 23 | | | | | | | bit 16 |

| U-0 | U-0 | U-0 | U-0 | U-0 | S/HC-1 | R/W/HC-0 | S/HC-0 |
|---|---|---|---|---|---|---|---|
| — | — | — | — | — | FRESET(3) | TXREQ(2) | UINC |
| bit 15 | | | | | | | bit 8 |

| R-1 | U-0 | U-0 | R/W-0 | U-0 | R/W-0 | U-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| TXEN | — | — | TXATIE | — | TXQEIE | — | TXQNIE |
| bit 7 | | | | | | | bit 0 |

| **Legend:** | | | |
|---|---|---|---|
| R = Readable bit | W = Writable bit | U = Unimplemented bit, read as '0' | |
| -n = Value at POR | '1' = Bit is set | '0' = Bit is cleared | x = Bit is unknown |

bit 31-29 **PLSIZE[2:0]**: Payload Size bits(1)
000 = 8 data bytes
001 = 12 data bytes
010 = 16 data bytes
011 = 20 data bytes
100 = 24 data bytes
101 = 32 data bytes
110 = 48 data bytes
111 = 64 data bytes

bit 28-24 **FSIZE[4:0]**: FIFO Size bits(1)
0_0000 = FIFO is 1 Message deep
0_0001 = FIFO is 2 Messages deep
0_0010 = FIFO is 3 Messages deep
...
1_1111 = FIFO is 32 Messages deep

bit 23 **Unimplemented:** Read as '0'

bit 22-21 **TXAT[1:0]**: Retransmission Attempts bits
This feature is enabled when CiCON.RTXAT is set.
00 = Disable retransmission attempts
01 = Three retransmission attempts
10 = Unlimited number of retransmission attempts
11 = Unlimited number of retransmission attempts

bit 20-16 **TXPRI[4:0]**: Message Transmit Priority bits
00000 = Lowest Message Priority
...
11111 = Highest Message Priority

bit 15-11 **Unimplemented:** Read as '0'

**Note 1:** These bits can only be modified in Configuration mode.
**2:** This bit is updated when a message completes (or aborts) or when the FIFO is reset.
**3:** FRESET is set while in Configuration mode and is automatically cleared in Normal mode.


© 2019-2020 Microchip Technology Inc.

**REGISTER 3-26: CiTXQCON – TRANSMIT QUEUE CONTROL REGISTER (CONTINUED)**

bit 10 **FRESET**: FIFO Reset bit(3)
1 = FIFO will be reset when bit is set; cleared by hardware when FIFO was reset. User should wait until this bit is clear before taking any action.
0 = No effect

bit 9 **TXREQ**: Message Send Request bit(2)
1 = Requests sending a message; the bit will automatically clear when all the messages queued in the TXQ are successfully sent.
0 = Clearing the bit to '0' while set ('1') will request a message abort.

bit 8 **UINC**: Increment Head bit
When this bit is set, the FIFO head will increment by a single message.

bit 7 **TXEN**: TX Enable
1 = Transmit Message Queue. This bit always reads as '1'.

bit 6-5 **Unimplemented:** Read as '0'

bit 4 **TXATIE**: Transmit Attempts Exhausted Interrupt Enable bit
1 = Enable interrupt
0 = Disable interrupt

bit 3 **Unimplemented:** Read as '0'

bit 2 **TXQEIE**: Transmit Queue Empty Interrupt Enable bit
1 = Interrupt enabled for TXQ empty
0 = Interrupt disabled for TXQ empty

bit 1 **Unimplemented:** Read as '0'

bit 0 **TXQNIE**: Transmit Queue Not Full Interrupt Enable bit
1 = Interrupt enabled for TXQ not full
0 = Interrupt disabled for TXQ not full

**Note 1:** These bits can only be modified in Configuration mode.
**2:** This bit is updated when a message completes (or aborts) or when the FIFO is reset.
**3:** FRESET is set while in Configuration mode and is automatically cleared in Normal mode.

**REGISTER 3-27: CiTXQSTA – TRANSMIT QUEUE STATUS REGISTER**

| U-0 | U-0 | U-0 | U-0 | U-0 | U-0 | U-0 | U-0 |
|---|---|---|---|---|---|---|---|
| — | — | — | — | — | — | — | — |
| bit 31 | | | | | | | bit 24 |

| U-0 | U-0 | U-0 | U-0 | U-0 | U-0 | U-0 | U-0 |
|---|---|---|---|---|---|---|---|
| — | — | — | — | — | — | — | — |
| bit 23 | | | | | | | bit 16 |

| U-0 | U-0 | U-0 | R-0 | R-0 | R-0 | R-0 | R-0 |
|---|---|---|---|---|---|---|---|
| — | — | — | TXQCI[4:0](1) | | | | |
| bit 15 | | | | | | | bit 8 |

| HS/C-0 | HS/C-0 | HS/C-0 | HS/C-0 | U-0 | R-1 | U-0 | R-1 |
|---|---|---|---|---|---|---|---|
| TXABT(2)(3) | TXLARB (2)(3) | TXERR(2)(3) | TXATIF | — | TXQEIF | — | TXQNIF |
| bit 7 | | | | | | | bit 0 |

| **Legend:** | | | |
|---|---|---|---|
| R = Readable bit | W = Writable bit | U = Unimplemented bit, read as '0' | |
| -n = Value at POR | '1' = Bit is set | '0' = Bit is cleared | x = Bit is unknown |

bit 3-13 **Unimplemented:** Read as '0'

bit 12-8 **TXQCI[4:0]:** Transmit Queue Message Index bits(1)
A read of this register will return an index to the message that the FIFO will next attempt to transmit.

bit 7 **TXABT**: Message Aborted Status bit(2)(3)
1 = Message was aborted
0 = Message completed successfully

bit 6 **TXLARB**: Message Lost Arbitration Status bit(2)(3)
1 = Message lost arbitration while being sent
0 = Message did not loose arbitration while being sent

bit 5 **TXERR**: Error Detected During Transmission bit(2)(3)
1 = A bus error occurred while the message was being sent
0 = A bus error did not occur while the message was being sent

bit 4 **TXATIF**: Transmit Attempts Exhausted Interrupt Pending bit
1 = Interrupt pending
0 = Interrupt Not pending

bit 3 **Unimplemented:** Read as '0'

bit 2 **TXQEIF**: Transmit Queue Empty Interrupt Flag bit
1 = TXQ is empty
0 = TXQ is not empty, at least 1 message queued to be transmitted

bit 1 **Unimplemented:** Read as '0'

bit 0 **TXQNIF**: Transmit Queue Not Full Interrupt Flag bit
1 = TXQ is not full
0 = TXQ is full

**Note 1:** TXQCI[4:0] gives a zero-indexed value to the message in the TXQ. If the TXQ is 4 messages deep (FSIZE = 5'h03) TXQCI will take on a value of 0 to 3 depending on the state of the TXQ.

**2:** This bit is cleared when TXREQ is set or by writing a 0 using the SPI.

**3:** This bit is updated when a message completes (or aborts) or when the TXQ is reset.


© 2019-2020 Microchip Technology Inc.

**REGISTER 3-28: CiTXQUA – TRANSMIT QUEUE USER ADDRESS REGISTER**

| R-x | R-x | R-x | R-x | R-x | R-x | R-x | R-x |
|---|---|---|---|---|---|---|---|
| TXQUA[31:24] | | | | | | | |
| bit 31 | | | | | | | bit 24 |

| R-x | R-x | R-x | R-x | R-x | R-x | R-x | R-x |
|---|---|---|---|---|---|---|---|
| TXQUA[23:16] | | | | | | | |
| bit 23 | | | | | | | bit 16 |

| R-x | R-x | R-x | R-x | R-x | R-x | R-x | R-x |
|---|---|---|---|---|---|---|---|
| TXQUA[15:8] | | | | | | | |
| bit 15 | | | | | | | bit 8 |

| R-x | R-x | R-x | R-x | R-x | R-x | R-x | R-x |
|---|---|---|---|---|---|---|---|
| TXQUA[7:0] | | | | | | | |
| bit 7 | | | | | | | bit 0 |

| **Legend:** | | | |
|---|---|---|---|
| R = Readable bit | W = Writable bit | U = Unimplemented bit, read as '0' | |
| -n = Value at POR | '1' = Bit is set | '0' = Bit is cleared | x = Bit is unknown |

bit 31-0 **TXQUA[31:0]:** TXQ User Address bits
A read of this register will return the address where the next message is to be written (TXQ head).

**Note 1:** This register is not guaranteed to read correctly in Configuration mode and should only be accessed when the module is not in Configuration mode.

**REGISTER 3-29: CiFIFOCONm – FIFO CONTROL REGISTER m, (m = 1 TO 31)**

| R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| PLSIZE[2:0](1) | | | FSIZE[4:0](1) | | | | |
| bit 31 | | | | | | | bit 24 |

| U-0 | R/W-1 | R/W-1 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| — | TXAT[1:0] | | TXPRI[4:0] | | | | |
| bit 23 | | | | | | | bit 16 |

| U-0 | U-0 | U-0 | U-0 | U-0 | S/HC-1 | R/W/HC-0 | S/HC-0 |
|---|---|---|---|---|---|---|---|
| — | — | — | — | — | FRESET(3) | TXREQ(2) | UINC |
| bit 15 | | | | | | | bit 8 |

| R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| TXEN(1) | RTREN | RXTSEN(1) | TXATIE | RXOVIE | TFERFFIE | TFHRFHIE | TFNRFNIE |
| bit 7 | | | | | | | bit 0 |

| **Legend:** | | | |
|---|---|---|---|
| R = Readable bit | W = Writable bit | U = Unimplemented bit, read as '0' | |
| -n = Value at POR | '1' = Bit is set | '0' = Bit is cleared | x = Bit is unknown |

bit 31-29 **PLSIZE[2:0]:** Payload Size bits(1)
000 = 8 data bytes
001 = 12 data bytes
010 = 16 data bytes
011 = 20 data bytes
100 = 24 data bytes
101 = 32 data bytes
110 = 48 data bytes
111 = 64 data bytes

bit 28-24 **FSIZE[4:0]:** FIFO Size bits(1)
0_0000 = FIFO is 1 Message deep
0_0001 = FIFO is 2 Messages deep
0_0010 = FIFO is 3 Messages deep
...
1_1111 = FIFO is 32 Messages deep

bit 23 **Unimplemented:** Read as '0'

bit 22-21 **TXAT[1:0]:** Retransmission Attempts bits
This feature is enabled when CiCON.RTXAT is set.
00 = Disable retransmission attempts
01 = Three retransmission attempts
10 = Unlimited number of retransmission attempts
11 = Unlimited number of retransmission attempts

bit 20-16 **TXPRI[4:0]:** Message Transmit Priority bits
00000 = Lowest Message Priority
...
11111 = Highest Message Priority

**Note 1:** These bits can only be modified in Configuration mode.
**2:** This bit is updated when a message completes (or aborts) or when the FIFO is reset.
**3:** FRESET is set while in Configuration mode and is automatically cleared in Normal mode.


© 2019-2020 Microchip Technology Inc.

**REGISTER 3-29: CiFIFOCONm – FIFO CONTROL REGISTER m, (m = 1 TO 31) (CONTINUED)**

| | |
|---|---|
| bit 15-11 | **Unimplemented:** Read as '0' |
| bit 10 | **FRESET:** FIFO Reset bit(3)<br>1 = FIFO will be reset when bit is set; cleared by hardware when FIFO was reset. User should wait until this bit is clear before taking any action.<br>0 = No effect |
| bit 9 | **TXREQ:** Message Send Request bit(2)<br>TXEN = 1 (FIFO configured as a Transmit FIFO)<br>1 = Requests sending a message; the bit will automatically clear when all the messages queued in the FIFO are successfully sent.<br>0 = Clearing the bit to '0' while set ('1') will request a message abort.<br>TXEN = 0 (FIFO configured as a Receive FIFO)<br>This bit has no effect. |
| bit 8 | **UINC:** Increment Head/Tail bit<br>TXEN = 1 (FIFO configured as a Transmit FIFO)<br>When this bit is set, the FIFO head will increment by a single message.<br>TXEN = 0 (FIFO configured as a Receive FIFO)<br>When this bit is set, the FIFO tail will increment by a single message. |
| bit 7 | **TXEN:** TX/RX FIFO Selection bit(1)<br>1 = Transmit FIFO<br>0 = Receive FIFO |
| bit 6 | **RTREN**: Auto RTR Enable bit<br>1 = When a remote transmit is received, TXREQ will be set.<br>0 = When a remote transmit is received, TXREQ will be unaffected. |
| bit 5 | **RXTSEN**: Received Message Time Stamp Enable bit(1)<br>1 = Capture time stamp in received message object in RAM.<br>0 = Do not capture time stamp. |
| bit 4 | **TXATIE**: Transmit Attempts Exhausted Interrupt Enable bit<br>1 = Enable interrupt<br>0 = Disable interrupt |
| bit 3 | **RXOVIE**: Overflow Interrupt Enable bit<br>1 = Interrupt enabled for overflow event<br>0 = Interrupt disabled for overflow event |
| bit 2 | **TFERFFIE**: Transmit/Receive FIFO Empty/Full Interrupt Enable bit<br>TXEN = 1 (FIFO configured as a Transmit FIFO)<br>Transmit FIFO Empty Interrupt Enable<br>1 = Interrupt enabled for FIFO empty<br>0 = Interrupt disabled for FIFO empty<br>TXEN = 0 (FIFO configured as a Receive FIFO)<br>Receive FIFO Full Interrupt Enable<br>1 = Interrupt enabled for FIFO full<br>0 = Interrupt disabled for FIFO full |

**Note 1:** These bits can only be modified in Configuration mode.

**2:** This bit is updated when a message completes (or aborts) or when the FIFO is reset.

**3:** FRESET is set while in Configuration mode and is automatically cleared in Normal mode.

**REGISTER 3-29: CiFIFOCONm – FIFO CONTROL REGISTER m, (m = 1 TO 31) (CONTINUED)**

bit 1 **TFHRFHIE**: Transmit/Receive FIFO Half Empty/Half Full Interrupt Enable bit
TXEN = 1 (FIFO configured as a Transmit FIFO)
Transmit FIFO Half Empty Interrupt Enable
1 = Interrupt enabled for FIFO half empty
0 = Interrupt disabled for FIFO half empty
TXEN = 0 (FIFO configured as a Receive FIFO)
Receive FIFO Half Full Interrupt Enable
1 = Interrupt enabled for FIFO half full
0 = Interrupt disabled for FIFO half full

bit 0 **TFNRFNIE**: Transmit/Receive FIFO Not Full/Not Empty Interrupt Enable bit
TXEN = 1 (FIFO configured as a Transmit FIFO)
Transmit FIFO Not Full Interrupt Enable
1 = Interrupt enabled for FIFO not full
0 = Interrupt disabled for FIFO not full
TXEN = 0 (FIFO configured as a Receive FIFO)
Receive FIFO Not Empty Interrupt Enable
1 = Interrupt enabled for FIFO not empty
0 = Interrupt disabled for FIFO not empty

**Note 1:** These bits can only be modified in Configuration mode.
**2:** This bit is updated when a message completes (or aborts) or when the FIFO is reset.
**3:** FRESET is set while in Configuration mode and is automatically cleared in Normal mode.


© 2019-2020 Microchip Technology Inc.

**REGISTER 3-30: CiFIFOSTAm – FIFO STATUS REGISTER m, (m = 1 TO 31)**

| U-0 | U-0 | U-0 | U-0 | U-0 | U-0 | U-0 | U-0 |
|---|---|---|---|---|---|---|---|
| — | — | — | — | — | — | — | — |
| bit 31 | | | | | | | bit 24 |

| U-0 | U-0 | U-0 | U-0 | U-0 | U-0 | U-0 | U-0 |
|---|---|---|---|---|---|---|---|
| — | — | — | — | — | — | — | — |
| bit 23 | | | | | | | bit 16 |

| U-0 | U-0 | U-0 | R-0 | R-0 | R-0 | R-0 | R-0 |
|---|---|---|---|---|---|---|---|
| — | — | — | FIFOCI[4:0](1) | | | | |
| bit 15 | | | | | | | bit 8 |

| HS/C-0 | HS/C-0 | HS/C-0 | HS/C-0 | HS/C-0 | R-0 | R-0 | R-0 |
|---|---|---|---|---|---|---|---|
| TXABT(2)(3) | TXLARB (2)(3) | TXERR(2)(3) | TXATIF | RXOVIF | TFERFFIF | TFHRFHIF | TFNRFNIF |
| bit 7 | | | | | | | bit 0 |

| **Legend:** | | | |
|---|---|---|---|
| R = Readable bit | W = Writable bit | U = Unimplemented bit, read as '0' | |
| -n = Value at POR | '1' = Bit is set | '0' = Bit is cleared | x = Bit is unknown |

bit 31-13 **Unimplemented:** Read as '0'

bit 12-8 **FIFOCI[4:0]:** FIFO Message Index bits(1)

TXEN = 1 (FIFO is configured as a Transmit FIFO)
A read of this bit field will return an index to the message that the FIFO will next attempt to transmit.
TXEN = 0 (FIFO is configured as a Receive FIFO)
A read of this bit field will return an index to the message that the FIFO will use to save the next message

bit 7 **TXABT:** Message Aborted Status bit(2)(3)

1 = Message was aborted
0 = Message completed successfully

bit 6 **TXLARB:** Message Lost Arbitration Status bit(2)(3)

1 = Message lost arbitration while being sent
0 = Message did not lose arbitration while being sent

bit 5 **TXERR:** Error Detected During Transmission bit(2)(3)

1 = A bus error occurred while the message was being sent
0 = A bus error did not occur while the message was being sent

bit 4 **TXATIF:** Transmit Attempts Exhausted Interrupt Pending bit

TXEN = 1 (FIFO is configured as a Transmit FIFO)
1 = Interrupt pending
0 = Interrupt not pending
TXEN = 0 (FIFO is configured as a Receive FIFO)
Read as '0'

**Note 1:** FIFOCI[4:0] gives a zero-indexed value to the message in the FIFO. If the FIFO is 4 messages deep (FSIZE = 5'h03) FIFOCI will take on a value of 0 to 3 depending on the state of the FIFO.

**2:** This bit is cleared when TXREQ is set or by writing a 0 using the SPI.

**3:** This bit is updated when a message completes (or aborts) or when the FIFO is reset.

**REGISTER 3-30: CiFIFOSTAm – FIFO STATUS REGISTER m, (m = 1 TO 31) (CONTINUED)**

bit 3 **RXOVIF:** Receive FIFO Overflow Interrupt Flag bit
TXEN = 1 (FIFO is configured as a Transmit FIFO)
Unused, Read as '0'
TXEN = 0 (FIFO is configured as a Receive FIFO)
1 = Overflow event has occurred
0 = No overflow event has occurred

bit 2 **TFERFFIF:** Transmit/Receive FIFO Empty/Full Interrupt Flag bit
TXEN = 1 (FIFO is configured as a Transmit FIFO)
Transmit FIFO Empty Interrupt Flag
1 = FIFO is empty
0 = FIFO is not empty; at least one message queued to be transmitted
TXEN = 0 (FIFO is configured as a Receive FIFO)
Receive FIFO Full Interrupt Flag
1 = FIFO is full
0 = FIFO is not full

bit 1 **TFHRFHIF:** Transmit/Receive FIFO Half Empty/Half Full Interrupt Flag bit
TXEN = 1 (FIFO is configured as a Transmit FIFO)
Transmit FIFO Half Empty Interrupt Flag
1 = FIFO is ≤ half full
0 = FIFO is > half full
TXEN = 0 (FIFO is configured as a Receive FIFO)
Receive FIFO Half Full Interrupt Flag
1 = FIFO is ≥ half full
0 = FIFO is < half full

bit 0 **TFNRFNIF:** Transmit/Receive FIFO Not Full/Not Empty Interrupt Flag bit
TXEN = 1 (FIFO is configured as a Transmit FIFO)
Transmit FIFO Not Full Interrupt Flag
1 = FIFO is not full
0 = FIFO is full
TXEN = 0 (FIFO is configured as a Receive FIFO)
Receive FIFO Not Empty Interrupt Flag
1 = FIFO is not empty, contains at least one message
0 = FIFO is empty

**Note 1:** FIFOCI[4:0] gives a zero-indexed value to the message in the FIFO. If the FIFO is 4 messages deep (FSIZE = 5'h03) FIFOCI will take on a value of 0 to 3 depending on the state of the FIFO.
**2:** This bit is cleared when TXREQ is set or by writing a 0 using the SPI.
**3:** This bit is updated when a message completes (or aborts) or when the FIFO is reset.


© 2019-2020 Microchip Technology Inc.

**REGISTER 3-31: CiFIFOUAm – FIFO USER ADDRESS REGISTER m, (m = 1 TO 31)**

| R-x | R-x | R-x | R-x | R-x | R-x | R-x | R-x |
|---|---|---|---|---|---|---|---|
| FIFOUA[31:24] | | | | | | | |
| bit 31 | | | | | | | bit 24 |

| R-x | R-x | R-x | R-x | R-x | R-x | R-x | R-x |
|---|---|---|---|---|---|---|---|
| FIFOUA[23:16] | | | | | | | |
| bit 23 | | | | | | | bit 16 |

| R-x | R-x | R-x | R-x | R-x | R-x | R-x | R-x |
|---|---|---|---|---|---|---|---|
| FIFOUA[15:8] | | | | | | | |
| bit 15 | | | | | | | bit 8 |

| R-x | R-x | R-x | R-x | R-x | R-x | R-x | R-x |
|---|---|---|---|---|---|---|---|
| FIFOUA[7:0] | | | | | | | |
| bit 7 | | | | | | | bit 0 |

| **Legend:** | | | |
|---|---|---|---|
| R = Readable bit | W = Writable bit | U = Unimplemented bit, read as '0' | |
| -n = Value at POR | '1' = Bit is set | '0' = Bit is cleared | x = Bit is unknown |

bit 31-0 **FIFOUA[31:0]:** FIFO User Address bits

TXEN = 1 (FIFO is configured as a Transmit FIFO)
A read of this register will return the address where the next message is to be written (FIFO head).
TXEN = 0 (FIFO is configured as a Receive FIFO)
A read of this register will return the address where the next message is to be read (FIFO tail).

**Note 1:** This bit is not guaranteed to read correctly in Configuration mode and should only be accessed when the module is not in Configuration mode.

**REGISTER 3-32: CiFLTCONm – FILTER CONTROL REGISTER m, (m = 0 TO 7)**

| R/W-0 | U-0 | U-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| FLTEN3 | — | — | F3BP[4:0](1) | | | | |
| bit 31 | | | | | | | bit 24 |

| R/W-0 | U-0 | U-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| FLTEN2 | — | — | F2BP[4:0](1) | | | | |
| bit 23 | | | | | | | bit 16 |

| R/W-0 | U-0 | U-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| FLTEN1 | — | — | F1BP[4:0](1) | | | | |
| bit 15 | | | | | | | bit 8 |

| R/W-0 | U-0 | U-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| FLTEN0 | — | — | F0BP[4:0](1) | | | | |
| bit 7 | | | | | | | bit 0 |

| **Legend:** | | | |
|---|---|---|---|
| R = Readable bit | W = Writable bit | U = Unimplemented bit, read as '0' | |
| -n = Value at POR | '1' = Bit is set | '0' = Bit is cleared | x = Bit is unknown |

bit 31 **FLTEN3:** Enable Filter 3 to Accept Messages bit
1 = Filter is enabled
0 = Filter is disabled

bit 30-29 **Unimplemented:** Read as '0'

bit 28-24 **F3BP[4:0]:** Pointer to FIFO when Filter 3 hits bits(1)
1_1111 = Message matching filter is stored in FIFO 31
1_1110 = Message matching filter is stored in FIFO 30
.....
0_0010 = Message matching filter is stored in FIFO 2
0_0001 = Message matching filter is stored in FIFO 1
0_0000 = Reserved FIFO 0 is the TX Queue and cannot receive messages

bit 23 **FLTEN[2]:** Enable Filter 2 to Accept Messages bit
1 = Filter is enabled
0 = Filter is disabled

bit 22-21 **Unimplemented:** Read as '0'

bit 20-16 **F2BP[4:0]:** Pointer to FIFO when Filter 2 hits bits(1)
1_1111 = Message matching filter is stored in FIFO 31
1_1110 = Message matching filter is stored in FIFO 30
.....
0_0010 = Message matching filter is stored in FIFO 2
0_0001 = Message matching filter is stored in FIFO 1
0_0000 = Reserved FIFO 0 is the TX Queue and cannot receive messages

bit 15 **FLTEN1:** Enable Filter 1 to Accept Messages bit
1 = Filter is enabled
0 = Filter is disabled

bit 14-13 **Unimplemented:** Read as '0'

**Note 1:** This bit can only be modified if the corresponding filter is disabled (FLTEN = 0).


© 2019-2020 Microchip Technology Inc.

**REGISTER 3-32: CiFLTCONm – FILTER CONTROL REGISTER m, (m = 0 TO 7) (CONTINUED)**

bit 12-8 **F1BP[4:0]:** Pointer to FIFO when Filter 1 hits bits(1)

1_1111 = Message matching filter is stored in FIFO 31
1_1110 = Message matching filter is stored in FIFO 30
.....
0_0010 = Message matching filter is stored in FIFO 2
0_0001= Message matching filter is stored in FIFO 1
0_0000 = Reserved FIFO 0 is the TX Queue and cannot receive messages

bit 7 **FLTEN[0]:** Enable Filter 0 to Accept Messages bit

1 = Filter is enabled
0 = Filter is disabled

bit 6-5 **Unimplemented:** Read as '0'

bit 4-0 **F0BP[4:0]:** Pointer to FIFO when Filter 0 hits bits(1)

1_1111 = Message matching filter is stored in FIFO 31
1_1110 = Message matching filter is stored in FIFO 30
.....
0_0010 = Message matching filter is stored in FIFO 2
0_0001 = Message matching filter is stored in FIFO 1
0_0000 = Reserved FIFO 0 is the TX Queue and cannot receive messages

**Note 1:** This bit can only be modified if the corresponding filter is disabled (FLTEN = 0).

**REGISTER 3-33: CiFLTOBJm – FILTER OBJECT REGISTER m,(m = 0 TO 31)**

| U-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| — | EXIDE | SID11 | EID[17:13] | | | | |
| bit 31 | | | | | | | bit 24 |

| R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| EID[12:5] | | | | | | | |
| bit 23 | | | | | | | bit 16 |

| R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| EID[4:0] | | | | | SID[10:8] | | |
| bit 15 | | | | | | | bit 8 |

| R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| SID[7:0] | | | | | | | |
| bit 7 | | | | | | | bit 0 |

| **Legend:** | | | |
|---|---|---|---|
| R = Readable bit | W = Writable bit | U = Unimplemented bit, read as '0' | |
| -n = Value at POR | '1' = Bit is set | '0' = Bit is cleared | x = Bit is unknown |

bit 31 **Unimplemented:** Read as '0'

bit 30 **EXIDE:** Extended Identifier Enable bit
If MIDE = 1:
1 = Match only messages with extended identifier
0 = Match only messages with standard identifier

bit 29 **SID11:** Standard Identifier filter bit

bit 28-11 **EID[17:0]:** Extended Identifier filter bits
In DeviceNet mode, these are the filter bits for the first 18 data bits

bit 10-0 **SID[10:0]:** Standard Identifier filter bits

**Note 1:** This register can only be modified when the filter is disabled(CiFLTCON.FLTENm = 0).


© 2019-2020 Microchip Technology Inc.

**REGISTER 3-34: CiMASKm – MASK REGISTER m, (m = 0 TO 31)**

| U-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| — | MIDE | MSID11 | MEID[17:13] | | | | |
| bit 31 | | | | | | | bit 24 |

| R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| MEID[12:5] | | | | | | | |
| bit 23 | | | | | | | bit 16 |

| R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| MEID[4:0] | | | | | MSID[10:8] | | |
| bit 15 | | | | | | | bit 8 |

| R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 |
|---|---|---|---|---|---|---|---|
| MSID[7:0] | | | | | | | |
| bit 7 | | | | | | | bit 0 |

| **Legend:** | | | |
|---|---|---|---|
| R = Readable bit | W = Writable bit | U = Unimplemented bit, read as ‘0’ | |
| -n = Value at POR | ‘1’ = Bit is set | ‘0’ = Bit is cleared | x = Bit is unknown |

bit 31 **Unimplemented:** Read as ‘0’

bit 30 **MIDE:** Identifier Receive mode bit
1 = Match only message types (standard or extended ID) that correspond to EXIDE bit in filter
0 = Match both standard and extended message frames if filters match

bit 29 **MSID11:** Standard Identifier Mask bit

bit 28-11 **MEID[17:0]:** Extended Identifier Mask bits
In DeviceNet mode, these are the mask bits for the first 18 data bits

bit 10-0 **MSID[10:0]:** Standard Identifier Mask bits

**NOTES:**


© 2019-2020 Microchip Technology Inc.

## 3.3 Message Memory

The MCP2518FD device contains a 2 KB RAM that is used to store message objects. There are three different kinds of message objects:

- Table 3-5: Transmit Message Objects used by the TXQ and by TX FIFOs.
- Table 3-6: Receive Message Objects used by RX FIFOs.
- Table 3-7: TEF objects.

Figure 3-2 illustrates how message objects are mapped into RAM. The number of message objects for the TEF, the TXQ, and for each FIFO is configurable. Only the message objects for FIFO2 are shown in detail. The number of data bytes per message object (payload) is individually configurable for the TXQ and each FIFO.

FIFOs and message objects can only be configured in Configuration mode.

The TEF objects are allocated first. Space in RAM will only be reserved if CiCON.STEF = 1.

Next the TXQ objects are allocated. Space in RAM will only be reserved if CiCON.TXQEN = 1.

Next the message objects for FIFO1 through FIFO31 are allocated.

This highly flexible configuration results in an efficient usage of the RAM.

The addresses of the message objects depend on the selected configuration. The application does not have to calculate the addresses. The User Address field provides the address of the next message object to read from or write to.

### 3.3.1 RAM ECC

The RAM is protected with an Error Correction Code (ECC). The ECC logic supports Single Error Detection (SEC) and Double Error Detection (DED).

SEC/DED requires seven parity bits in addition to the 32 data bits.

Figure 3-3 shows the block diagram of the ECC logic.

#### 3.3.1.1 ECC Enable and Disable

The ECC logic can be enabled by setting ECCCON.ECCEN. When ECC is enabled, the data written to the RAM is encoded, and the data read from RAM is decoded.

When the ECC logic is disabled, the data is written to RAM, the parity bits are taken from ECCCON.PARITY. This enables the testing of the ECC logic by the user. During a read the parity bits are stripped out and the data is read back unchanged.

#### 3.3.1.2 RAM Write

During a RAM write, the Encoder calculates the parity bits and adds the parity bits to the input data.

#### 3.3.1.3 RAM READ

During a RAM read, the Decoder checks the output data from RAM for consistency and removes the parity bits. It corrects single bit errors and detects double bit errors.

**FIGURE 3-2: MESSAGE MEMORY ORGANIZATION**

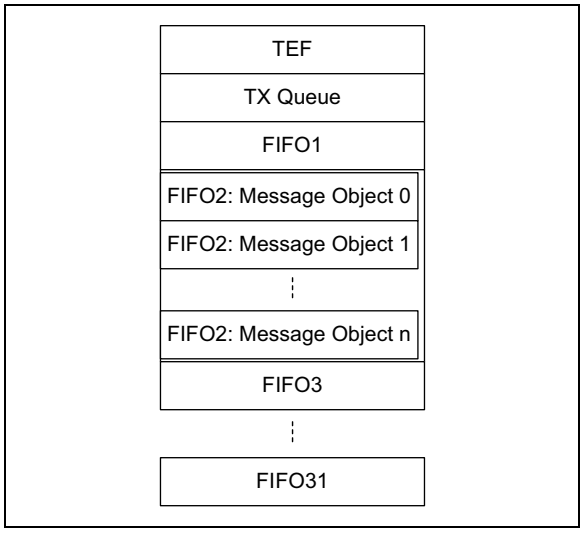



**FIGURE 3-3: ECC LOGIC**

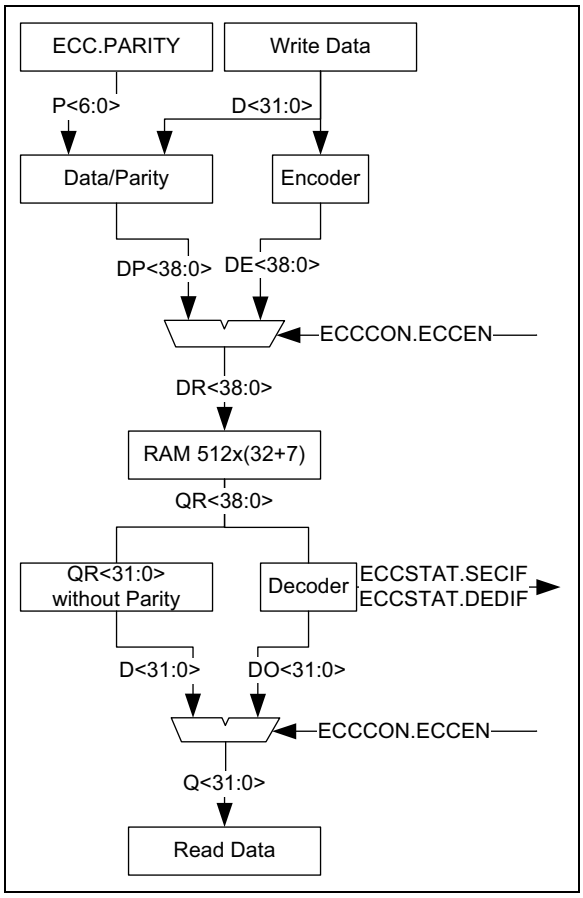


**TABLE 3-5: TRANSMIT MESSAGE OBJECT (TXQ AND TX FIFO)**

| Word | | Bit 31/23/15/7 | Bit 30/22/14/6 | Bit 29/21/13/5 | Bit 28/20/12/4 | Bit 27/19/11/3 | Bit 26/18/10/2 | Bit 25/17/9/1 | Bit 24/16/8/0 |
|---|---|---|---|---|---|---|---|---|---|
| T0 | 31:24 | — | — | SID11 | EID[17:13] | | | | |
| | 23:16 | EID[12:5] | | | | | | | |
| | 15:8 | EID[4:0] | | | | | SID[10:8] | | |
| | 7:0 | SID[7:0] | | | | | | | |
| T1 | 31:24 | SEQ[22:15] | | | | | | | |
| | 23:16 | SEQ[14:7] | | | | | | | |
| | 15:8 | SEQ[6:0] | | | | | | | ESI |
| | 7:0 | FDF | BRS | RTR | IDE | DLC[3:0] | | | |
| T2 (1) | 31:24 | Transmit Data Byte 3 | | | | | | | |
| | 23:16 | Transmit Data Byte 2 | | | | | | | |
| | 15:8 | Transmit Data Byte 1 | | | | | | | |
| | 7:0 | Transmit Data Byte 0 | | | | | | | |
| T3 | 31:24 | Transmit Data Byte 7 | | | | | | | |
| | 23:16 | Transmit Data Byte 6 | | | | | | | |
| | 15:8 | Transmit Data Byte 5 | | | | | | | |
| | 7:0 | Transmit Data Byte 4 | | | | | | | |
| Ti | 31:24 | Transmit Data Byte n | | | | | | | |
| | 23:16 | Transmit Data Byte n-1 | | | | | | | |
| | 15:8 | Transmit Data Byte n-2 | | | | | | | |
| | 7:0 | Transmit Data Byte n-3 | | | | | | | |

bit T0.31-30 **Unimplemented:** Read as '0'

bit T0.29 **SID11:** In FD mode the standard ID can be extended to 12 bit using r1

bit T0.28-11 **EID[17:0]:** Extended Identifier

bit T0.10-0 **SID[10:0]:** Standard Identifier

bit T1.31-9 **SEQ[22:0]:** Sequence to keep track of transmitted messages in Transmit Event FIFO

bit T1.8 **ESI:** Error Status Indicator

In CAN to CAN gateway mode (CiCON.ESIGM=1), the transmitted ESI flag is a "logical OR" of T1.ESI and error passive state of the CAN controller;

In normal mode ESI indicates the error status

1 = Transmitting node is error passive

0 = Transmitting node is error active

bit T1.7 **FDF:** FD Frame; distinguishes between CAN and CAN FD formats

bit T1.6 **BRS:** Bit Rate Switch; selects if data bit rate is switched

bit T1.5 **RTR:** Remote Transmission Request; not used in CAN FD

bit T1.4 **IDE:** Identifier Extension Flag; distinguishes between base and extended format

bit T1.3-0 **DLC[3:0]:** Data Length Code

**Note 1:** Data Bytes 0-n: payload size is configured individually in control register (CiFIFOCONm.PLSIZE[2:0]).


© 2019-2020 Microchip Technology Inc.

**TABLE 3-6: RECEIVE MESSAGE OBJECT**

| Word | | Bit 31/23/15/7 | Bit 30/22/14/6 | Bit 29/21/13/5 | Bit 28/20/12/4 | Bit 27/19/11/3 | Bit 26/18/10/2 | Bit 25/17/9/1 | Bit 24/16/8/0 |
|---|---|---|---|---|---|---|---|---|---|
| R0 | 31:24 | — | — | SID11 | EID[17:13] | | | | |
| | 23:16 | EID[12:5] | | | | | | | |
| | 15:8 | EID[4:0] | | | | | SID[10:8] | | |
| | 7:0 | SID[7:0] | | | | | | | |
| R1 | 31:24 | — | — | — | — | — | — | — | — |
| | 23:16 | — | — | — | — | — | — | — | — |
| | 15:8 | FILHIT[4:0] | | | | | — | — | ESI |
| | 7:0 | FDF | BRS | RTR | IDE | DLC[3:0] | | | |
| R2 (2) | 31:24 | RXMSGTS[31:24] | | | | | | | |
| | 23:16 | RXMSGTS[23:16] | | | | | | | |
| | 15:8 | RXMSGTS[15:8] | | | | | | | |
| | 7:0 | RXMSGTS[7:0] | | | | | | | |
| R3 (1) | 31:24 | Receive Data Byte 3 | | | | | | | |
| | 23:16 | Receive Data Byte 2 | | | | | | | |
| | 15:8 | Receive Data Byte 1 | | | | | | | |
| | 7:0 | Receive Data Byte 0 | | | | | | | |
| R4 | 31:24 | Receive Data Byte 7 | | | | | | | |
| | 23:16 | Receive Data Byte 6 | | | | | | | |
| | 15:8 | Receive Data Byte 5 | | | | | | | |
| | 7:0 | Receive Data Byte 4 | | | | | | | |
| Ri | 31:24 | Receive Data Byte n | | | | | | | |
| | 23:16 | Receive Data Byte n-1 | | | | | | | |
| | 15:8 | Receive Data Byte n-2 | | | | | | | |
| | 7:0 | Receive Data Byte n-3 | | | | | | | |

bit R0.31-30 **Unimplemented:** Read as ‘x’

bit R0.29 **SID[11]:** In FD mode the standard ID can be extended to 12 bit using r1

bit R0.28-11 **EID[17:0]:** Extended Identifier

bit R0.10-0 **SID[10:0]:** Standard Identifier

bit R1.31-16 **Unimplemented:** Read as ‘x’

bit R1.15-11 **FILTHIT[4:0]:** Filter Hit, number of filter that matched

bit R1.10-9 **Unimplemented:** Read as ‘x’

bit R1.8 **ESI:** Error Status Indicator

1 = Transmitting node is error passive

0 = Transmitting node is error active

bit R1.7 **FDF:** FD Frame; distinguishes between CAN and CAN FD formats

bit R1.6 **BRS:** Bit Rate Switch; indicates if data bit rate was switched

bit R1.5 **RTR:** Remote Transmission Request; not used in CAN FD

bit R1.4 **IDE:** Identifier Extension Flag; distinguishes between base and extended format

bit R1.3-0 **DLC[3:0]:** Data Length Code

bit R2.31-0 **RXMSGTS[31:0]:** Receive Message Time Stamp

**Note 1:** RXMOBJ: Data Bytes 0-n: payload size is configured individually in the FIFO control register (CiFIFOCONm.PLSIZE[2:0]).

**2:** R2 (RXMSGTS) only exits in objects where CiFIFOCONm.RXTSEN is set.

**TABLE 3-7: TRANSMIT EVENT FIFO OBJECT**

| Word | | Bit 31/23/15/7 | Bit 30/22/14/6 | Bit 29/21/13/5 | Bit 28/20/12/4 | Bit 27/19/11/3 | Bit 26/18/10/2 | Bit 25/17/9/1 | Bit 24/16/8/0 |
|---|---|---|---|---|---|---|---|---|---|
| TE0 | 31:24 | — | — | SID11 | EID[17:13] | | | | |
| | 23:16 | EID[12:5] | | | | | | | |
| | 15:8 | EID[4:0] | | | | | SID[10:8] | | |
| | 7:0 | SID[7:0] | | | | | | | |
| TE1 | 31:24 | SEQ[22:15] | | | | | | | |
| | 23:16 | SEQ[14:7] | | | | | | | |
| | 15:8 | SEQ[6:0] | | | | | | | ESI |
| | 7:0 | FDF | BRS | RTR | IDE | DLC[3:0] | | | |
| TE2 (1) | 31:24 | TXMSGTS[31:24] | | | | | | | |
| | 23:16 | TXMSGTS[23:16] | | | | | | | |
| | 15:8 | TXMSGTS[15:8] | | | | | | | |
| | 7:0 | TXMSGTS[7:0] | | | | | | | |

bit TE0.31-30 **Unimplemented:** Read as '0'

bit TE0.29 **SID11:** In FD mode the standard ID can be extended to 12 bit using r1

bit TE0.28-11 **EID[17:0]:** Extended Identifier

bit TE0.10-0 **SID[10:0]:** Standard Identifier

bit TE1.31-9 **SEQ[22:0]:** Sequence to keep track of transmitted messages

bit TE1.8 **ESI:** Error Status Indicator

1 = Transmitting node is error passive

0 = Transmitting node is error active

bit TE1.7 **FDF:** FD Frame; distinguishes between CAN and CAN FD formats

bit TE1.6 **BRS:** Bit Rate Switch; selects if data bit rate is switched

bit TE1.5 **RTR:** Remote Transmission Request; not used in CAN FD

bit TE1.4 **IDE:** Identifier Extension Flag; distinguishes between base and extended format

bit TE1.3-0 **DLC[3:0]:** Data Length Code

bit TE2.31-0 **TXMSGTS[31:0]:** Transmit Message Time Stamp (1)

**Note 1:** TE2 (TXMSGTS) only exits in objects where CiTEFCON.TEFTSEN is set.


© 2019-2020 Microchip Technology Inc.

## 4.0 SPI INTERFACE

The MCP2518FD device is designed to interface directly with a Serial Peripheral Interface port available on most microcontrollers. The SPI in the microcontroller must be configured in mode 0,0 or 1,1 in 8-bit operating mode.

SFR and Message Memory (RAM) are accessed using SPI instructions. Figure 4-1 illustrates the generic format of the SPI instructions (SPI mode 0,0). Each instruction starts with driving nCS low (falling edge on nCS). The 4-bit command and the 12-bit address are shifted into SDI on the rising edge of SCK. During a write instruction, data bits are shifted into SDI on the rising edge of SCK. During a read instruction, data bits are shifted out of SDO on the falling edge of SCK. One or more data bytes are transfered with one instruction. Data bits are updated on the falling edge of SCK and must be valid on the rising edge of SCK. Each instruction ends with driving nCS high (rising edge on nCS).

Refer to Figure 7-1 for detailed input and output timing for both mode 0,0 and mode 1,1.

Table 4-1 lists the SPI instructions and their format.

**Note 1:** The frequency of SCK has to be less than or equal to 0.85 * half the frequency of SYSCLK. This ensures that the synchronization between SCK and SYSCLK works correctly.

**2:** In order to minimize the Sleep current, the SDO pin of the MCP2518FD device must not be left floating while the device is in Sleep mode. This can be achieved by enabling a pull-up or pull-down resistor inside the MCU on the pin that is connected to the SDO pin, while the MCP2518FD device is in Sleep mode.

**FIGURE 4-1: SPI INSTRUCTION FORMAT**

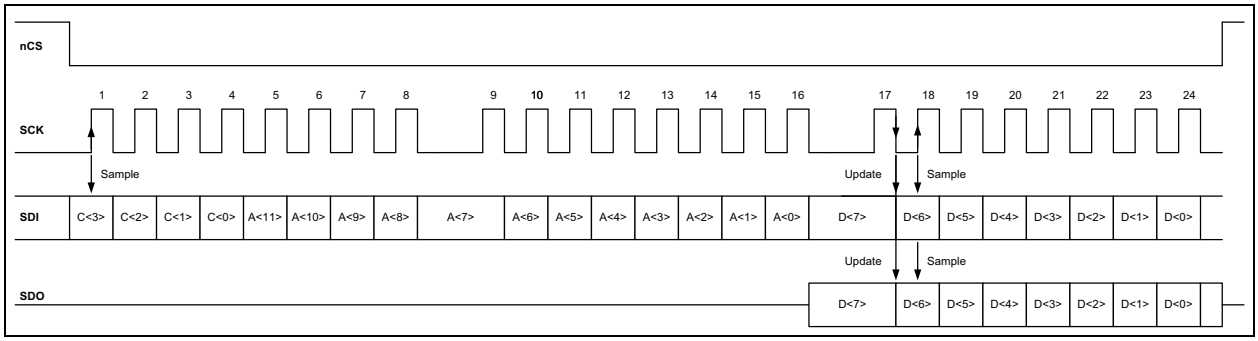



**TABLE 4-1: SPI INSTRUCTIONS**

| Name | Format | Description |
|---|---|---|
| RESET | C = 0b0000; A = 0x000 | Resets internal registers to default state; selects Configuration mode. |
| READ | C = 0b0011; A; D = SDO | Read SFR/RAM from address A. |
| WRITE | C = 0b0010; A; D = SDI | Write SFR/RAM to address A. |
| READ_CRC | C = 0b1011; A; N;<br>D = SDO; CRC = SDO | Read SFR/RAM from address A. N data bytes. Two bytes CRC.<br>CRC is calculated on C, A, N and D. |
| WRITE_CRC | C = 0b1010; A; N;<br>D = SDI; CRC = SDI | Write SFR/RAM to address A. N data bytes. Two bytes CRC.<br>CRC is calculated on C, A, N and D. |
| WRITE_SAFE | C = 0b1100; A;<br>D = SDI; CRC = SDI | Write SFR/RAM to address A. Check CRC before write. CRC is calculated on C, A and D. |

**Legend:** C = Command (4 bit), A = Address (12 bit), D = Data (1 to n bytes), N = Number of Bytes (1 byte), CRC (2 bytes)

## 4.1 SFR Access

The SFR access is byte-oriented. Any number of data bytes can be read or written with one instruction. The address is incremented by one automatically after every data byte. The address rolls over from 0xFFF to 0x000.

The following SPI instructions only show the different fields and their values. Every instruction follows the generic format illustrated in Figure 4-1.

### 4.1.1 RESET

Figure 4-2 illustrates the RESET instruction. The instruction starts with nCS going low. The Command (C[3:0] = 0b0000) is followed by the Address (A[11:0] = 0x000). The instruction ends when nCS goes high.

The RESET instruction should only be issued after the device enters Configuration mode. All SFR and State Machines are reset just like during a Power-on Reset (POR), and the device transitions immediately to Configuration mode.

The Message Memory is not changed.

The actual reset happens at the end of the instruction when nCS goes high.

### 4.1.2 SFR READ – READ

Figure 4-3 illustrates the READ instruction, while accessing SFR. The instruction starts with nCS going low. The Command (C[3:0] = 0b0011), is followed by the Address (A[11:0]). Afterwards, the data byte from address A (DB[A]) is shifted out, followed by data byte from address A+1 (DB[A+1]). Any number of data bytes can be read. The instruction ends when nCS goes high.

### 4.1.3 SFR WRITE – WRITE

Figure 4-4 illustrates the WRITE instruction, while accessing SFR. The instruction starts with nCS going low. The Command (C[3:0] = 0b0010), is followed by the Address (A[11:0]). Afterwards, the data byte is shifted into address A (DB[A]), next into address A+1 (DB[A+1]). Any number of data bytes can be written. The instruction ends when nCS goes high.

| **Note:** | The bit fields in the IOCON register must be written using the single data byte SFR WRITE instructions. |
|---|---|

Data bytes are written to the register with the falling edge on SCK following the 8th data bit.

**FIGURE 4-2: RESET INSTRUCTION**

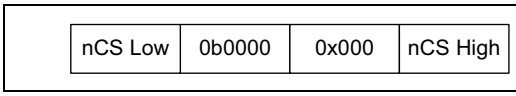



**FIGURE 4-3: SFR READ INSTRUCTION**

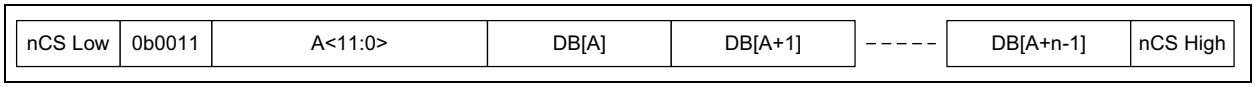
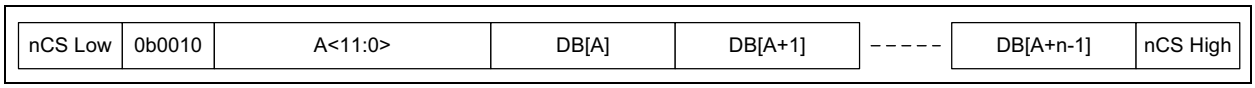



**FIGURE 4-4: SFR WRITE INSTRUCTION**





© 2019-2020 Microchip Technology Inc.

## 4.2 Message Memory Access

The Message Memory (RAM) access is word-oriented (4 bytes at a time). Any multiple of 4 data bytes can be read or written with one instruction. The address is incremented by one automatically after every data byte. The address rolls over from `0xBFF` to `0x400`.

Writes and Reads must be word-aligned. The lower two bits of the address are always assumed to be 0. It is not possible to do unaligned reads/writes.

The following SPI instructions only show the different fields and their values. Every instruction follows the generic format illustrated in Figure 4-1.

### 4.2.1 MESSAGE MEMORY READ – READ

Figure 4-5 illustrates the READ instruction, while accessing RAM. The instruction starts with nCS going low. The Command (C[3:0] = 0b0011), is followed by the Address (A[11:0]). Afterwards, the data byte from address A (DB[A]) is shifted out, followed by data byte from address A+1 (DB[A+1]). The instruction ends when nCS goes high.

Read commands from RAM must always read a multiple of 4 data bytes. A word is internally read from RAM after the address field, and after every fourth data byte read on the SPI. In case nCS goes high before a multiple of 4 data bytes is read on SDO, the incomplete read should be discarded by the microcontroller.

### 4.2.2 MESSAGE MEMORY WRITE – WRITE

Figure 4-6 illustrates the WRITE instruction, while accessing RAM. The instruction starts with nCS going low. The Command (C[3:0] = 0b0010), is followed by the Address (A[11:0]). Afterwards, the data byte is shifted into address A (DB[A]), next into address A+1 (DB[A+1]). The instruction ends when nCS goes high.

Write commands must always write a multiple of 4 data bytes. After every fourth data byte, with the falling edge on SCK, the RAM Word gets written. In case nCS goes high before a multiple of 4 data bytes is received on SDI, the data of the incomplete Word will not be written to RAM.

**FIGURE 4-5: MESSAGE MEMORY READ INSTRUCTION**

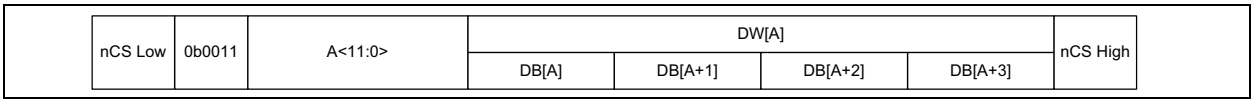
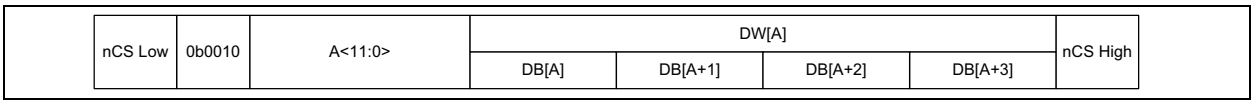



**FIGURE 4-6: MESSAGE MEMORY WRITE INSTRUCTION**



## 4.3 SPI Commands with CRC

In order to detect or avoid bit errors during SPI communication, SPI commands with CRC are available.

### 4.3.1 CRC CALCULATION

The CRC is calculated in parallel with the SPI shift register (see Figure 4-7).

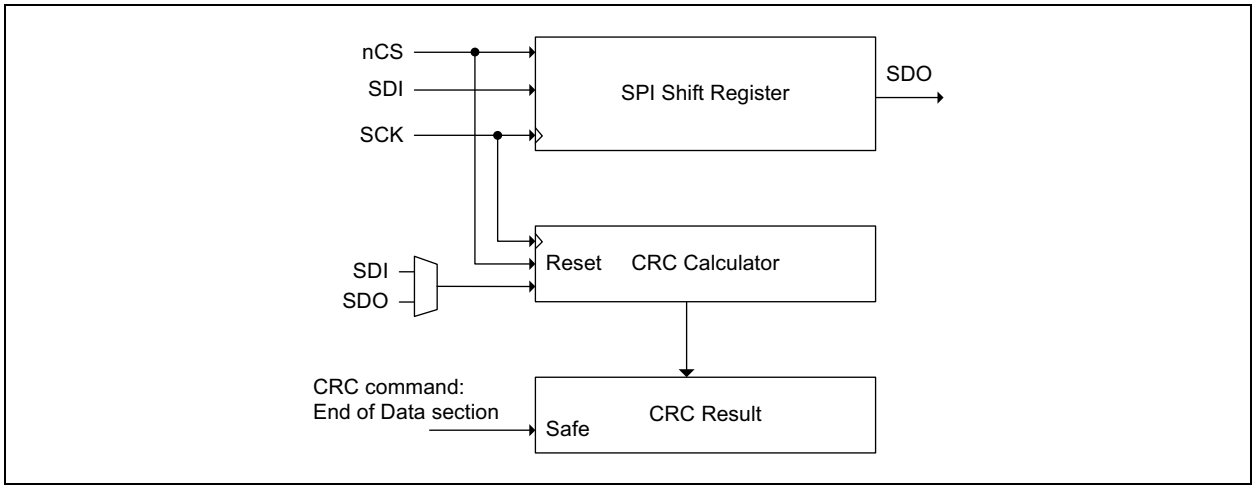
When nCS is asserted, the CRC calculator is reset to `0xFFFF`.

The result of the CRC calculation is available after the Data section of a CRC command. The result of the CRC calculation is written to the CRC register in case a CRC mismatch is detected. In case of a CRC mismatch, CRC.CRCERRIF is set.

The MCP2518FD device uses the following generator polynomial: CRC-16/USB (0x8005). CRC-16 detects all single and double-bit errors, all errors with an odd number of bits, all burst errors of length 16 or less, and most errors for longer bursts. This allows an excellent detection of SPI communication errors that can happen in the system, and heavily reduces the risk of miscommunication, even under noisy environments.

The maximum number of data bits is used while reading and writing TX or RX Message Objects. A RX Message Object with 64 Bytes of data + 12 Bytes ID and Time Stamp contains 76 Bytes or 608 bits. In comparison, USB data packets contain up to 1024 bits. CRC-16 has a Hamming Distance of 4 up to 1024 bits.

**FIGURE 4-7: CRC CALCULATION**





© 2019-2020 Microchip Technology Inc.

### 4.3.2 SFR READ WITH CRC – READ_CRC

Figure 4-8 illustrates the READ_CRC instruction, while accessing SFR. The instruction starts with nCS going low. The Command (C[3:0] = 0b1011), is followed by the Address (A[11:0]), and the number of data bytes (N[7:0]). Afterwards, the data byte from address A (DB[A]) is shifted out, followed by the data byte from address A+1 (DB[A+1]). Any number of data bytes can be read. Next the CRC is shifted out (CRC[15:0]). The instruction ends when nCS goes high.

The CRC is provided to the microcontroller. The microcontroller checks the CRC. No interrupt is generated on CRC mismatch during a READ_CRC command inside the MCP2518FD device.

If nCS goes high before the last byte of the CRC is shifted out, a CRC Form Error interrupt is generated: CRC.FERRIF.

### 4.3.3 SFR WRITE WITH CRC – WRITE_CRC

Figure 4-9 illustrates the WRITE_CRC instruction, while accessing SFR. The instruction starts with nCS going low. The Command (C[3:0] = 0b1010), is followed by the Address (A[11:0]), and the number of data bytes (N[7:0]). Afterwards, the data byte is shifted into address A (DB[A]), next into address A+1 (DB[A+1]). Any number of data bytes can be written. Next the CRC is shifted in (CRC[15:0]). The instruction ends when nCS goes high.

The SFR is written to the register after the data byte was shifted in on SDI, with the falling edge on SCK. Data bytes are written to the register before the CRC is checked.

The CRC is checked at the end of the write access. In case of a CRC mismatch, a CRC Error interrupt is generated: CRC.CRCERRIF.

If nCS goes high before the last byte of the CRC is shifted in, a CRC Form Error interrupt is generated: CRC.FERRIF.

**FIGURE 4-8: SFR READ WITH CRC INSTRUCTION**

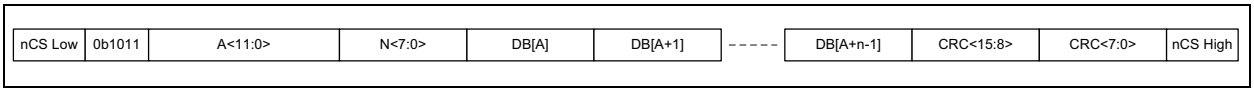
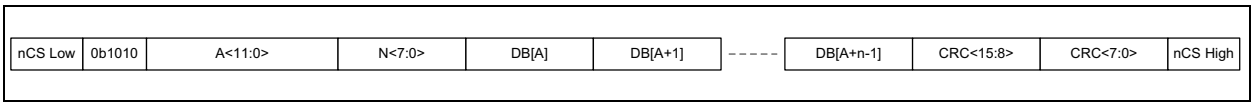



**FIGURE 4-9: SFR WRITE WITH CRC INSTRUCTION**




### 4.3.4 SFR WRITE SAFE WITH CRC – WRITE_SAFE

This instruction ensures that only correct data is written to the SFR.

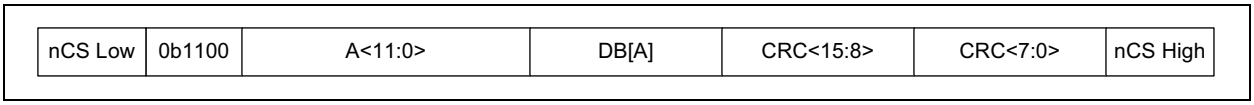
Figure 4-10 illustrates the WRITE_SAFE instruction, while accessing SFR. The instruction starts with nCS going low. The Command (C[3:0] = 0b1100), is followed by the Address (A[11:0]). Afterwards, one data byte is shifted into address A (DB[A]). Next the CRC (CRC[15:0]) is shifted in. The instruction ends when nCS goes high.

The data byte is only written to the SFR after the CRC is checked and if it matches.

If the CRC mismatches, the data byte is not written to the SFR and a CRC Error interrupt is generated: CRC.CRCERRIF.

If nCS goes high before the last byte of the CRC is shifted in, a CRC Form Error interrupt is generated: CRC.FERRIF.

**FIGURE 4-10: SFR WRITE SAFE WITH CRC INSTRUCTION**



### 4.3.5 MESSAGE MEMORY READ WITH CRC – READ_CRC

Figure 4-11 illustrates the READ_CRC instruction, while accessing RAM. The instruction starts with nCS going low. The Command (C[3:0] = 0b1011), is followed by the Address (A[11:0]), and the number of data Words (N[7:0]). Afterwards, the data byte from address A (DB[A]) is shifted out, followed by data byte from address A+1 (DB[A+1]). Next the CRC (CRC[15:0]) is shifted out. The instruction ends when nCS goes high.

Writes and Reads must be word-aligned. The lower two bits of the address are always assumed to be 0. It is not possible to do unaligned reads/writes.

Read commands should always read a multiple of 4 data bytes. A word is internally read from RAM after the “N” field, and after every fourth data byte read on the SPI. In case nCS goes high before a multiple of 4 data bytes are read on SDO, the incomplete read should be discarded by the microcontroller.

The CRC is provided to the microcontroller. The microcontroller checks the CRC. No interrupt is generated on CRC mismatch during a READ_CRC command inside the MCP2518FD device.

If nCS goes high before the last byte of the CRC is shifted out, a CRC Form Error interrupt is generated: CRC.FERRIF.

### 4.3.6 MESSAGE MEMORY WRITE WITH CRC – WRITE_CRC

Figure 4-12 illustrates the WRITE instruction accessing the RAM. The instruction starts with nCS going low. The Command (C[3:0] = 0b1010), is followed by the Address (A[11:0]), and the number of data Words (N[7:0]). Afterwards, the data byte is shifted into address A (DB[A]), next into address A+1 (DB[A+1]). Next the CRC (CRC[15:0]) is shifted in. The instruction ends when nCS goes high.

Write commands must always write a multiple of 4 data bytes. After every fourth data byte, with the falling edge on SCK, the RAM gets written. In case nCS goes high before a multiple of 4 data bytes is received on SDI, the data of the incomplete Word will not be written to RAM.

The CRC is checked at the end of the write access. In case of a CRC mismatch, a CRC interrupt is generated: CRC.CRCERRIF.

If nCS goes high before the last byte of the CRC is shifted in, a CRC interrupt is generated: CRC.FERRIF.

**FIGURE 4-11: MESSAGE MEMORY READ WITH CRC INSTRUCTION**

| nCS Low | 0b1011 | A<11:0> | N<7:0> | DW[A] | | | | CRC<15:8> | CRC<7:0> | nCS High |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | DB[A] | DB[A+1] | DB[A+2] | DB[A+3] | | | |

**FIGURE 4-12: MESSAGE MEMORY WRITE WITH CRC INSTRUCTION**

| nCS Low | 0b1010 | A<11:0> | N<7:0> | DW[A] | | | | CRC<15:8> | CRC<7:0> | nCS High |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | DB[A] | DB[A+1] | DB[A+2] | DB[A+3] | | | |

### 4.3.7 MESSAGE MEMORY WRITE SAFE WITH CRC – WRITE_SAFE

This instruction ensures that only correct data is written to RAM.

Figure 4-10 illustrates the WRITE_SAFE instruction, while accessing RAM. The instruction starts with nCS going low. The Command (C[3:0] = 0b1100), is followed by the Address (A[11:0]). Afterwards, the data byte is shifted into address A (DB[A]), next into address A+1 (DB[A+1]), A+2 (DB[A+2]), and A+3 (DB[A+3]). Next the CRC (CRC[15:0]) is shifted in. The instruction ends when nCS goes high.

The data word is only written to RAM after the CRC is checked and if it matches.

If the CRC mismatches, the data word is not written to RAM and a CRC Error interrupt is generated: CRC.CRCERRIF.

If nCS goes high before the last byte of the CRC is shifted in, a CRC interrupt is generated: CRC.FERRIF.

**FIGURE 4-13: MESSAGE MEMORY WRITE SAFE WITH CRC INSTRUCTION**

| nCS Low | 0b1100 | A<11:0> | DW[A] | | | | CRC<15:8> | CRC<7:0> | nCS High |
|---|---|---|---|---|---|---|---|---|---|
| | | | DB[A] | DB[A+1] | DB[A+2] | DB[A+3] | | | |


© 2019-2020 Microchip Technology Inc.

## 5.0 OSCILLATOR

Figure 5-1 shows the block diagram of the oscillator in the MCP2518FD device. The oscillator system generates the SYSCLK, which is used in the CAN FD Controller module and for RAM accesses. It is recommended by the CAN FD community to use either a 40 or 20 MHz SYSCLK.

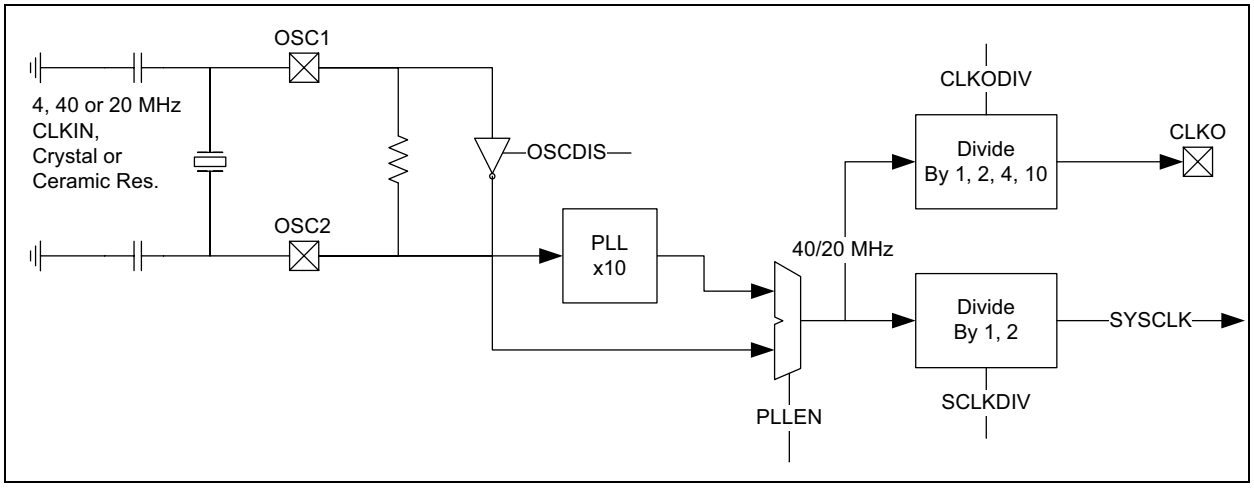
The time reference for clock generation can be an external 40, 20 or 4 MHz crystal, ceramic resonator or external clock.

The OSC register controls the oscillator. The PLL can be enabled to multiply the 4 MHz clock by 10.

The internal 40/20 MHz can be divided by two.

The internally generated clock can be divided and provided on the CLKO pin.

**FIGURE 5-1: MCP2518FD OSCILLATOR BLOCK DIAGRAM**



## 6.0 I/O CONFIGURATION

The IOCON register is used to configure the I/O pins:

- CLKO/SOF: select Clock Output or Start of Frame.
- TXCANOD: TXCAN can be configured as Push-Pull or as Open Drain output. Open Drain outputs allows the user to connect multiple controllers together to build a CAN network without using a transceiver.
- $\overline{\text{INT0}}$ and $\overline{\text{INT1}}$ can be configured as GPIO with similar registers as in the PIC microcontrollers or as Transmit and Receive interrupts.
- $\overline{\text{INT0}}$/GPIO0/XSTBY can also be used to automatically control the standby pin of the transceiver.
- INTOD: The interrupt pins can be configured as open-drain or push/pull outputs.

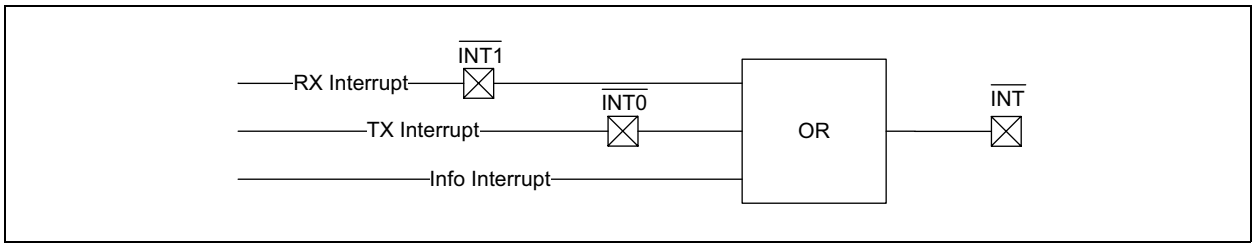
### 6.0.1 INTERRUPT PINS

The MCP2518FD device contains three different interrupt pins, see Figure 6-1:

- $\overline{\text{INT}}$ is asserted on any interrupt in the CiINT register (xIF & xIE), including the RX and TX interrupts.
- $\overline{\text{INT1}}$/GPIO1 can be configured as GPIO or RX interrupt pin (CiINT.RXIF & RXIE).
- $\overline{\text{INT0}}$/GPIO0 can be configured as GPIO or TX interrupt pin (CiINT.TXIF & TXIE).

All interrupt pins are active low.

**FIGURE 6-1: INTERRUPT PINS**





© 2019-2020 Microchip Technology Inc.

# 7.0 ELECTRICAL SPECIFICATIONS

## 7.1 Absolute Maximum Ratings†

$V_{DD}$ ..... -0.3V to 6.0V

DC Voltage at all I/O w.r.t GND ..... -0.3V to VDD + 0.3V

Virtual Junction Temperature, $T_{VJ}$ (IEC60747-1) ..... -40°C to +165°C

Soldering temperature of leads (10 seconds) ..... +300°C

ESD protection on all pins (IEC 801; Human Body Model) ..... ±4 kV

ESD protection on all pins (IEC 801; Machine Model) ..... ±400V

ESD protection on all pins (IEC 801; Charge Device Model) ..... ±750V

**† NOTICE:** Stresses above those listed under “Maximum ratings” may cause permanent damage to the device. This is a stress rating only and functional operation of the device at those or any other conditions above those indicated in the operational listings of this specification is not implied. Exposure to maximum rating conditions for extended periods may affect device reliability.

**TABLE 7-1: DC CHARACTERISTICS**

| DC Specifications | | Electrical Characteristics: Extended (E): $T_{AMB}$ = -40°C to +125°C; High (H): $T_{AMB}$ = -40°C to +150°C; $V_{DD}$ = 2.7V to 5.5V | | | | |
|---|---|---|---|---|---|---|
| **Sym.** | **Characteristic** | **Min.** | **Typ.** | **Max.** | **Units** | **Conditions/Comments** |
| **VDD Pin** | | | | | | |
| $V_{DD}$ | Voltage Range | 2.7 | — | 5.5 | V | RAM data retention guaranteed |
| $V_{PORH}$ | Power-on Reset Voltage | — | — | 2.65 | V | Highest voltage on $V_{DD}$ before device releases POR |
| $V_{PORL}$ | Power-on Reset Voltage | 2.2 | — | — | V | Lowest voltage on $V_{DD}$ before device asserts POR |
| $S_{VDD}$ | VDD Rise Rate to ensure POR | 0.05 | — | — | V/ms | **Note 1** |
| $I_{DD}$ | Supply Current | — | 15 | 20 | mA | 40 MHz SYSCLK, 20 MHz SPI activity |
| $I_{DDS}$ | Sleep Current | — | 15 | 60 | µA | Clock is stopped $T_{AMB}$ ≤ +85°C (**Note 1**) |
| | | — | — | 600 | — | Clock is stopped $T_{AMB}$ ≤ +150°C |
| $I_{DDLPM}$ | LPM Current | — | 4 | 10 | µA | Digital logic powered down |
| **Digital Input Pins** | | | | | | |
| $V_{IH}$ | High-Level Input Voltage | 0.7 $V_{DD}$ | — | $V_{DD}$ + 0.3 | V | |
| $V_{IL}$ | Low-Level Input Voltage | -0.3 | — | 0.3 $V_{DD}$ | V | |
| $V_{OSCPP}$ | OSC1 detection Voltage | 0.5 | — | — | V | Minimum peak-to-peak voltage on OSC1 pin (**Note 1**) |
| $I_{LI}$ | Input Leakage Current | | | | | |
| | OSC1 | -5 | — | +5 | µA | |
| | All other | -1 | — | +1 | µA | |
| **Digital Output Pins** | | | | | | |
| $V_{OH}$ | High-Level Output Voltage | $V_{DD}$ – 0.7 | — | — | V | $I_{OH}$ = -2 mA, $V_{DD}$ = 2.7V |
| $V_{OL}$ | Low-Level Output Voltage | | | | | |
| | TXCAN | — | — | 0.6 | V | $I_{OL}$ = 8 mA, $V_{DD}$ = 2.7V |
| | All other | — | — | 0.6 | V | $I_{OL}$ = 2 mA, $V_{DD}$ = 2.7V |

**Note 1:** Characterized; not 100% tested.


© 2019-2020 Microchip Technology Inc.

**TABLE 7-2: CLKOUT AND SOF AC CHARACTERISTICS**

| AC Specifications | | Electrical Characteristics: Extended (E): $T_{AMB}$ = -40°C to +125°C; High (H): $T_{AMB}$ = -40°C to +150°C; $V_{DD}$ = 2.7V to 5.5V | | | | |
|---|---|---|---|---|---|---|
| **Sym.** | **Characteristic** | **Min.** | **Typ.** | **Max.** | **Units** | **Conditions/Comments** |
| $T_{CLKOH}$ | CLKO Output High | 8 | — | — | ns | at 40 MHz (**Note 1**) |
| $T_{CLKOL}$ | CLKO Output Low | 8 | — | — | ns | **Note 1** |
| $T_{CLKOR}$ | CLKO Output Rise | — | — | 5 | ns | **Note 1** |
| $T_{CLKOF}$ | CLKO Output Fall | — | — | 5 | ns | **Note 1** |
| $T_{SOFH}$ | SOF Output High | — | 31 $T_{OSC}$ | — | ns | **Note 2** |
| $T_{SOFPD}$ | SOF Propagation Delay: RXCAN falling edge to SOF rising edge | — | 1 $T_{OSC}$ | — | ns | **Note 2** |

**Note 1:** Characterized; not 100% tested.

**2:** Design guidance only.

**TABLE 7-3: CRYSTAL OSCILLATOR AC CHARACTERISTICS**

| AC Specifications | | Electrical Characteristics: Extended (E): $T_{AMB}$ = -40°C to +125°C; High (H): $T_{AMB}$ = -40°C to +150°C; $V_{DD}$ = 2.7V to 5.5V | | | | |
|---|---|---|---|---|---|---|
| **Sym.** | **Characteristic** | **Min.** | **Typ.** | **Max.** | **Units** | **Conditions/Comments** |
| $F_{OSC1,CLKI}$ | OSC1 Input Frequency | 2 | 40 | 40 | MHz | External digital clock |
| $F_{OSC1,4M}$ | OSC1 Input Frequency | 4 - 0.5% | 4 | 4 + 0.5% | MHz | 4 MHz crystal/resonator (**Note 1**) |
| $F_{DRIFT}$ | SYSCLK frequency drift | — | — | 10 | ppm | Additional frequency drift of SYSCLK due to internal PLL at 4 MHz (**Note 1**) |
| $F_{OSC1,20M}$ | OSC1 Input Frequency | 20 - 0.5% | 20 | 20 + 0.5% | MHz | 20 MHz crystal/resonator (**Note 1**) |
| $F_{OSC1,40M}$ | OSC1 Input Frequency | 40 - 0.5% | 40 | 40 + 0.5% | MHz | 40 MHz crystal/resonator (**Note 1**) |
| $T_{OSC1}$ | $T_{OSC1}=1/F_{OSC1,x}$ | 25 | — | — | ns | |
| $T_{OSC1H}$ | OSC1 Input High | 0.45 * $T_{OSC}$ | — | 0.55 * $T_{OSC}$ | ns | **Note 1** |
| $T_{OSC1L}$ | OSC1 Input Low | 0.45 * $T_{OSC}$ | — | 0.55 * $T_{OSC}$ | ns | **Note 1** |
| $T_{OSC1R}$ | OSC1 Input Rise | — | — | 20 | ns | **Note 2** |
| $T_{OSC1F}$ | OSC1 Input Fall | — | — | 20 | ns | **Note 2** |
| $DC_{OSC1}$ | Duty Cycle on OSC1 | 45 | 50 | 55 | % | External clock duty cycle requirement (**Note 1**) |
| $T_{OSCSTAB}$ | Oscillator stabilization period | — | — | 3 | ms | From POR to final frequency (**Note 1**) |
| $T_{OSCSLEEP}$ | Oscillator stabilization from Sleep | — | — | 3 | ms | From Sleep to final frequency (**Note 1**) |
| $G_{M,4M}$ | Transconductance | 1470 | — | 2210 | µA/V | 4 MHz crystal (**Note 2**) |
| $G_{M,40M}$ | Transconductance | 2040 | — | 3060 | µA/V | 40 MHz crystal (**Note 2**) |

**Note 1:** Characterized; not 100% tested.

**2:** Design guidance only.

© 2019-2020 Microchip Technology Inc.

**TABLE 7-4: CAN BIT RATE**

| AC Specifications | | Electrical Characteristics: Extended (E): $T_{AMB}$ = -40°C to +125°C; High (H): $T_{AMB}$ = -40°C to +150°C; $V_{DD}$ = 2.7V to 5.5V | | | | |
|---|---|---|---|---|---|---|
| **Sym** | **Characteristic** | **Min** | **Typ** | **Max** | **Units** | **Conditions/Comments** |
| $BR_{NOM}$ | Nominal Bit Rate | 0.125 | 0.5 | 1 | Mbps | |
| $BR_{DATA}$ | Data Bit Rate | 0.5 | 2 | 8 | Mbps | $BR_{DATA} \geq BR_{NOM}$ |

**Note 1:** Tested bit rates. Device allows the configuration of more bit rates, including slower bit rates than the minimum stated.

**TABLE 7-5: CAN RX FILTER AC CHARACTERISTICS**

| AC Specifications | | Electrical Characteristics: Extended (E): $T_{AMB}$ = -40°C to +125°C; High (H): $T_{AMB}$ = -40°C to +150°C; $V_{DD}$ = 2.7V to 5.5V | | | | |
|---|---|---|---|---|---|---|
| **Sym.** | **Characteristic** | **Min.** | **Typ.** | **Max.** | **Units** | **Conditions/Comments** |
| $T_{PROP}$ | Filter propagation delay | — | 1 | — | ns | **Note 2** |
| $T_{FILTER}$ | Filter time | 50<br>80<br>130<br>225 | — | 100<br>140<br>220<br>390 | ns | $T00_{FILTER}$<br>$T01_{FILTER}$<br>$T10_{FILTER}$<br>$T11_{FILTER}$<br>**Note 3** |
| $T_{REVO-CERY}$ | Minimum high time on input for output to go high again | 5 | — | — | ns | **Note 2** |

**Note 1:** Characterized; not 100% tested.

**2:** Design guidance only.

**3:** Pulses on RXCAN shorter than the minimum TFILTER time will be ignored; pulses longer than the maximum TFILTER time will wake-up the device.


© 2019-2020 Microchip Technology Inc.

**TABLE 7-6: SPI AC CHARACTERISTICS**

| AC Specifications | | | Electrical Characteristics: Extended (E): $T_{AMB}$ = -40°C to +125°C; High (H): $T_{AMB}$ = -40°C to +150°C, $V_{DD}$ = 2.7V to 5.5V | | | | |
|---|---|---|---|---|---|---|---|
| **Param.** | **Sym.** | **Characteristic** | **Min.** | **Typ.** | **Max.** | **Units** | **Conditions** |
| | $F_{SCK}$ | SCK Input Frequency | — | — | 17 | MHz | **Note 3** |
| | $T_{SCK}$ | SCK Period, TSCK=1/FSCK | 59 | — | — | ns | **Note 3** |
| 1 | $T_{SCKH}$ | SCK High Time | 20 | — | — | ns | |
| 2 | $T_{SCKL}$ | SCK Low Time | 20 | — | — | ns | |
| 3 | $T_{SCKR}$ | SCK Rise Time | — | — | 100 | ns | **Note 2** |
| 4 | $T_{SCKF}$ | SCK Fall Time | — | — | 100 | ns | **Note 2** |
| 5 | $T_{CS2SCK}$ | nCS ↓ to SCK ↑ | $T_{SCK}$/2 | — | — | ns | |
| 6 | $T_{SCK2CS}$ | SCK ↑ to nCS ↑ | $T_{SCK}$ | — | — | ns | |
| 7 | $T_{SDI2SCK}$ | SDI Setup: SDI ↕ to SCK ↑ | 5 | — | — | ns | |
| 8 | $T_{SCK2SDI}$ | SDI Hold: SCK ↑ to SDI ↕ | 5 | — | — | ns | |
| 9 | $T_{SCK2SDO}$ | SDO Valid: SCK ↓ to SDO ↕ | — | — | 20 | ns | $C_{LOAD}$ = 50 pF |
| 10 | $T_{CS2SDOZ}$ | SDO High Z: nCS ↑ to SDO Z | — | — | 2 $T_{SCK}$ | ns | $C_{LOAD}$ = 50 pF |
| 11 | $T_{CSD}$ | nCS ↑ to nCS ↓ | $T_{SCK}$ | — | — | ns | **Note 2** |

**Note 1:** Characterized; not 100% tested.

**2:** Design guidance only.

**3:** FSCK must be less than or equal to 0.85*(FSYSCLK/2).

**FIGURE 7-1: SPI I/O TIMING**

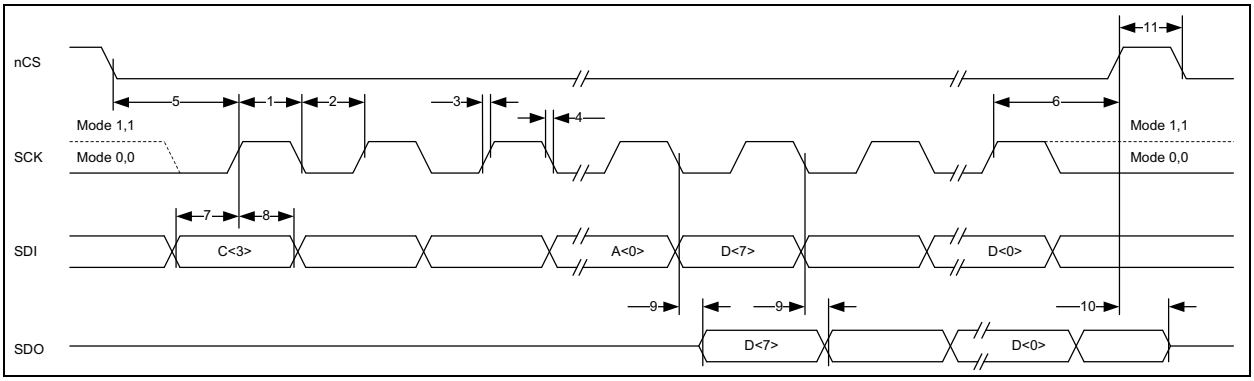



**TABLE 7-7: TEMPERATURE SPECIFICATIONS**

| Parameters | Sym. | Min. | Typ. | Max. | Units | Conditions |
|---|---|---|---|---|---|---|
| **Temperature Ranges** | | | | | | |
| Operating Temperature Range | $T_A$ | -40 | — | +150 | °C | |
| Storage Temperature Range | $T_A$ | -55 | — | +150 | °C | |
| **Thermal Package Resistance** | | | | | | |
| Thermal Resistance for SOIC-14 | $\theta_{JA}$ | — | +110 | — | °C/W | |
| Thermal Resistance for DFN-14 | $\theta_{JA}$ | — | +45 | — | °C/W | |

**NOTES:**


© 2019-2020 Microchip Technology Inc.

## 8.0 TYPICAL PERFORMANCE CURVES

**Note:** The graphs and tables provided following this note are a statistical summary based on a limited number of samples and are provided for informational purposes only. The performance characteristics listed herein are not tested or guaranteed. In some graphs or tables, the data presented may be outside the specified operating range (for example, outside specified power supply range) and therefore outside the warranted range.

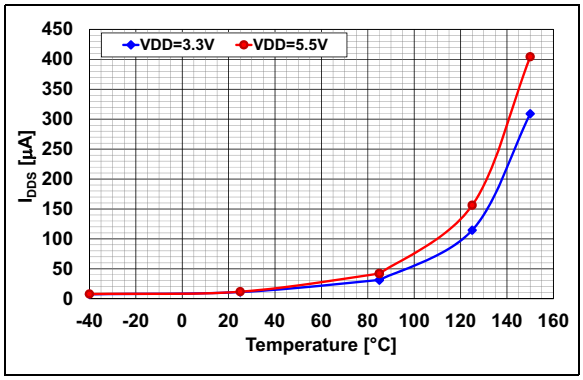



***FIGURE 8-1:*** *Average IDDS vs. Temperature.*

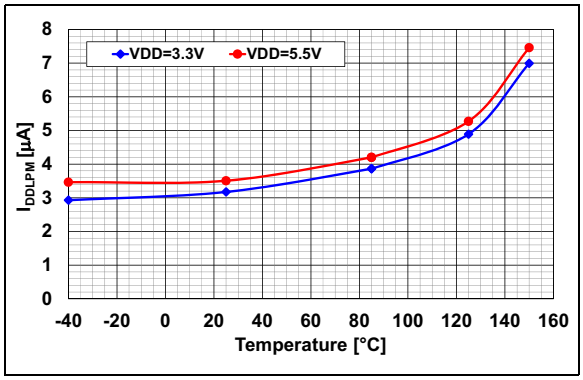



***FIGURE 8-2:*** *Average IDDLPM vs. Temperature.*

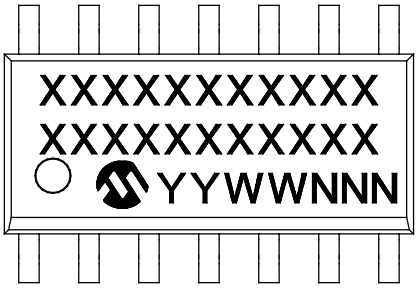
**NOTES:**


© 2019-2020 Microchip Technology Inc.

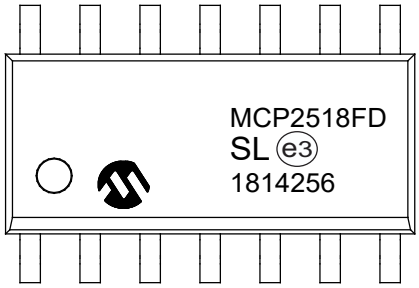
## 9.0 PACKAGING INFORMATION

### 9.1 Package Marking Information

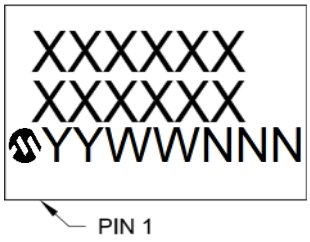
14-Lead SOIC



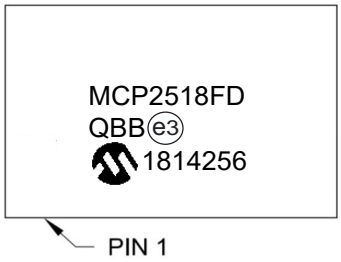

Example:




14-Lead VDFN




Example:

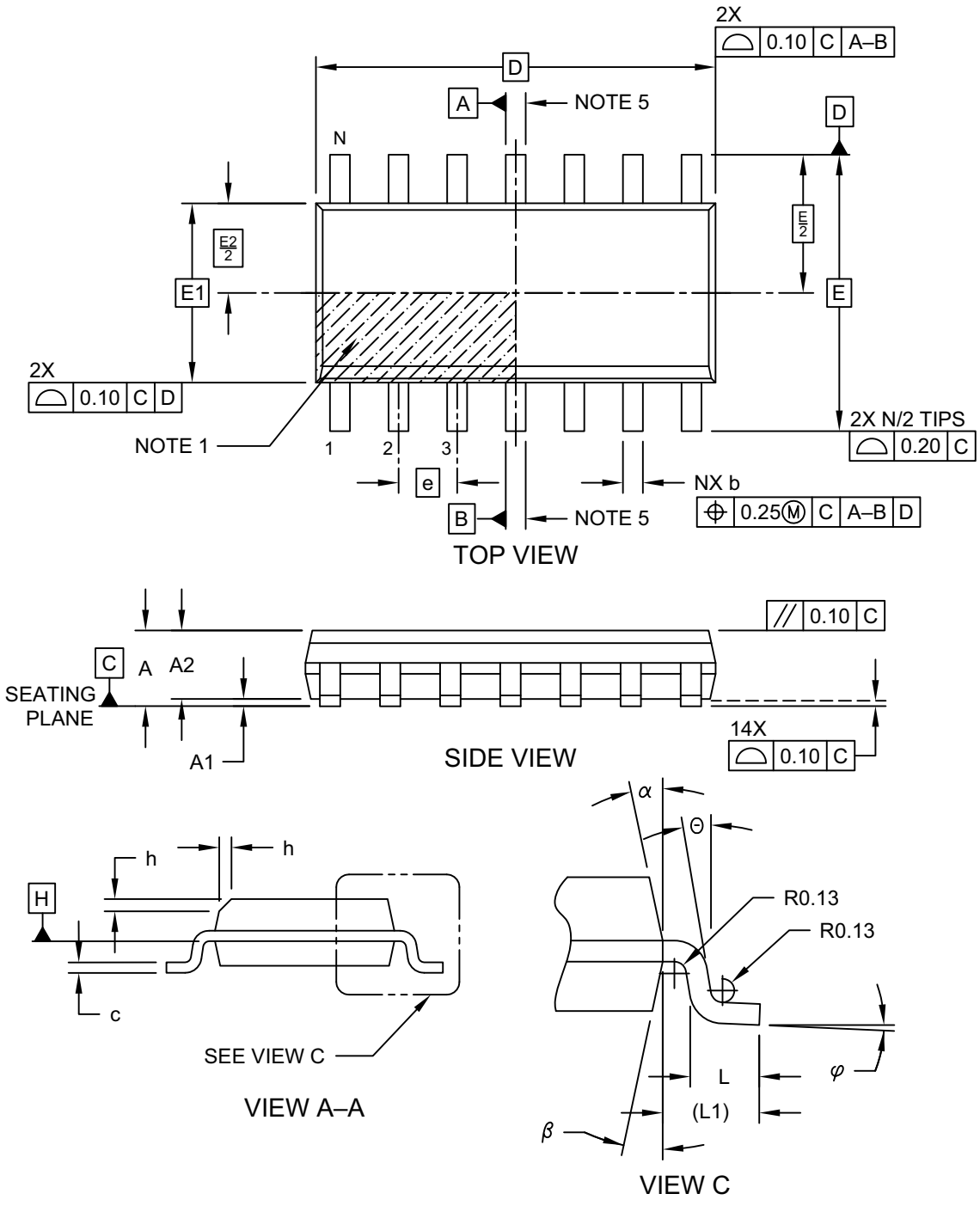



| | | |
|---|---|---|
| **Legend:** | XX...X | Customer-specific information |
| | Y | Year code (last digit of calendar year) |
| | YY | Year code (last 2 digits of calendar year) |
| | WW | Week code (week of January 1 is week ‘01’) |
| | NNN | Alphanumeric traceability code |
| | (e3) | Pb-free JEDEC® designator for Matte Tin (Sn) |
| | * | This package is Pb-free. The Pb-free JEDEC designator ((e3)) can be found on the outer packaging for this package. |

**Note**: In the event the full Microchip part number cannot be marked on one line, it will be carried over to the next line, thus limiting the number of available characters for customer-specific information.

## 14-Lead Plastic Small Outline (SL) - Narrow, 3.90 mm Body [SOIC]

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging

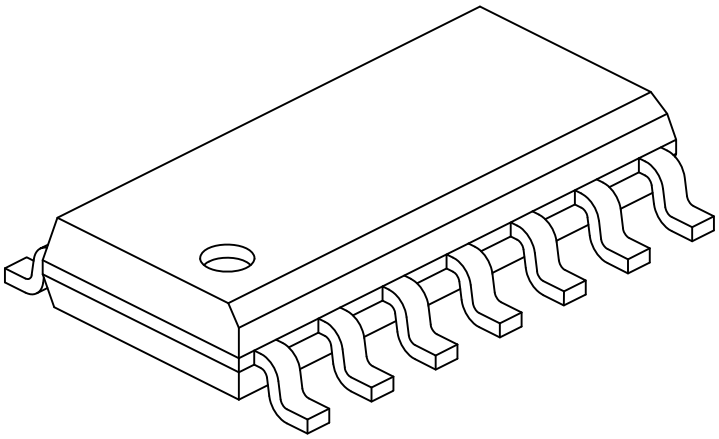



Microchip Technology Drawing No. C04-065-SL Rev D Sheet 1 of 2


© 2019-2020 Microchip Technology Inc.

## 14-Lead Plastic Small Outline (SL) - Narrow, 3.90 mm Body [SOIC]

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



| Units | | MILLIMETERS | | |
|---|---|---|---|---|
| Dimension Limits | | MIN | NOM | MAX |
| Number of Pins | N | | 14 | |
| Pitch | e | | 1.27 BSC | |
| Overall Height | A | - | - | 1.75 |
| Molded Package Thickness | A2 | 1.25 | - | - |
| Standoff § | A1 | 0.10 | - | 0.25 |
| Overall Width | E | | 6.00 BSC | |
| Molded Package Width | E1 | | 3.90 BSC | |
| Overall Length | D | | 8.65 BSC | |
| Chamfer (Optional) | h | 0.25 | - | 0.50 |
| Foot Length | L | 0.40 | - | 1.27 |
| Footprint | L1 | | 1.04 REF | |
| Lead Angle | Θ | 0° | - | - |
| Foot Angle | φ | 0° | - | 8° |
| Lead Thickness | c | 0.10 | - | 0.25 |
| Lead Width | b | 0.31 | - | 0.51 |
| Mold Draft Angle Top | α | 5° | - | 15° |
| Mold Draft Angle Bottom | β | 5° | - | 15° |

Notes:

1. Pin 1 visual index feature may vary, but must be located within the hatched area.
2. § Significant Characteristic
3. Dimension D does not include mold flash, protrusions or gate burrs, which shall not exceed 0.15 mm per end. Dimension E1 does not include interlead flash or protrusion, which shall not exceed 0.25 mm per side.
4. Dimensioning and tolerancing per ASME Y14.5M

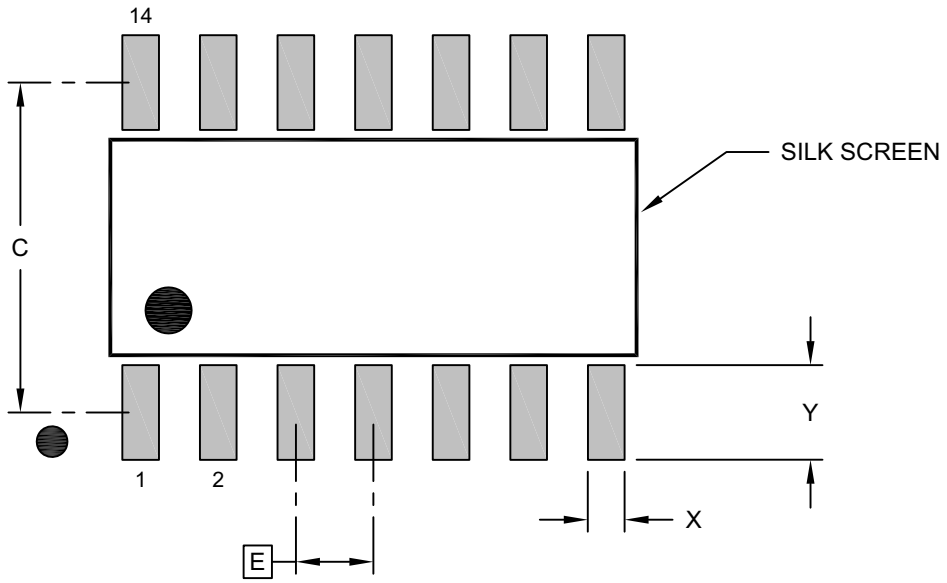
   BSC: Basic Dimension. Theoretically exact value shown without tolerances.

   REF: Reference Dimension, usually without tolerance, for information purposes only.
5. Datums A & B to be determined at Datum H.

Microchip Technology Drawing No. C04-065-SL Rev D Sheet 2 of 2

### 14-Lead Plastic Small Outline (SL) - Narrow, 3.90 mm Body [SOIC]

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging




RECOMMENDED LAND PATTERN

| Units | | MILLIMETERS | | |
|---|---|---|---|---|
| Dimension Limits | | MIN | NOM | MAX |
| Contact Pitch | E | | 1.27 BSC | |
| Contact Pad Spacing | C | | 5.40 | |
| Contact Pad Width (X14) | X | | | 0.60 |
| Contact Pad Length (X14) | Y | | | 1.55 |

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M

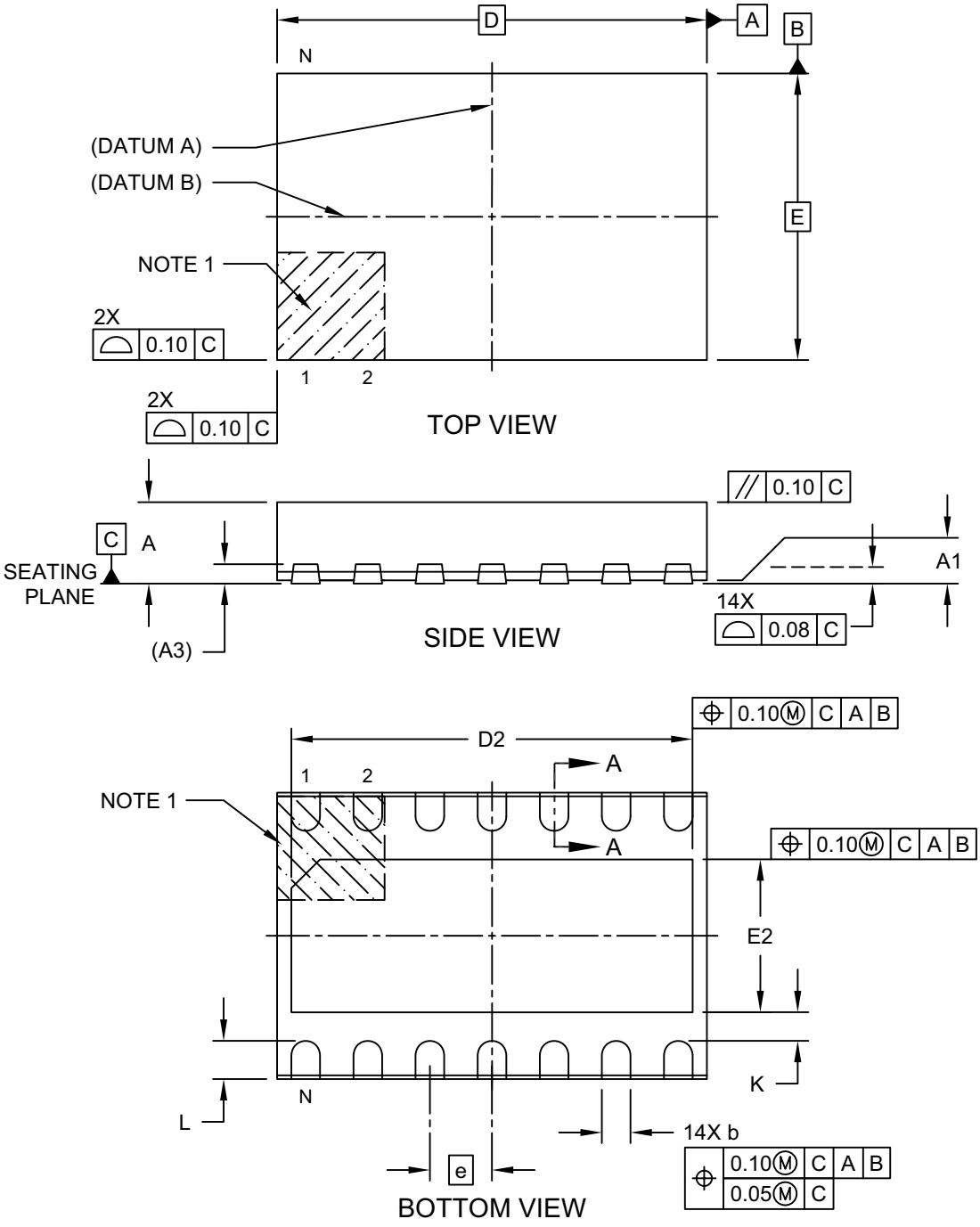
   BSC: Basic Dimension. Theoretically exact value shown without tolerances.

Microchip Technology Drawing No. C04-2065-SL Rev D


© 2019-2020 Microchip Technology Inc.

**14-Lead Very Thin Plastic Dual Flat, No Lead Package (QBB) - 4.5x3 mm Body [VDFN] With 1.6x4.2 mm Exposed Pad and Stepped Wettable Flanks**

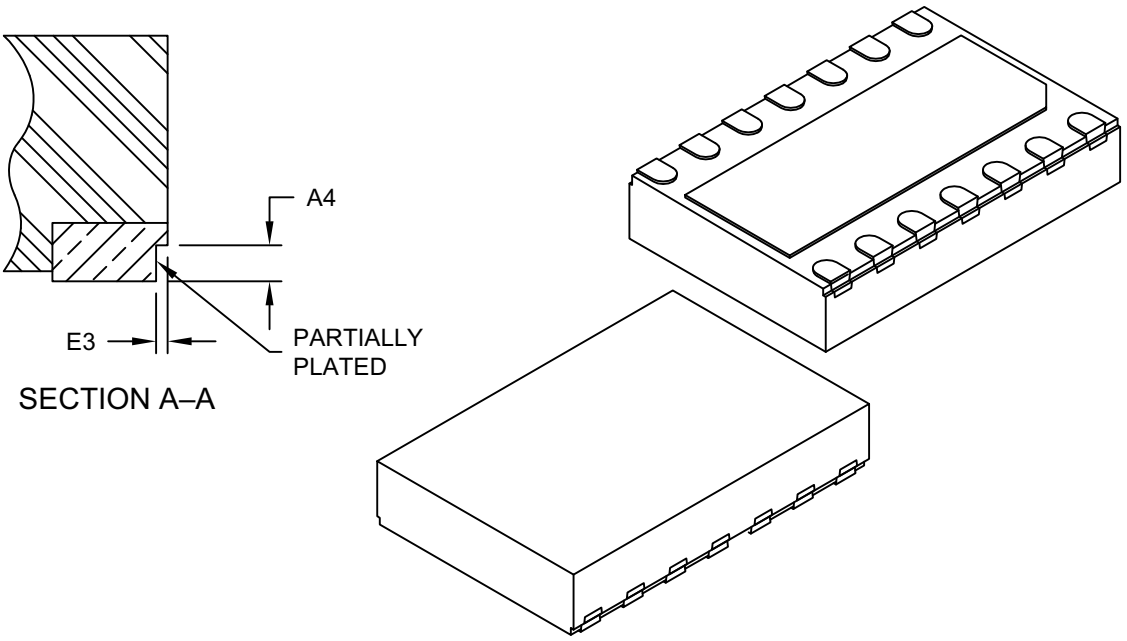
**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging




Microchip Technology Drawing C04-21361 Rev B Sheet 1 of 2

### 14-Lead Very Thin Plastic Dual Flat, No Lead Package (QBB) - 4.5x3 mm Body [VDFN] With 1.6x4.2 mm Exposed Pad and Stepped Wettable Flanks

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging




| Units | | MILLIMETERS | | |
|---|---|---|---|---|
| Dimension Limits | | MIN | NOM | MAX |
| Number of Terminals | N | | 14 | |
| Pitch | e | | 0.65 BSC | |
| Overall Height | A | 0.80 | 0.85 | 0.90 |
| Standoff | A1 | 0.00 | 0.03 | 0.05 |
| Terminal Thickness | A3 | | 0.203 REF | |
| Overall Length | D | | 4.50 BSC | |
| Exposed Pad Length | D2 | 4.15 | 4.20 | 4.25 |
| Overall Width | E | | 3.00 BSC | |
| Exposed Pad Width | E2 | 1.50 | 1.60 | 1.70 |
| Terminal Width | b | 0.27 | 0.32 | 0.37 |
| Terminal Length | L | 0.35 | 0.40 | 0.45 |
| Terminal-to-Exposed-Pad | K | 0.20 | - | - |
| Wettable Flank Step Cut Depth | A4 | 0.10 | 0.13 | 0.15 |
| Wettable Flank Step Cut Width | E3 | - | - | 0.04 |

Notes:

1. Pin 1 visual index feature may vary, but must be located within the hatched area.
2. Package is saw singulated
3. Dimensioning and tolerancing per ASME Y14.5M

   BSC: Basic Dimension. Theoretically exact value shown without tolerances.

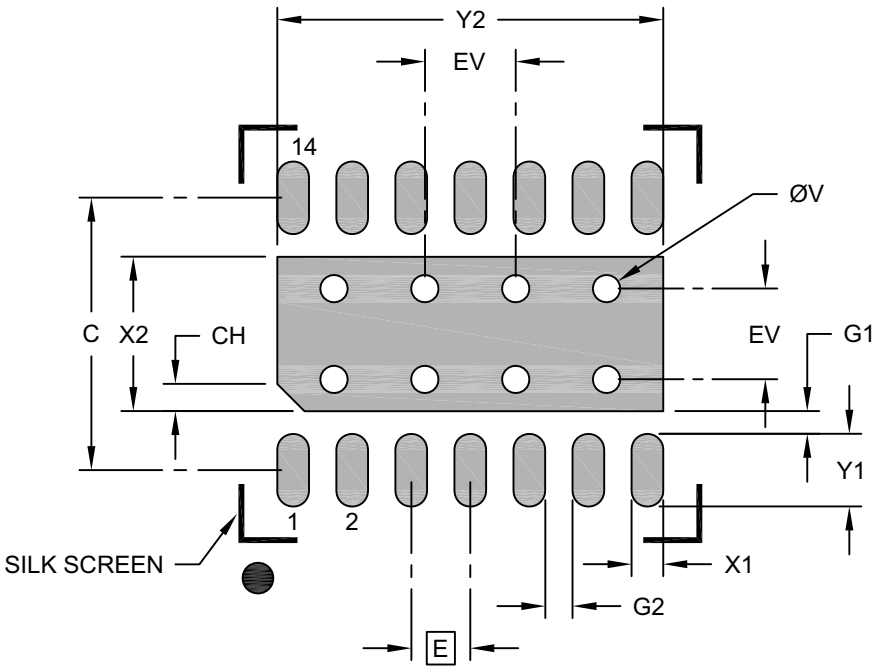
   REF: Reference Dimension, usually without tolerance, for information purposes only.

Microchip Technology Drawing C04-21361 Rev B Sheet 2 of 2


© 2019-2020 Microchip Technology Inc.

### 14-Lead Very Thin Plastic Dual Flat, No Lead Package (QBB) - 4.5x3 mm Body [VDFN] With 1.6x4.2 mm Exposed Pad and Stepped Wettable Flanks

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging




RECOMMENDED LAND PATTERN

| Units | | MILLIMETERS | | |
|---|---|---|---|---|
| Dimension Limits | | MIN | NOM | MAX |
| Contact Pitch | E | | 0.65 BSC | |
| Optional Center Pad Width | X2 | | | 1.70 |
| Optional Center Pad Length | Y2 | | | 4.25 |
| Contact Pad Spacing | C | | 3.00 | |
| Contact Pad Width (X14) | X1 | | | 0.35 |
| Contact Pad Length (X14) | Y1 | | | 0.80 |
| Pin 1 Index Chamfer | CH | | 0.30 | |
| Contact Pad to Center Pad (X14) | G1 | 0.20 | | |
| Contact Pad to Center Pad (X12) | G2 | 0.20 | | |
| Thermal Via Diameter | V | | 0.30 | |
| Thermal Via Pitch | EV | | 1.00 | |

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M

   BSC: Basic Dimension. Theoretically exact value shown without tolerances.

2. For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-23361 Rev B

**NOTES:**


© 2019-2020 Microchip Technology Inc.

# APPENDIX A: REVISION HISTORY

## Revision B (December 2020)

- Added AEC-Q100 qualification to **Special Features** section.
- Updated **Table 3-1**.
- Updated **Register 3-2**.
- Updated **Register 3-4**.
- Updated **Register 3-5**.
- Updated **Register 3-7**.
- Updated **Register 3-23**.
- Updated **Section 3.3.1, RAM ECC**.
- Updated **Section 4.1, SFR Access**.
- Updated **Section 4.1.3, SFR WRITE – WRITE**.
- Updated **Table 7-6**.

## Revision A (April 2019)

- Original release of this document

## APPENDIX B: CAN FD CONFORMANCE

The MCP2518FD passed the CAN FD conformance tests specified in ISO 16845-1:2016.

ISO 11898-1:2015 lists non-mandatory features. Table B-1 clarifies which optional features are implemented.

**TABLE B-1: ISO OPTIONAL FEATURES**

| No. | Optional Feature | Implemented |
|---|---|---|
| 1 | FD frame format | Yes |
| 2 | Disabling of frame formats | Yes. Classical CAN frame format. |
| 3 | Limited LLC frames | No. Full range of IDs and DLCs implemented. |
| 4 | No transmission of frames including padding bytes | N/A. See No. 3. |
| 5 | LLC Abort interface | Yes |
| 6 | ESI and BRS bit values | Yes |
| 7 | Method to provide MAC data consistency | Yes |
| 8 | Time and time triggering | Start of Frame output. |
| 9 | Time stamping | Yes. 32 bit TBC. |
| 10 | Bus monitoring mode | Yes |
| 11 | Handle | Yes |
| 12 | Restricted operation | Yes |
| 13 | Separate prescalers for nominal bits and for data bits | Yes |
| 14 | Disabling of automatic retransmission | Yes |
| 15 | Maximum number of retransmissions | Yes. One, 3 or unlimited. |
| 16 | Disabling of protocol exception event on res bit detected recessive | Yes. Selectable. |
| 17 | PCS_Status | No |
| 18 | Edge filtering during the bus integration state | Yes. Selectable. |
| 19 | Time resolution for SSP placement | Yes. 128 $T_Q$. Measured, manual or disabled. |
| 20 | FD_T/R message | TX and RX interrupts. |


© 2019-2020 Microchip Technology Inc.

## PRODUCT IDENTIFICATION SYSTEM

To order or obtain information, e.g., on pricing or delivery, refer to the factory or the listed sales office.

| PART NO. | X[1] | -X | /XX | XXX |
|---|---|---|---|---|
| Device | Tape and Reel Option | Temperature Range | Package | Qualification |

| | | | |
|---|---|---|---|
| **Device:** | MCP2518FD: CAN FD Controller | | |
| **Tape and Reel Option:** | T | = | Tape and Reel[1] |
| **Temperature Range:** | E | = | -40°C to +125°C (Extended) |
| | H | = | -40°C to +150°C (High) |
| **Package:** | SL | = | Plastic SOIC (150 mil Body), 14-Lead |
| | QBB | = | Plastic VDFN (4.5 x 3 mm Body), 14-Lead with 1.6 x 4.2 mm Exposed Pad and Stepped Wettable Flanks |
| **Qualification** | Blank | = | Standard qualification |
| | VAO | = | Automotive Qualification |

**Examples:**

a) MCP2518FDT-E/SL = Tape and Reel, Extended Temperature, Plastic SOIC (150 mil Body), 14-Lead

b) MCP2518FDT-H/SL = Tape and Reel, High Temperature, Plastic SOIC (150 mil Body), 14-Lead

c) MCP2518FDT- E/QBB = Tape and Reel, Extended Temperature, Plastic VDFN (4.5 x 3 mm Body) 14-Lead with 1.6 x 4.2 mm Exposed Pad and Stepped Wettable Flanks

d) MCP2518FDT-H/QBB = Tape and Reel, High Temperature, Plastic VDFN (4.5 x 3 mm Body), 14-Lead, with 1.6 x 4.2 mm, Exposed Pad and Stepped Wettable Flanks

e) MCP2518FDT-H/QBBVAO = Tape and Reel, High Temperature, Plastic VDFN (4.5 x 3 mm Body), 14-Lead, with 1.6 x 4.2 mm, Exposed Pad and Stepped Wettable Flanks, Automotive Qualified

**Note 1:** Tape and Reel identifier only appears in the catalog part number description. This identifier is used for ordering purposes and is not printed on the device package. Check with your Microchip Sales Office for package availability with the Tape and Reel option.

**NOTES:**


© 2019-2020 Microchip Technology Inc.

**Note the following details of the code protection feature on Microchip devices:**

- Microchip products meet the specifications contained in their particular Microchip Data Sheet.
- Microchip believes that its family of products is secure when used in the intended manner and under normal conditions.
- There are dishonest and possibly illegal methods being used in attempts to breach the code protection features of the Microchip devices. We believe that these methods require using the Microchip products in a manner outside the operating specifications contained in Microchip's Data Sheets. Attempts to breach these code protection features, most likely, cannot be accomplished without violating Microchip's intellectual property rights.
- Microchip is willing to work with any customer who is concerned about the integrity of its code.
- Neither Microchip nor any other semiconductor manufacturer can guarantee the security of its code. Code protection does not mean that we are guaranteeing the product is "unbreakable." Code protection is constantly evolving. We at Microchip are committed to continuously improving the code protection features of our products. Attempts to break Microchip's code protection feature may be a violation of the Digital Millennium Copyright Act. If such acts allow unauthorized access to your software or other copyrighted work, you may have a right to sue for relief under that Act.

Information contained in this publication is provided for the sole purpose of designing with and using Microchip products. Information regarding device applications and the like is provided only for your convenience and may be superseded by updates. It is your responsibility to ensure that your application meets with your specifications.

THIS INFORMATION IS PROVIDED BY MICROCHIP "AS IS". MICROCHIP MAKES NO REPRESENTATIONS OR WARRANTIES OF ANY KIND WHETHER EXPRESS OR IMPLIED, WRITTEN OR ORAL, STATUTORY OR OTHERWISE, RELATED TO THE INFORMATION INCLUDING BUT NOT LIMITED TO ANY IMPLIED WARRANTIES OF NON-INFRINGEMENT, MERCHANTABILITY, AND FITNESS FOR A PARTICULAR PURPOSE OR WARRANTIES RELATED TO ITS CONDITION, QUALITY, OR PERFORMANCE.

IN NO EVENT WILL MICROCHIP BE LIABLE FOR ANY INDIRECT, SPECIAL, PUNITIVE, INCIDENTAL OR CONSEQUENTIAL LOSS, DAMAGE, COST OR EXPENSE OF ANY KIND WHATSOEVER RELATED TO THE INFORMATION OR ITS USE, HOWEVER CAUSED, EVEN IF MICROCHIP HAS BEEN ADVISED OF THE POSSIBILITY OR THE DAMAGES ARE FORESEEABLE. TO THE FULLEST EXTENT ALLOWED BY LAW, MICROCHIP'S TOTAL LIABILITY ON ALL CLAIMS IN ANY WAY RELATED TO THE INFORMATION OR ITS USE WILL NOT EXCEED THE AMOUNT OF FEES, IF ANY, THAT YOU HAVE PAID DIRECTLY TO MICROCHIP FOR THE INFORMATION. Use of Microchip devices in life support and/or safety applications is entirely at the buyer's risk, and the buyer agrees to defend, indemnify and hold harmless Microchip from any and all damages, claims, suits, or expenses resulting from such use. No licenses are conveyed, implicitly or otherwise, under any Microchip intellectual property rights unless otherwise stated.

*For information regarding Microchip's Quality Management Systems, please visit www.microchip.com/quality.*

**Trademarks**

The Microchip name and logo, the Microchip logo, Adaptec, AnyRate, AVR, AVR logo, AVR Freaks, BesTime, BitCloud, chipKIT, chipKIT logo, CryptoMemory, CryptoRF, dsPIC, FlashFlex, flexPWR, HELDO, IGLOO, JukeBlox, KeeLoq, Kleer, LANCheck, LinkMD, maXStylus, maXTouch, MediaLB, megaAVR, Microsemi, Microsemi logo, MOST, MOST logo, MPLAB, OptoLyzer, PackeTime, PIC, picoPower, PICSTART, PIC32 logo, PolarFire, Prochip Designer, QTouch, SAM-BA, SenGenuity, SpyNIC, SST, SST Logo, SuperFlash, Symmetricom, SyncServer, Tachyon, TempTrackr, TimeSource, tinyAVR, UNI/O, Vectron, and XMEGA are registered trademarks of Microchip Technology Incorporated in the U.S.A. and other countries.

APT, ClockWorks, The Embedded Control Solutions Company, EtherSynch, FlashTec, Hyper Speed Control, HyperLight Load, IntelliMOS, Libero, motorBench, mTouch, Powermite 3, Precision Edge, ProASIC, ProASIC Plus, ProASIC Plus logo, Quiet-Wire, SmartFusion, SyncWorld, Temux, TimeCesium, TimeHub, TimePictra, TimeProvider, Vite, WinPath, and ZL are registered trademarks of Microchip Technology Incorporated in the U.S.A.

Adjacent Key Suppression, AKS, Analog-for-the-Digital Age, Any Capacitor, AnyIn, AnyOut, BlueSky, BodyCom, CodeGuard, CryptoAuthentication, CryptoAutomotive, CryptoCompanion, CryptoController, dsPICDEM, dsPICDEM.net, Dynamic Average Matching, DAM, ECAN, EtherGREEN, In-Circuit Serial Programming, ICSP, INICnet, Inter-Chip Connectivity, JitterBlocker, KleerNet, KleerNet logo, memBrain, Mindi, MiWi, MPASM, MPF, MPLAB Certified logo, MPLIB, MPLINK, MultiTRAK, NetDetach, Omniscient Code Generation, PICDEM, PICDEM.net, PICkit, PICtail, PowerSmart, PureSilicon, QMatrix, REAL ICE, Ripple Blocker, SAM-ICE, Serial Quad I/O, SMART-I.S., SQI, SuperSwitcher, SuperSwitcher II, Total Endurance, TSHARC, USBCheck, VariSense, ViewSpan, WiperLock, Wireless DNA, and ZENA are trademarks of Microchip Technology Incorporated in the U.S.A. and other countries.

SQTP is a service mark of Microchip Technology Incorporated in the U.S.A.

The Adaptec logo, Frequency on Demand, Silicon Storage Technology, and Symmcom are registered trademarks of Microchip Technology Inc. in other countries.

GestIC is a registered trademark of Microchip Technology Germany II GmbH & Co. KG, a subsidiary of Microchip Technology Inc., in other countries.

All other trademarks mentioned herein are property of their respective companies.

© 2019-2020, Microchip Technology Incorporated, All Rights Reserved.

ISBN: 978-1-5224-7119-6

# Worldwide Sales and Service

## AMERICAS

**Corporate Office**
2355 West Chandler Blvd.
Chandler, AZ 85224-6199
Tel: 480-792-7200
Fax: 480-792-7277
Technical Support:
http://www.microchip.com/support
Web Address:
www.microchip.com

**Atlanta**
Duluth, GA
Tel: 678-957-9614
Fax: 678-957-1455

**Austin, TX**
Tel: 512-257-3370

**Boston**
Westborough, MA
Tel: 774-760-0087
Fax: 774-760-0088

**Chicago**
Itasca, IL
Tel: 630-285-0071
Fax: 630-285-0075

**Dallas**
Addison, TX
Tel: 972-818-7423
Fax: 972-818-2924

**Detroit**
Novi, MI
Tel: 248-848-4000

**Houston, TX**
Tel: 281-894-5983

**Indianapolis**
Noblesville, IN
Tel: 317-773-8323
Fax: 317-773-5453
Tel: 317-536-2380

**Los Angeles**
Mission Viejo, CA
Tel: 949-462-9523
Fax: 949-462-9608
Tel: 951-273-7800

**Raleigh, NC**
Tel: 919-844-7510

**New York, NY**
Tel: 631-435-6000

**San Jose, CA**
Tel: 408-735-9110
Tel: 408-436-4270

**Canada - Toronto**
Tel: 905-695-1980
Fax: 905-695-2078

## ASIA/PACIFIC

**Australia - Sydney**
Tel: 61-2-9868-6733

**China - Beijing**
Tel: 86-10-8569-7000

**China - Chengdu**
Tel: 86-28-8665-5511

**China - Chongqing**
Tel: 86-23-8980-9588

**China - Dongguan**
Tel: 86-769-8702-9880

**China - Guangzhou**
Tel: 86-20-8755-8029

**China - Hangzhou**
Tel: 86-571-8792-8115

**China - Hong Kong SAR**
Tel: 852-2943-5100

**China - Nanjing**
Tel: 86-25-8473-2460

**China - Qingdao**
Tel: 86-532-8502-7355

**China - Shanghai**
Tel: 86-21-3326-8000

**China - Shenyang**
Tel: 86-24-2334-2829

**China - Shenzhen**
Tel: 86-755-8864-2200

**China - Suzhou**
Tel: 86-186-6233-1526

**China - Wuhan**
Tel: 86-27-5980-5300

**China - Xian**
Tel: 86-29-8833-7252

**China - Xiamen**
Tel: 86-592-2388138

**China - Zhuhai**
Tel: 86-756-3210040

## ASIA/PACIFIC

**India - Bangalore**
Tel: 91-80-3090-4444

**India - New Delhi**
Tel: 91-11-4160-8631

**India - Pune**
Tel: 91-20-4121-0141

**Japan - Osaka**
Tel: 81-6-6152-7160

**Japan - Tokyo**
Tel: 81-3-6880- 3770

**Korea - Daegu**
Tel: 82-53-744-4301

**Korea - Seoul**
Tel: 82-2-554-7200

**Malaysia - Kuala Lumpur**
Tel: 60-3-7651-7906

**Malaysia - Penang**
Tel: 60-4-227-8870

**Philippines - Manila**
Tel: 63-2-634-9065

**Singapore**
Tel: 65-6334-8870

**Taiwan - Hsin Chu**
Tel: 886-3-577-8366

**Taiwan - Kaohsiung**
Tel: 886-7-213-7830

**Taiwan - Taipei**
Tel: 886-2-2508-8600

**Thailand - Bangkok**
Tel: 66-2-694-1351

**Vietnam - Ho Chi Minh**
Tel: 84-28-5448-2100

## EUROPE

**Austria - Wels**
Tel: 43-7242-2244-39
Fax: 43-7242-2244-393

**Denmark - Copenhagen**
Tel: 45-4485-5910
Fax: 45-4485-2829

**Finland - Espoo**
Tel: 358-9-4520-820

**France - Paris**
Tel: 33-1-69-53-63-20
Fax: 33-1-69-30-90-79

**Germany - Garching**
Tel: 49-8931-9700

**Germany - Haan**
Tel: 49-2129-3766400

**Germany - Heilbronn**
Tel: 49-7131-72400

**Germany - Karlsruhe**
Tel: 49-721-625370

**Germany - Munich**
Tel: 49-89-627-144-0
Fax: 49-89-627-144-44

**Germany - Rosenheim**
Tel: 49-8031-354-560

**Israel - Ra'anana**
Tel: 972-9-744-7705

**Italy - Milan**
Tel: 39-0331-742611
Fax: 39-0331-466781

**Italy - Padova**
Tel: 39-049-7625286

**Netherlands - Drunen**
Tel: 31-416-690399
Fax: 31-416-690340

**Norway - Trondheim**
Tel: 47-7288-4388

**Poland - Warsaw**
Tel: 48-22-3325737

**Romania - Bucharest**
Tel: 40-21-407-87-50

**Spain - Madrid**
Tel: 34-91-708-08-90
Fax: 34-91-708-08-91

**Sweden - Gothenberg**
Tel: 46-31-704-60-40

**Sweden - Stockholm**
Tel: 46-8-5090-4654

**UK - Wokingham**
Tel: 44-118-921-5800
Fax: 44-118-921-5820


© 2019-2020 Microchip Technology Inc.